

# ***In silico*, nearly half of junction-spanning 5-6-5 gapmer designs across 38 modelled NR4A3 fusion junctions of five extraskelatal myxoid chondrosarcoma partner genes pair a wild-type parent gene over a ten-base-pair duplex through the catalytic gap, and, as an identity between base counts, a longer gap trades gap-level margin against parent-paired gap DNA**

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## **ABSTRACT**

In extraskelatal myxoid chondrosarcoma (EMC), an ultra-rare sarcoma, a variable partner gene fuses to *NR4A3*, creating a junction present in no normal transcript. An antisense gapmer could in principle cleave it and spare both parents; none is reported in the literature retrieved here. Of 190 junction-spanning designs (16-mers, 5-6-5, a six-nucleotide DNA gap) across the 38 in-frame junctions of five modelled partners, 87 pair their catalytic gap against one of six mature parents searched over ten or more contiguous base pairs, 61 of those 87 against wild-type *NR4A3* (59 at one site) and 85 against their own parents. That is a rate over designs: 35 of 38 junctions have one clearing that screen. Scrambles, the weakest null here, reach 6.2% against 45.8% observed; chimeras at real exon termini of the same two transcripts, almost never reported, reach 40.6%, so no excess specific to this disease's breakpoints is resolved. Ten base pairs is adopted, not measured and not separated: at seven, 175 of 190 pair a parent and 9 of 38 junctions clear, but so does that null, to 91.4% against the 92.1% observed there; at six, the gap's own length, 181 of 190; and across cuts of six to thirteen the excess over the strongest null changes sign four times (§2.5). A longer gap quiets the transcript screens, partly by construction at a fixed mismatch budget: parent-clean designs per junction rise from 2.7 to 6.7 while the liable count stays flat, and gap-level margin is won only by conceding parent-paired gap DNA. Search depth moves it too: six of nine designs clean at the default ceiling are not at ten times it. Three designs survive every screen, two at any parent-duplex threshold; none at a junction any patient is reported to carry, so every named reagent carries loads. The work is computational: nothing was synthesised or tested; nothing asserts efficacy, safety, delivery or clinical readiness. Every sequence named is a research reagent, not for administration to any person or animal. Two leads carry off-target loads and longest parent runs of eight and nine base pairs, so a cut of eight condemns both: 5'-GGGCATATCATCAAAC-3' at *EWSR1* exon 12 and 5'-GGGCATATCTTGTGTG-3' at *TAF15* exon 6, with two more for coverage; and three not to be used: designs the mature-parent screen clears or cannot read, pairing their whole gap against unspliced wild-type *NR4A3*. The pipeline is released for breakpoints outside this panel.

**Keywords** antisense oligonucleotide; gapmer; RNase-H1; fusion transcript; NR4A3; extraskelatal myxoid chondrosarcoma; oligonucleotide design pipeline; off-target screening

## **1 · Introduction**

EMC is defined in the large majority of cases by an in-frame fusion of *EWSR1* to the orphan nuclear receptor *NR4A3*,<sup>1</sup> with *TAF15* a substantial minority and *TCF12* and *TFG* rare.<sup>2</sup> *FUS::NR4A3* is reported in a recent series that identified it by sequencing in two of five variant EMCs.<sup>3</sup> Next-generation sequencing of six EMCs finds few recurrent secondary mutations beyond the fusion,<sup>4</sup> so it is to a first approximation the single clonal driver. That is an absence of co-mutations in six tumours rather than a dependency: no study cited here tests whether an EMC cell requires the fusion for growth or survival, and no knockdown, dependency screen or xenograft in this disease is cited anywhere in this paper, so the target's necessity to the tumour is assumed throughout and is not shown. In the *EWSR1* and *TCF12* junction types described, the predicted product joins the partner's amino-terminal transactivation domain to essentially the entire *NR4A3* protein, including its nuclear-receptor DNA-binding domain.<sup>1,5</sup> Those two sources are the extent of the claim: a junction at *NR4A3* exon 2 lies upstream of the start codon and does not

describe the same product at all (§2.6). The disease is ultra-rare: it accounts for approximately 1–3% of soft-tissue sarcomas, in a course a review of it describes as typically indolent yet metastatic.<sup>6</sup> No incidence, prevalence, annual case count or survival figure for this disease is read from any source here, so nothing in this paper sizes the population a reagent would serve in absolute terms.

That driver is currently untargeted. Surgery with clear margins is the backbone of localised disease, and for advanced disease no clinically validated agent directly targets *NR4A3*.<sup>6</sup> The largest EMC-specific prospective study, a single-arm phase 2 of pazopanib in centrally confirmed *NR4A3*-translocated disease, returned four objective responses in 22 evaluable patients;<sup>7</sup> anthracycline-based chemotherapy returned four in ten evaluable patients in a molecularly confirmed retrospective series of eleven, a result that series presents as running counter to the prior record.<sup>8</sup> The review describes a low objective response rate to anthracycline-based chemotherapy,<sup>6</sup> which the retrospective series just cited runs counter to, and the pazopanib report opens on low sensitivity to cytotoxic

chemotherapy generally;<sup>7</sup> none of the three publishes a response rate by line of therapy. Every figure in this paragraph is an objective response rate, and no progression-free survival, disease-control rate, duration of response or overall survival figure is read from any of these reports here, so the paragraph bounds what has been reported to shrink tumours and is not a characterisation of how these agents are used or of how long patients live. The population a fusion-directed agent would address is close to the whole disease: across 58 molecularly confirmed cases, 79% carried *EWSR1::NR4A3*, 16% *TAF15::NR4A3* and 3% *TCF12::NR4A3*.<sup>9</sup>

The chimeric mRNA offers a discrimination handle its protein product does not. The *NR4A3* ligand-binding domain is retained near-intact and identical in sequence to wild-type *NR4A3*, so a ligand that engages it cannot distinguish fusion from wild type. The breakpoint junction can: it is a contiguous stretch of sequence present in no normal transcript and absent from both parent transcripts, so selectivity can in principle be enforced by base-pairing rather than by protein conformation. It is the target of every design here.

Targeting a fusion breakpoint with an oligonucleotide is not new. The approach has a continuous lineage from 1991,<sup>10</sup> including RNase-H-dependent antisense at a sarcoma fusion breakpoint in 1997,<sup>11</sup> and the fusion-exclusivity rationale was stated as a general principle in 2005.<sup>12</sup> Parental sparing has been reported at four fusions, two with a readout on the wild-type parent transcript and two asserting junction specificity without one.<sup>13–16</sup> A bi-shRNA lipoplex against the *EWSR1::FLI1* junction reached preclinical justification,<sup>17</sup> and an *N*-acetylgalactosamine (GalNAc)-conjugated junction siRNA in fibrolamellar hepatocellular carcinoma passed the delivery gate in a rare fusion-driven cancer, reaching durable growth inhibition in patient-derived xenografts.<sup>18</sup> That last precedent does not transfer to this disease and is cited for the junction, not for the route: GalNAc conjugates enter cells through the asialoglycoprotein receptor, which is a liver receptor, and the report's own basis for using it is that fibrolamellar tumours retain that receptor at hepatocyte levels. An extraskelatal soft-tissue sarcoma satisfies neither premise, and no delivery route is proposed anywhere in this paper. The contribution here is the indication rather than the modality: across 5,153 unique records retrieved from Europe PMC, four mention *EWSR1::NR4A3* at title or abstract level, and none is an oligonucleotide study. Those four resolve to three papers, one of them indexed twice; that de-duplication is recorded in the working record travelling with the archive rather than in the retrieval artefact, which holds the record identifiers and the counts.

Two questions follow that the field has not asked of this disease. The first is whether specificity sorts by partner at all. No junction-directed oligonucleotide study has compared specificity across fusion partners: each precedent addresses one fusion (a bi-shRNA lipoplex against the *EWSR1::FLI1* junction in Ewing sarcoma, antisense oligonucleotides against *NAB2::STAT6* in solitary fibrous tumour; §4.1), and none asks whether the partner changes what is achievable. In this disease the partner varies, and partner identity may not be clinically inert: across the two series of antiangiogenic tyrosine-kinase inhibition in advanced EMC that report a partner breakdown at all, no objective response is reported in a *TAF15* patient, on a *TAF15* arm of three to five patients whose Wilson upper bound remains compatible with equal response.<sup>7,19</sup> The two series are not shown to be independent cohorts, which is why the count is a

range and why they are not pooled: the smaller ran at an institution that was a site of the larger trial, under the same senior investigator, so the same patients may appear in both and the distinct *TAF15* arm may be as few as three. Neither primary report's retrieved record states the per-arm denominators, and the sunitinib record states the partner split only qualitatively. Both full texts are paywalled here, so both may state more than their records show. Both denominators are read from published reviews of the two series.<sup>20,21</sup> The second is whether a junction oligonucleotide must be bespoke per patient, or whether one sequence can serve more than one fusion, which decides whether the deployable artefact for an ultra-rare disease is a stock reagent or a panel.

This paper answers both from sequence, sets out where a computational screen stops being able to answer them, and ends by naming the oligonucleotides to synthesise, the controls that make a knockdown result interpretable, and the pre-registrable threshold that would falsify, at its top margin, the ranking every candidate here is ordered by (§4). The design and screening procedure is released with the artefacts, so a candidate can be designed by the same route for a breakpoint the panel does not carry (§4.5). Box 1 collects the sequences, the cautions and the terms of art the sections below use before the Methods (§6) define them.

### Box 1 · Sequences, cautions and the void condition

**Research use only.** Every sequence in this box is a research reagent for laboratory investigation only and must not be administered to any person or animal; the operative statement is in Declarations and governs this manuscript throughout, not any one section. Do not order an oligonucleotide by copying it out of this PDF: the canonical record is `fusion-junction-aso-sequences.csv` and `fusion-junction-aso-sequences.fasta` (§6). Before any oligonucleotide is ordered, the breakpoint of the cell line or tumour sample used as the test article must be established at nucleotide resolution by RNA sequencing (§4.4): every design here is specific to the exon pair or pairs it was tiled at, and none is valid for an unverified junction. That is a laboratory precondition for an experiment and not a clinical indication to sequence a patient: no sequence named here may be administered to anyone, so no sequencing result about a person changes what is available to them.

Every line below points at a fuller statement in the section cited, and none of it is argued here.

**The terms of art (§6).** A *seam* is a donor exon joined to an acceptor exon, and each seam is tiled by several designs. Each design is one *register* of its junction, one way of sliding the 16-mer while the breakpoint still falls inside the six-nucleotide DNA *catalytic gap* RNase-H1 cleaves within; the 5-6-5 geometry admits five per junction. A design's *gap-level margin* is the count of junction-unique bases inside that geometry's gap on the shorter side of the breakpoint: the number of bases a wild-type parent would have to match by coincidence in order to pair the whole gap. The gap-level margin is the axis this paper's central negative is stated on; where a table instead ranks by predicted liability, or by the conventional design rules of §2.10, that table's caption says so.

**Exon numbers here are transcript exon indices, not coding-exon indices,** counted from the transcript 5' end of the model §6 names, including non-coding exons. The two conventions differ for *TCF12*, *TFG* and *NR4A3*; the numbers used here are what a breakpoint is matched to a design by, and §6 gives the accession for

every gene so a report numbered the other way can be reconciled. The consequence is not cosmetic: at one *EWSR1* exon-13 donor this paper's *NR4A3* exon-3 and exon-2 acceptors carry five registers each with no molecule in common, and two registers at each carry `do_not_order`, for different reasons (§2.5, §2.6). An acceptor exon number read under the wrong convention therefore selects a different reagent.

A *near-match* is a transcript window pairing a design at 14 or more of its 16 positions, and is *gap-paired* where the six gap positions are themselves paired, which is the class RNase-H1 could cleave; a gap-paired near-match on the sense strand is what this paper calls a *gap-paired sense-strand match*. That name replaced *cleavage risk* on 2026-08-19, because the earlier term named a catalytic outcome for what is a sequence observation. No screen here predicts cleavage (§5); each grades hybridisation only. A design is *clean* where a complete hit list at a stated search depth returns no sense-strand near-match, and a design's *load* is its predicted off-target burden counted as near-matches. A design is *liable* where a wild-type parent pairs its whole catalytic gap over a contiguous run reaching a stated length, ten base pairs unless another is named, and the count of liable designs is what §2.5 reports at each length it reads the same screen at. That length is a whole-duplex count and not the enzyme's unit: the figure ten is taken from is a range of 7 to 10 RNA:DNA hybridised nucleotides, and a 5-6-5 design holds six of those against every target, below that range in every design, so four or more pairs of any ten-base-pair run counted here are locked-wing pairs. The count is a stratification of the panel at a stated cut and not a prediction of cleavage (§5, §6). Counts from that hit list come at one of two search depths, a default alignment ceiling and a tenfold deeper one (a hitlist of 50 alignments per query, of which 15 are retained, against 500 with retention raised to match so that no list is truncated; §6), and each is reported with the depth it was taken at. Five screens are applied, numbered here as in §6: (1) the alignment screen, (2) the exhaustive transcript scan, (3) the pre-mRNA screen, (4) the mature-parent screen and (5) the genome scan.

**The two lead reagents to synthesise (§4.1, which names two more for coverage).** 5'-GGGCATATCATCAAC-3' at *EWSR1* exon 12 joined to *NR4A3* exon 3, carrying 123 gap-paired sense-strand near-matches at six gene loci at the deeper search ceiling, of which 82 hits fall on computationally predicted gene models rather than curated ones (§5), together with a sense-strand near-match in wild-type *TAF15* precursor RNA (§4.3); and 5'-GGGCATATCTTGTGTG-3' at *TAF15* exon 6, carrying 8 such near-matches at five loci and no sense-strand pre-mRNA site. Both hold the top gap-level margin of 3, and neither has been synthesised or tested. Neither clears the parent screen with room to spare: their longest parent runs through the gap are eight and nine base pairs against a criterion of ten, so at any cut of eight or below, which is inside the range §5 bounds that criterion by, both fall inside the liability class this paper's central negative is about, and at nine the *TAF15* reagent alone does (§4.1). Between them they address the published breakpoints of roughly two thirds of molecularly confirmed cases (39.9% to 82.8% on the breakpoint denominators alone, and optimistic by an unmeasured amount because one arm is priced 3 of 3 on a three-tumour series, the four bounds §4.1 sets on it running in different directions rather than all one way; §4.1, §5) on the coverage ladder: the reagent set ordered so that each added reagent, each *rung*, carries the cumulative share of published breakpoints the set through that rung addresses, with a *bound* row

giving what that share would be if every remaining breakpoint of a partner were covered (§4.1).

**Three designs not to be used (§2.6).** 5'-CAGTGGGCTCTCCACG-3' and 5'-GCAGTGGGCTCTCCAC-3' at *EWSR1* exon 13 joined to *NR4A3* exon 2, and 5'-TGATGAGGGCCTTGTG-3' at *TAF15* exon 6 joined to the *NR4A3* intron-2 cryptic exon. Two of the three cleared the spliced-cDNA parent screen and the third's seam that screen cannot address, so its record reads `not_screened` rather than `clear`; and each pairs its whole catalytic gap against the patient's own un-rearranged *NR4A3* allele, which that screen cannot see.

**A second class not to be ordered, and it is much larger (§2.5).**

Any design a wild-type parent pairs through the whole catalytic gap at ten base pairs or more surrenders the advantage the modality has, and that is this paper's central negative rather than a side finding: 87 of the 190 panel designs, and 249 of the 780 records the canonical file holds, the records whose `mature_parent_duplex_through_gap_bp` column reaches ten, which is the column the count is reproducible from. They are marked `■` in Tables 3 and 4 and carry `do_not_order` in the canonical file. The three designs of the paragraph above carry `do_not_order` for the other reason, so 252 records carry the flag in all. **■ IS THAT VERDICT, WHEREVER IT APPEARS.** A sequence printed with `■` beside it, in a table, in a figure, or inline in the body text, is one this paper says must not be ordered or used; the marker means the same thing in prose as it does in a table, and it is never decorative. Tables 3 and 4 mark `■` every design of this class that they print, but between them they print only a small minority of it, because they are per-junction and per-screen selections rather than a census, so an absence from those tables is not a clearance and the canonical file is the only complete record. Five of the nine designs §2.4 names as carrying no sense-strand near-match are in this class. An unmarked row is not a clearance: the marker is set at ten base pairs, and at seven, 175 of the 190 pair a parent through the whole gap. Neither verdict can be read off a sequence by eye. Two of the three designs §2.6 forbids, and one of the two lead reagents above, each sit a single base from a design at the same junction carrying the opposite verdict:

5'-CAGTGGGCTCTCCACG-3' `■` beside  
 5'-AGTGGGCTCTCCACGG-3', 5'-TGATGAGGGCCTTGTG-3'  
`■` beside 5'-GATGAGGGCCTTGTGT-3', and the orderable  
 5'-GGGCATATCTTGTGTG-3' beside  
 5'-AGGGCATATCTTGTGT-3' `■`. The canonical file carries that pairing for every design that has one, in its `near_identical_design_with_a_different_verdict` column.

**The cell line (§3).** No *NR4A3* fusion is detectable in H-EMC-SS on the public record, which is a filtered caller's silence and bounds detection at a sensitivity nothing here quantifies rather than establishing absence, so no reagent named here can be assumed testable in it without sequencing its junction first. That is not a statement that the line is misidentified.

**The replicate floor and the void condition (§4.4).** Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose-response in the same wells, and the cut is 5.0, taken as a convention rather than measured for this comparison (§4.4). Three biological replicates are a floor and not a target: above a replicate standard deviation of about 0.65 on the natural-log scale, no observed ratio at or above one can place the upper limit of a two-sided 95% interval below that cut at three

replicates, so the test can fail only where the reagent is anti-selective and the test is otherwise void rather than negative. A *void* test is one that cannot fail, which is a different outcome from one that fails to falsify, and voidness is a property of the assay's variance rather than of the design. The controls, the assay placement and the limit-of-quantification condition without which the ratio is not reportable are in §4.4.

#### **A design procedure for a breakpoint outside this panel (§4.5).**

The pipeline that produced the 190 designs is released with the artefacts, and §4.5 gives its input, the five screens a new design must clear, and the limit on what it produces: a candidate, not a validated reagent.

## **2 · Results**

The results are ordered for a laboratory deciding what to make. How far any of the counts below can be trusted is stated in §5, which bounds all of them.

**How the counts are denominated.** Six numbers recur below and they are not interchangeable. 231 is the number of donor-exon by

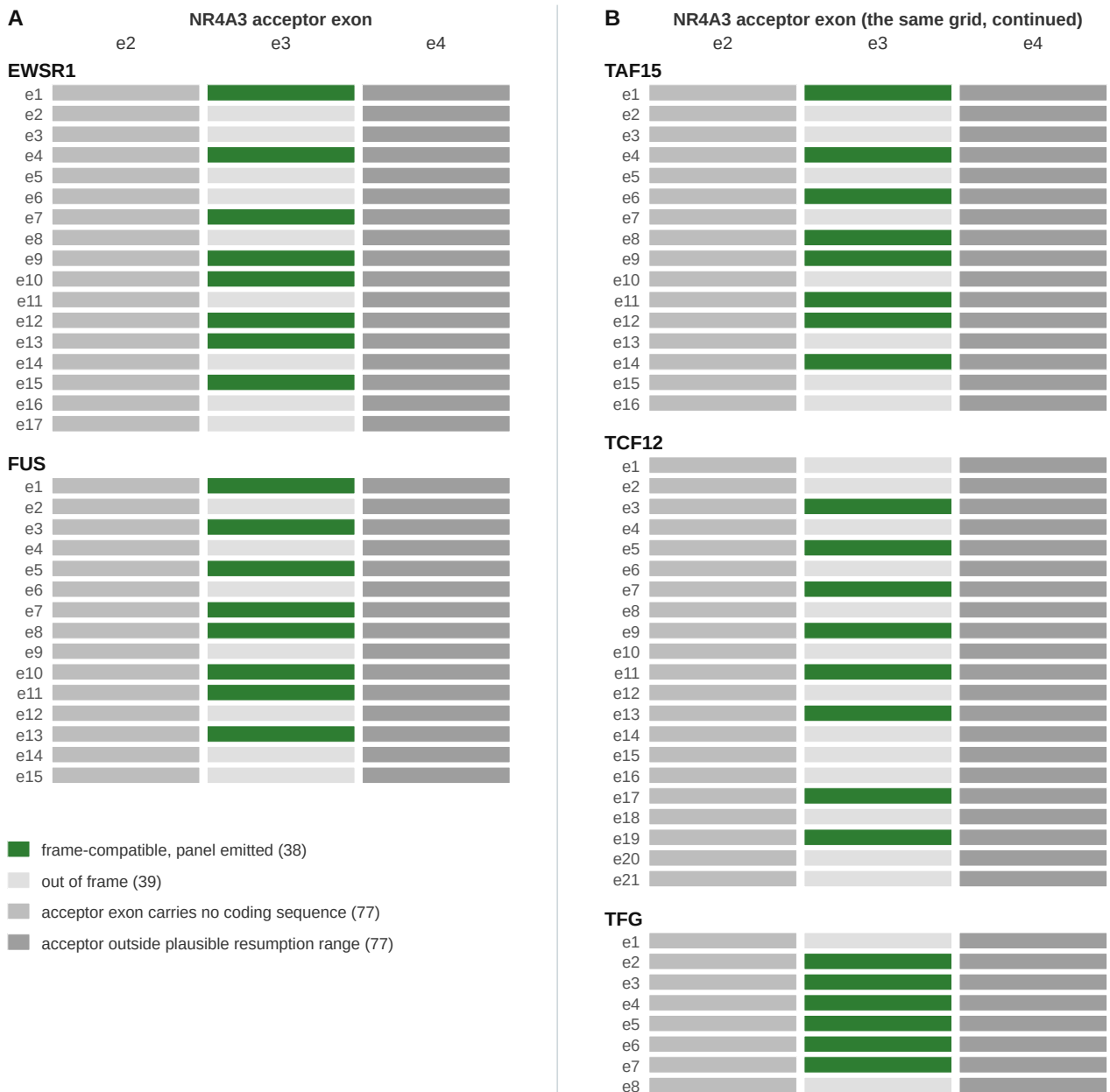
acceptor-exon pairs graded for frame. 38 are the in-frame junctions among them. Those 38 carry 190 design records, which are 176 distinct molecules, because nine of the 16-mers span more than one partner's junction and are recorded once per junction. Of the 190, 183 have a returned specificity screen; the other seven failed at the remote service, which matters because a free energy needs only a sequence where a screen needs a query that came back. 187 is the count re-screened at the tenfold deeper ceiling. Each result below names the denominator it uses.

### **2.1 · The reading frame as the bound on junction space**

Grading all 231 pairs of a donor exon with NR4A3 exons 2-4 across the five partners returns 38 in-frame junctions (Table 1, Figure 1). The refusals are structural. NR4A3 exon 2 carries no coding sequence and is refused in every pair, and an exon-4 acceptor would delete the NR4A3 DNA-binding domain that every reported EMC chimera retains. All the variance therefore sits in the exon-3 column. Within that column, being in frame reduces to a single arithmetic condition, a donor coding phase of 1, which is necessary and sufficient across those 77 exon-3 rows but only necessary across all 231.

### Frame compatibility across the NR4A3 fusion junction space

231 donor-exon  $\times$  acceptor-exon pairs graded over 5 partners; 38 are frame-compatible, and all of them lie in one acceptor column. The grid runs down A and continues in B; the three acceptor columns repeat in both.



Frame compatibility is an arithmetic property of exon structure. This is not a claim about which junctions patients carry: breakpoint recurrence is a clinical observation, and no partner-and-exon-resolved series exists for most of this space.

**Figure 1. Reading-frame compatibility across the NR4A3 fusion junction space.** All 231 donor-exon  $\times$  acceptor-exon pairs across *EWSR1*, *FUS*, *TAF15*, *TCF12* and *TFG* — named here in the order the panels draw them — graded against the frame condition. The grid is one grid, drawn as two side-by-side continuations to fit the page, so the three acceptor columns appear twice; a pair is not duplicated by appearing in the second block. Green marks the 38 frame-compatible pairs the panel is emitted at; the three refusal classes are greys, lightest to darkest: out of frame (39), acceptor exon carries no coding sequence (77), and acceptor outside the plausible resumption range (77). The panel repeats this key and these counts beside the grid, so it can be read without this caption. Rows are donor exons grouped by partner; columns are *NR4A3* acceptor exons 2 to 4, the declared acceptor window, and not all eight exons of the transcript — the 231 pairs are every donor exon against that window rather than against the whole gene. Two acceptor columns are refused in every pair for structural reasons, so the 38 in-frame junctions lie in a single column. The five partners are those for which transcript models are held here; *PGR*, the sixth partner §2.6 reports a single seam at, is not modelled and has no row.



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**Research use only, and not for administration to any person or animal.** Every oligonucleotide sequence named in these tables is a research reagent intended solely for laboratory investigation. None is a medicine or a candidate drug, none has been synthesised or tested by anyone, and none may be administered to any human being or animal, compounded for such use, or supplied to any person for such use. Custom oligonucleotide synthesis is commercially available, so the restriction is on use rather than on access. A table row is not a recommendation, and the full statement is in the main text's Declarations.

**Chemistry — what the geometry column and Table 7's columns denote.** Every design named in these tables is a gapmer of locked-nucleic-acid (LNA) wings around a central DNA gap, on a phosphorothioate backbone, and each geometry is written wing-gap-wing: 5-6-5 is five LNA nucleotides, a six-nucleotide DNA gap and five LNA nucleotides; 5-8-5 and 5-10-5 the same five-nucleotide LNA wings around gaps of eight and ten. That architecture is what the main text's Methods specify, and it is the chemistry every screen, every design rule and every duplex figure in these tables assumes. The bare base sequence of a row ordered as unmodified DNA is a different molecule, and nothing reported here is about it.

**Do not order these three sequences.** The main text names them as NOT to be carried forward — each pairs its whole catalytic gap against the patient's own un-rearranged *NR4A3* allele — and none of them is in any row of these tables: 5'-CAGTGGGCTCTCCACG-3', 5'-GCAGTGGGCTCTCCAC-3', 5'-TGATGAGGGCCTTG-3'. They are printed here because excluding them by description leaves a reader with a transcribed sequence nothing to check it against, and the margin for a slip is small: the closest sequence these tables DO print shares a run of 15 contiguous bases with one of the three, whose own length is 16.

**⚠ THAT LIST IS NOT THE WHOLE DO-NOT-ORDER SET, AND THESE TABLES ARE NOT A WHITELIST.** A further fifteen rows of these tables — 14 distinct molecules, since one is printed twice — are marked  in their own row and are also NOT to be ordered, for the same reason against a wild-type parent that is not always *NR4A3*. An unmarked row is not a clearance either; see the  note under Table 3.

**Table 1. The in-frame junction space across five NR4A3 fusion partners.** Every donor-exon × *NR4A3*-acceptor-exon pair was graded against the frame condition before any design was emitted. Every design counted here is at the 5-6-5 geometry — a 16-mer, the architecture the chemistry note above expands — so this table sizes one design space and not the modality. The gap-level margin is the count of bases inside the six-nucleotide catalytic gap that no wild-type parent carries at that position, taken on the shorter side of the junction; it runs from 1 to 3, the geometry's ceiling of half the gap rounded down, and it is a count of bases and not a score. Frame compatibility is an arithmetic property of exon structure and is not a claim about which junctions patients carry. A fusion-specific design is one whose catalytic gap contains at least one base that no wild-type parent carries at that position, so its gap-level margin is one or more. Both middle columns are printed because they are different conditions and only one of them is arithmetic: in-frame is a property of the exon lengths, while a fusion-specific design existing at the seam is a property of the sequence, and it can only ever be the smaller of the two. They agree in every one of the 5 partner rows here, which is a measurement and not a duplicated cell. “GC range of those designs (%)” and “best gap-level margin” are over that partner's fusion-specific designs: every one of the 38 in-frame junctions admits five, one per junction-spanning register, so a partner's range is over its in-frame count times five. The margin column is a maximum, so a row reading 3 says the partner has a design at the geometry's ceiling — half the gap rounded down, which Table 7 gives — and nothing about how many.

| Table 1. The in-frame junction space across five NR4A3 fusion partners |             |                   |          |                                |                               |                       |
|------------------------------------------------------------------------|-------------|-------------------|----------|--------------------------------|-------------------------------|-----------------------|
| 5' partner                                                             | donor exons | exon pairs graded | in-frame | with ≥1 fusion-specific design | GC range of those designs (%) | best gap-level margin |
| <i>EWSR1</i>                                                           | 17          | 51                | 8        | 8                              | 37.5–75.0                     | 3                     |
| <i>FUS</i>                                                             | 15          | 45                | 8        | 8                              | 31.2–62.5                     | 3                     |
| <i>TAF15</i>                                                           | 16          | 48                | 8        | 8                              | 31.2–68.8                     | 3                     |
| <i>TCF12</i>                                                           | 21          | 63                | 8        | 8                              | 25.0–56.2                     | 3                     |
| <i>TFG</i>                                                             | 8           | 24                | 6        | 6                              | 25.0–50.0                     | 3                     |
| all 5 partners                                                         | —           | 231               | 38       | 38                             | —                             | —                     |

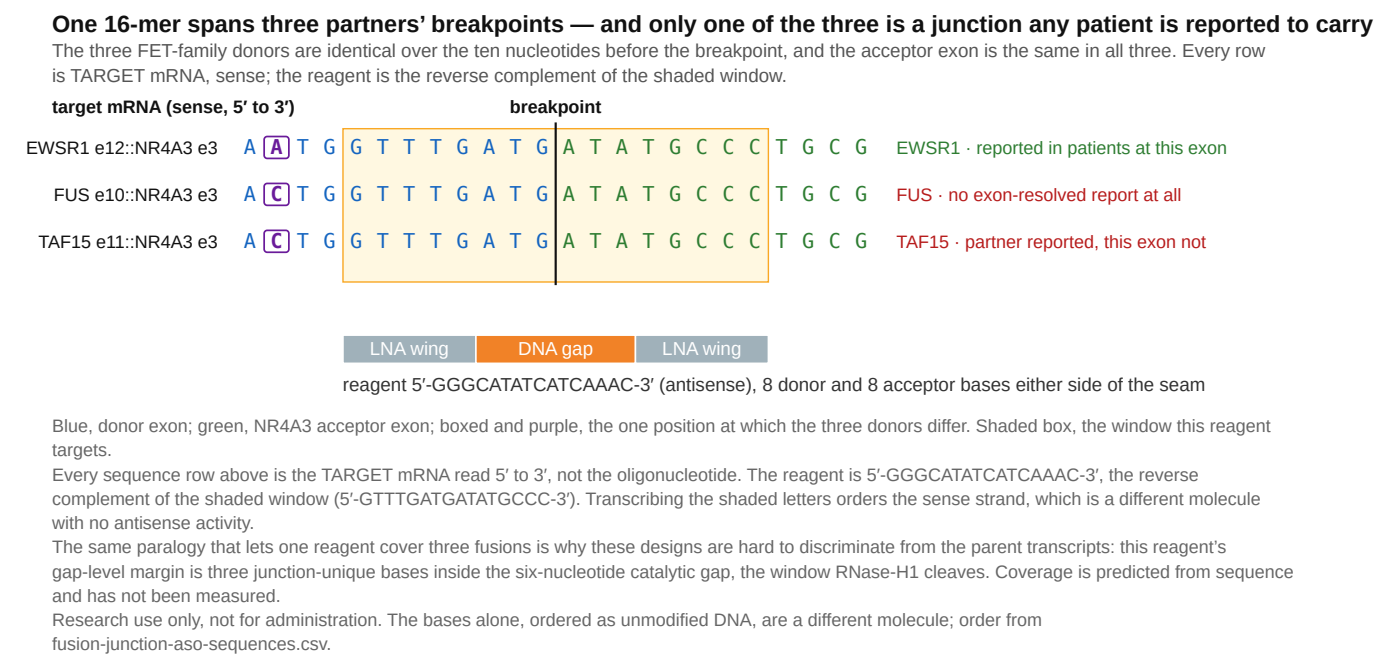
Among these, *EWSR1* exon 12 joined to *NR4A3* exon 3 is the junction reported most often: type 1 in 10 of the 15 *EWSR1*-rearranged tumours of an 18-case series.<sup>22</sup> Designs at this junction therefore correspond to the largest documented patient group. The type numbers are each source's own shorthand for a donor-acceptor exon pair and are used here only where a source uses them; every junction in this paper is named by its exon pair, so no type number needs to be resolved in order to read a design.

No design at any of the 38 junctions is a perfect complement of any of the six parent transcripts. That check excluded none of the 190 and could not have: a junction-spanning window cannot occur intact in a parent. It is a guard against an error in the screen, not a specificity result. GC runs 25.0–75.0% across partners, with 132 of the 190 inside the conventional 40–60% band. Finding candidate

sequences is therefore not the constraint on this modality in this disease.

2.2 · Cross-partner coverage by a single oligonucleotide

Nine designs span more than one partner's junction exactly, and all nine draw from *EWSR1*, *TAF15* and *FUS* (Figure 2). Five cover the same three-partner set, differing only in register across the junction. The best by gap-level margin is 5'-GGGCATATCATCAAAC-3' (43.8% GC, gap-level margin 3), which divides eight donor and eight acceptor bases at the junction of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, and occurs in none of the six wild-type parent transcripts. The basis is sequence identity: the three donors are identical over the ten bases immediately 5' of their breakpoints, diverging at the eleventh. No design draws more than ten donor bases, which is what makes the coverage arithmetically possible.



**Figure 2. One 16-mer spans three partners' breakpoints.** The junction windows of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, aligned at the breakpoint. Blue, donor exon; green, acceptor exon; positions at which the three donors differ are boxed as well as coloured, for greyscale and colour-blind readers. The shaded box is the target window of 5'-GGGCATATCATCAAAC-3', with the 5-6-5 locked-nucleic-acid (LNA)/DNA/LNA gapmer architecture below it; the note beneath the panel gives its gap-level margin of three. The three donors are the FET family — *FUS*, *EWSR1* and *TAF15* — and they are identical over the ten nucleotides before the breakpoint, which is what makes one oligonucleotide junction-spanning at all three junctions. Two of the three drawn junctions are not reported in any patient: the exon-resolved *TAF15* breakpoints in this disease are at exon 6 rather than exon 11, and no exon-resolved *FUS* breakpoint has been published at all (§2.2). This is a statement about sequence, not a claim that one reagent serves three patient groups. Coverage is predicted from sequence and has not been measured.

In one respect the published data contradict the clinical reading of this result. The only exon-resolved *TAF15::NR4A3* breakpoints reported in EMC are at exon 6, not exon 11. The primary report of the variant fusion places the breakpoint there,<sup>23</sup> and in a series of 18 EMCs all three *TAF15*-rearranged tumours carried exon 6 joined to *NR4A3* exon 3.<sup>22</sup> The exon-6 junction shares a single donor base with the exon-11 junction, so this oligonucleotide cannot engage the *TAF15* junction that patients are reported to carry.

That junction is itself in-frame and yields five fusion-specific designs (43.8–50.0% GC), all five screened and orientation-filtered. Every one of them retains a sense-strand near-match spanning the catalytic gap. At the tenfold deeper ceiling, where every hit list is complete, those recount to three gene loci at best, and five for the design its gap-level margin ranks first (Table 2); three of those five

are annotated only as predicted gene models, which is a property of the deep hit list rather than a column of any table here. Two of the five nonetheless return no exact and no single-mismatch match on the exhaustive transcript scan.

So the one *TAF15* junction with a published breakpoint is designable and is not among the cleaner junctions, while the junction the multi-partner result rests on has no reported patient. For *FUS* no exon-resolved EMC breakpoint has been published at all. The three-partner result is therefore a statement about FET-family (*FUS*, *EWSR1*, *TAF15*) sequence architecture and a hypothesis about junctions not yet observed. It is not a claim that one reagent serves three patient groups. Testing it requires breakpoint sequencing of archival *TAF15*- and *FUS*-positive cases.

**Table 2. The best available design at each in-frame junction that has one — 35 of the 38.** Table 4 selects across the panel and Table 3 selects within each junction by gap-level margin; this table selects within each junction by parent liability, which is the question a patient's fusion poses. Every row here is an *NR4A3* exon 3 acceptor seam: the acceptor half is the same in all 38 of them, so what a row varies is the donor and not which acceptor a patient's fusion joins to. The seams at another acceptor are graded out of this panel rather than screened and found wanting, and they are reported elsewhere: the four *NR4A3* exon-2 acceptor seams — EWSR1 e7::NR4A3 e2, EWSR1 e13::NR4A3 e2, PGR e2::NR4A3 e2 and TAF15 e6::NR4A3 e2 — are in Table 5 and §2.6, and the patient-derived models §3 names carry that acceptor rather than this one; the TAF15 e6 :: NR4A3 intron-2 cryptic exon seam reaches these tables only through the design the do-not-order list above names. Rows are in ascending order of partner, then of donor exon number, then of *NR4A3* acceptor exon number. The gap-level margin is the count of bases inside the six-nucleotide catalytic gap that no wild-type parent carries at that position, taken on the shorter side of the junction; it runs from 1 to 3, the geometry's ceiling of half the gap rounded down, and it is a count of bases and not a score. A design's parent liability is the length of the longest contiguous duplex any of the six wild-type parent transcripts — *EWSR1*, *FUS*, *NR4A3*, *TAF15*, *TCF12* and *TFG* — forms through its whole catalytic gap, searched in the mature (spliced) parent transcript in the forward orientation only, and it disqualifies at ten base pairs; the “longest parent duplex through the gap” column is that same length for the design the row names. Designs are ranked by parent liability first, since sparing the wild-type parents is what the modality exists for, then by pre-mRNA sites, then by distinct gene loci, with ties broken on gap-level margin rather than on raw hit counts. Nothing was re-screened: every field is joined from a screen already reported above. The denominator of the “designs clearing the parent screen” column is not a parent-screen figure: it is how many of that junction's designs RETURNED a deep alignment screen, which is the screen supplying every rank key here. Each seam of this panel is tiled by five junction-spanning registers (Table 7), and 3 of the 190 deep submissions failed at the remote service, so a design whose submission failed is absent from its junction's row set and appears in neither half of the cell — the only reason a denominator here reads below five, and an identity this generator checks before building the table. The parent screen itself is offline and exhaustive over every design, so the numerator alone is what it decided. Whether a junction has a published exon-resolved breakpoint is reported separately from specificity and never folded into the ranking — “published” means an exon-resolved EMC breakpoint is reported for that exon pair in prose; “published (deposit)” that the exon is resolved instead by a deposited chimeric record, which §2.3 describes; “exon not reported” that a breakpoint of that partner is resolved at a different exon, by either route; and “none published” that no breakpoint of that partner is resolved to an exon anywhere on the public record. *TCF12* and *TFG* are both the deposit case, and they are the same case: one seam each, resolved by a deposited chimeric record — AF289510.1 and AY532911.1 — rather than by an exon named in prose, so each carries one “published (deposit)” row and reads “exon not reported” elsewhere. That the two are graded alike is not a statement that their evidence is equally strong: the *TFG* record rests on one primary sequence plus four patent sequences that are one family from one group, with no report behind any of them, which §2.3 states and this column does not. “None published” is absence of evidence: EMC case reports usually name the partner gene without sequencing to nucleotide resolution. Gap-paired near-matches are at the tenfold deeper alignment ceiling, where every hit list is complete. RefSeq lists one accession per annotated variant, so the left half of that cell counts transcript accession RECORDS and not distinct genes, which is why the collapsed locus count is printed beside it: the two differ by a median of 6.0 records per locus over the 31 rows whose cell carries no censoring mark, and most at EWSR1 e15::NR4A3 e3, whose 33 records are one locus. Across the deep screens' uncensored designs, taken over every stored near-match rather than the gap-paired subset, the collapse runs at a median of 5.52 records per locus and a maximum of 28.0; §5 gives the same figure at the DEFAULT ceiling, which is smaller and does not describe a column read at the deeper one. The genome column is the observed number of gap-paired sites at ≤2 mismatches over the number expected for an arbitrary 16-mer, so 1.00 is chance. It is counted EITHER ORIENTATION against an either-orientation null, unlike the sense-filtered near-match columns beside it, so it includes sites on the strand an antisense oligonucleotide cannot pair; §2.7 gives the share for the lead. A junction with no design clearing the parent screen is reported as such rather than given a best row, and Table 3 marks those junctions too, since Table 3 ranks by margin instead and does print a sequence at each of them. **Clearing the parent screen means one thing and one thing only: no wild-type parent pairs the whole catalytic gap at ten base pairs or more.** That ten-base-pair cut is a criterion this work ADOPTS rather than measures, and the reading moves with it: at seven base pairs 175 of the 190 panel designs pair a parent through the whole gap against 87 at ten, 181 do so at some length, and only nine of the 38 seams have a design that clears against 35 at ten (§2.9). A cell below the cut is a reading at one cut and nothing wider. No row of this table reaches the criterion — the column's highest reading is 9 base pairs — so no row carries the  Tables 3 and 4 use, and an unmarked row here is not a clearance on any wider ground.  The recommendation is a reading at that one cut, and what moves with the cut is AVAILABILITY more than identity. Re-ranked on the same screened designs by the same keys, the seams that still have a best available design number 31 of the 38 at nine base pairs, all of them the molecule printed here; 23 of the 38 at eight base pairs, six of them a different molecule; 9 of the 38 at seven base pairs, two of them a different molecule. Nothing was re-screened to say that; only the liability test moved, and a seam that loses its row loses a reagent rather than gaining a worse one.  At sixteen junctions Table 3 names a DIFFERENT DESIGN from this table, because the two tables rank on different keys. At seven of those sixteen one of the pair is condemned and the other is not, and at all seven of those seven the two share fifteen of their 16 contiguous bases — one register apart. At a further three junctions Table 3 names a condemned design where this table names none at all, which is not a disagreement but is also not a clearance. Check the junction AND the whole sequence against `fusion-junction-aso-sequences.csv` before ordering either. The last column is a conventional design audit, computed for whichever design this table names from the same artefact and by the same code Table 4 uses. The four conventional antisense design rules audited are GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, and no CpG dinucleotide. It is reported beside the ranking and is never folded into it: the two orderings select different molecules, which is the disagreement §2.10 is about, and this is the table one reagent is chosen from. A design the audit does not cover would read “not audited” rather than blank, since a blank in a rules column reads as breaking none. Every sequence in this table is an antisense 16-mer, tiled in the 5-6-5 locked-nucleic-acid (LNA)/DNA/LNA architecture on a phosphorothioate backbone that §6 specifies; the bases alone, ordered as unmodified DNA, are a different molecule. Research use only: not for administration to any person or animal (Declarations). The canonical machine-readable copy of every sequence is `fusion-junction-aso-sequences.csv`, which is what to order from rather than this PDF.



**Table 2. The best available design at each in-frame junction that has one — 35 of the 38 · every row is read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction            | exon-resolved breakpoint | designs clearing the parent screen | best available design  | gap-level margin | longest parent duplex through the gap (bp) | sense-strand gap-paired near-matches at the deeper ceiling (transcripts → loci) | genome-wide gap-paired load, observed/expected | conventional rules failed                                  |
|---------------------|--------------------------|------------------------------------|------------------------|------------------|--------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------------|
| EWSR1 e1::NR4A3 e3  | exon not reported        | 2 of 5                             | 5'-GGGCATATCCGTGGAC-3' | 3                | 0                                          | 0 → 0                                                                           | 0.07                                           | GC outside 40–60%, contains a CpG                          |
| EWSR1 e4::NR4A3 e3  | exon not reported        | 4 of 5                             | 5'-AGGGCATATCAGTGGG-3' | 2                | 6                                          | 9 → 3                                                                           | 0.71                                           | none                                                       |
| EWSR1 e7::NR4A3 e3  | exon not reported        | 5 of 5                             | 5'-CAGGGCATATTCTGCT-3' | 1                | 8                                          | 14 → 5                                                                          | 1.08                                           | none                                                       |
| EWSR1 e9::NR4A3 e3  | exon not reported        | 4 of 5                             | 5'-GGCATATCACCAGGCT-3' | 2                | 7                                          | 15 → 4                                                                          | 0.73                                           | none                                                       |
| EWSR1 e10::NR4A3 e3 | exon not reported        | 3 of 5                             | 5'-GGCATATCTAGATCAA-3' | 2                | 7                                          | 70 → 4                                                                          | 0.61                                           | GC outside 40–60%                                          |
| EWSR1 e12::NR4A3 e3 | published                | 4 of 5                             | 5'-GGGCATATCATCAAAC-3' | 3                | 8                                          | 123 → 6                                                                         | 0.62                                           | none                                                       |
| EWSR1 e13::NR4A3 e3 | published                | 3 of 5                             | 5'-GGGCATATCTCCACGG-3' | 3                | 8                                          | 24 → 2                                                                          | 0.29                                           | GC outside 40–60%, contains a CpG                          |
| EWSR1 e15::NR4A3 e3 | exon not reported        | 4 of 5                             | 5'-GGGCATATCCGGGGGC-3' | 3                | 0                                          | 33 → 1                                                                          | 0.06                                           | GC outside 40–60%, homopolymer run of four, contains a CpG |
| FUS e1::NR4A3 e3    | none published           | 2 of 5                             | 5'-GGGCATATCGTTTGAG-3' | 3                | 8                                          | 5 → 2                                                                           | 0.12                                           | contains a CpG                                             |
| FUS e3::NR4A3 e3    | none published           | 5 of 5                             | 5'-GGCATATTGTCTGGC-3'  | 2                | 7                                          | 18 → 3                                                                          | 0.76                                           | none                                                       |
| FUS e5::NR4A3 e3    | none published           | 2 of 3                             | 5'-GGGCATATCTCCACCT-3' | 3                | 8                                          | 33 → 4                                                                          | 0.65                                           | none                                                       |
| FUS e7::NR4A3 e3    | none published           | 4 of 5                             | 5'-GGCATATCACAAATT-3'  | 2                | 7                                          | 10 → 3                                                                          | 0.92                                           | GC outside 40–60%                                          |
| FUS e8::NR4A3 e3    | none published           | 3 of 5                             | 5'-AGGGCATATCGGAGTC-3' | 2                | 0                                          | 0 → 0                                                                           | 0.06                                           | contains a CpG                                             |
| FUS e10::NR4A3 e3   | none published           | 4 of 5                             | 5'-GGGCATATCATCAAAC-3' | 3                | 8                                          | 123 → 6                                                                         | 0.62                                           | none                                                       |
| FUS e11::NR4A3 e3   | none published           | 3 of 5                             | 5'-GGGCATATCTCCTCGC-3' | 3                | 9                                          | 30 → 1                                                                          | 0.22                                           | GC outside 40–60%, contains a CpG                          |
| FUS e13::NR4A3 e3   | none published           | 4 of 5                             | 5'-GGGCATATCCATGTGA-3' | 3                | 7                                          | 2 → 2                                                                           | 0.75                                           | none                                                       |
| TAF15 e1::NR4A3 e3  | exon not reported        | 2 of 5                             | 5'-GGGCATATCCGACATG-3' | 3                | 0                                          | 0 → 0                                                                           | 0.04                                           | contains a CpG                                             |
| TAF15 e4::NR4A3 e3  | exon not reported        | 2 of 5                             | 5'-GGCATATCTGACTGAC-3' | 2                | 8                                          | 12 → 2                                                                          | 0.69                                           | none                                                       |

**Table 2. The best available design at each in-frame junction that has one — 35 of the 38 — every row is read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction            | exon-resolved breakpoint | designs clearing the parent screen | best available design  | gap-level margin | longest parent duplex through the gap (bp) | sense-strand gap-paired near-matches at the deeper ceiling (transcripts → loci) | genome-wide gap-paired load, observed/expected | conventional rules failed                  |
|---------------------|--------------------------|------------------------------------|------------------------|------------------|--------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------|
| TAF15 e6::NR4A3 e3  | published                | 1 of 5                             | 5'-GGGCATATCTGTGTG-3'  | 3                | 9                                          | 8 → 5                                                                           | 0.60                                           | none                                       |
| TAF15 e8::NR4A3 e3  | exon not reported        | 4 of 5                             | 5'-GGGCATATCACCAAAA-3' | 3                | 7                                          | 36 → 3                                                                          | 0.86                                           | homopolymer run of four                    |
| TAF15 e9::NR4A3 e3  | exon not reported        | 2 of 5                             | 5'-AGGGCATATCAGCATC-3' | 2                | 6                                          | 6 → 2                                                                           | 1.13                                           | none                                       |
| TAF15 e11::NR4A3 e3 | exon not reported        | 4 of 5                             | 5'-GGGCATATCATCAAAC-3' | 3                | 8                                          | 123 → 6                                                                         | 0.62                                           | none                                       |
| TAF15 e12::NR4A3 e3 | exon not reported        | 3 of 5                             | 5'-GGGCATATCTGCCGC-3'  | 3                | 6                                          | 4 → 2                                                                           | 0.04                                           | GC outside 40–60%, contains a CpG          |
| TAF15 e14::NR4A3 e3 | exon not reported        | 0 of 5                             | —                      | —                | —                                          | —                                                                               | —                                              | —                                          |
| TCF12 e3::NR4A3 e3  | exon not reported        | 0 of 5                             | —                      | —                | —                                          | —                                                                               | —                                              | —                                          |
| TCF12 e5::NR4A3 e3  | published (deposit)      | 4 of 5                             | 5'-GGGCATATCCATCAGA-3' | 3                | 7                                          | 17 → 1                                                                          | 0.56                                           | none                                       |
| TCF12 e7::NR4A3 e3  | exon not reported        | 2 of 5                             | 5'-GGGCATATCAAGCGCT-3' | 3                | 8                                          | 0 → 0                                                                           | 0.26                                           | contains a CpG                             |
| TCF12 e9::NR4A3 e3  | exon not reported        | 1 of 5                             | 5'-GGGCATATCTTGATA-3'  | 3                | 8                                          | 39 → 6                                                                          | 0.71                                           | none                                       |
| TCF12 e11::NR4A3 e3 | exon not reported        | 3 of 5                             | 5'-GGCATATCTAGAATGC-3' | 2                | 8                                          | 4 → 3                                                                           | 0.69                                           | none                                       |
| TCF12 e13::NR4A3 e3 | exon not reported        | 2 of 5                             | 5'-GGCATATCTGTGAGAG-3' | 2                | 7                                          | 6 → 5                                                                           | 1.08                                           | none                                       |
| TCF12 e17::NR4A3 e3 | exon not reported        | 3 of 5                             | 5'-GGGCATATCTCTATAA-3' | 3                | 7                                          | 14 → 5                                                                          | 0.83                                           | GC outside 40–60%                          |
| TCF12 e19::NR4A3 e3 | exon not reported        | 2 of 5                             | 5'-GGGCATATCTCTGACT-3' | 3                | 8                                          | 75 → 3                                                                          | 0.82                                           | none                                       |
| TFG e2::NR4A3 e3    | exon not reported        | 0 of 4                             | —                      | —                | —                                          | —                                                                               | —                                              | —                                          |
| TFG e3::NR4A3 e3    | exon not reported        | 4 of 5                             | 5'-GGGCATATCAAATAAT-3' | 3                | 8                                          | 18 → 6                                                                          | 1.35                                           | GC outside 40–60%                          |
| TFG e4::NR4A3 e3    | exon not reported        | 1 of 5                             | 5'-AGGGCATATCATTTTC-3' | 2                | 7                                          | 82 → 12                                                                         | 1.77                                           | GC outside 40–60%, homopolymer run of four |
| TFG e5::NR4A3 e3    | exon not reported        | 2 of 5                             | 5'-GGCATATCTGAAACCT-3' | 2                | 8                                          | 34 → 7                                                                          | 1.38                                           | none                                       |
| TFG e6::NR4A3 e3    | exon not reported        | 3 of 5                             | 5'-GGGCATATCTTCAATC-3' | 3                | 9                                          | 29 → 3                                                                          | 0.73                                           | none                                       |

**Table 2. The best available design at each in-frame junction that has one — 35 of the 38 — every row is read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction            | exon-resolved breakpoint | designs clearing the parent screen | best available design  | gap-level margin | longest parent duplex through the gap (bp) | sense-strand gap-paired near-matches at the deeper ceiling (transcripts → loci) | genome-wide gap-paired load, observed/expected | conventional rules failed |
|---------------------|--------------------------|------------------------------------|------------------------|------------------|--------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------|---------------------------|
| TFG<br>e7::NR4A3 e3 | published<br>(deposit)   | 2 of 5                             | 5'-GGCATATCTGAATACT-3' | 2                | 9                                          | 24 → 6                                                                          | 1.24                                           | GC outside 40–60%         |

### 2.3 · The non-FET partners: coverage and specificity

*TCF12* and *TFG* are the partners in this panel that are not FET-family proteins, and neither appears in any of the nine multi-partner designs: all nine draw only on *EWSR1*, *TAF15* and *FUS*. *TCF12* reaches multi-partner coverage only under a relaxed criterion that tolerates mismatches in the oligonucleotide wings. That check had little power to fail, because a donor sharing the last bases before the breakpoint by coincidence passes it as readily as a paralogue does, so it does not separate FET paralogy from incidental exon homology. The stronger evidence for paralogy is that the remaining four of the nine, which span two partners rather than three, are also FET-only.

Specificity does not sort by partner on the one existence statistic tested, and does on the one other axis this section prints. Taking at each junction the lowest count any of its designs achieves after the orientation filter (a per-design minimum, which is in the released per-junction screens and not in Table 3, whose row is that junction's highest-margin design and not its cleanest) every one of the five partners has at least one junction whose cleanest design carries no sense-strand near-match across the catalytic gap at the default search ceiling: three of eight at both *TCF12* and *FUS*, two of eight at *EWSR1*, one of eight at *TAF15* and one of six at *TFG*. At the tenfold deeper ceiling that becomes four partners: each of *EWSR1*, *FUS*, *TAF15* and *TCF12* keeps one such junction and *TFG* keeps none, its single default-depth zero returning 29 across the gap when searched deeper. On this one existence statistic, then, no partner is uniformly clean or uniformly dirty, and which exon a fusion breaks at bears on specificity as well as which gene it breaks into. That is the limit of what was tested: no comparison between partners was performed, and one axis printed here does sort by partner. Table 2's genome-wide gap-paired load runs at a mean of 1.30 of chance across *TFG*'s five scored junctions — Table 2 gives a best row only at the 35 junctions where some design clears the parent screen (§2.7) — four of them above chance, against partner means of 0.51 to 0.71 taken over seven or eight junctions each, and three rows above chance among the other thirty junctions; over the whole corpus the same axis reads 2.23 across *TFG*'s thirty designs against 0.74 across the other 146 of the 176 distinct molecules. Those two means are not comparable summaries: the *TFG* distribution has a standard deviation of 3.41, larger than its own mean, and one design at 20 times chance, so its median is 1.53 while the other 146 sit at a median of 0.74 with a standard deviation of 0.44. On medians the contrast is 2.1-fold rather than 3.0-fold. That axis is reported and it is confounded rather than explained: the null behind it assumes uniform bases, so it tracks base composition, and *TFG*'s designs average 39.0% GC against 50.6% elsewhere. Non-*TFG* designs at matched composition (the 46 of the other 146 carrying no more than seven G or C bases of 16, the band 29 of *TFG*'s 30 designs fall in) run at 1.04. That restriction is one-sided — the 1.04 is over matched non-

*TFG* designs while the 2.23 is over all thirty *TFG* designs — and restricting both sides gives 2.28 against 1.04. Matching on composition therefore removes 20% of the difference in means one-sided, 0.74 to 1.04 against 2.23, and 17% with both sides restricted, so part of the excess is GC and not partner; how large a part depends on the scale chosen, and both figures are on the plain difference in means. On medians, the scale this paragraph moved to above, matching removes a third rather than a fifth: the matched non-*TFG* median is 1.00 against *TFG*'s 1.53, so the median contrast falls from 2.1-fold to 1.5-fold and does not close. The matching rule is stated because the answer moves with it too: taken instead over the whole span of GC that *TFG*'s designs cover, the same figure is 0.89. Neither reading leaves the partner effect intact, and neither is a composition-matched null: the one built for the mature-parent screen was not run for the genome scan.

The same tension the *TAF15* result carries applies to *TCF12*, and in the same direction. The one published *TCF12::NR4A3* breakpoint describes a chimera retaining the first 108 *TCF12* residues,<sup>5</sup> and names no exon; the same authors deposited the chimeric cDNA, and that deposit resolves the junction to the nucleotide. GenBank AF289510.1 carries two chromosome-tagged source features that meet at the junction. Mapped against the transcript models used here, the donor side ends at *TCF12* exon 5 and at no other exon, the acceptor side begins at *NR4A3* exon 3 and at no other exon, and the twelve bases either side of the seam are identical to the seam the panel was designed on. The base-level assignment, its chromosome-tagged coordinates and the translation check that reproduces the deposit's own recorded protein are in the released artefact. That junction is in-frame and designable, and its best-margin design retains 17 gap-paired near-matches at the deeper ceiling, every one of them a variant of a single locus (Table 2). That locus is the curated *PIK3CG*, which is read from the deep hit list; Table 2 carries no gene column. None of the four *TCF12* designs with no sense-strand near-match is at that exon. So for *TCF12* as for *TAF15*, the junction a patient is reported to carry is designable and is not among the clean ones, while the clean junctions have no reported patient. What remains unmeasured at *TCF12* is not the exon but the distribution: one *TCF12*-rearranged tumour has been sequenced at this junction, and it is the tumour the junction was defined by. No search of the nucleotide or read archives returns a second *TCF12::NR4A3* sequence, and what that leaves the coverage estimate resting on is stated with the estimate (§4.1).

The same archive search resolves *TFG*, the other non-FET partner, in the same way and to the same limit. No paper places a *TFG::NR4A3* breakpoint at an exon, but a deposited chimeric mRNA record does (GenBank AY532911.1, annotated as a *TFG-NR4A3* fusion protein), and it lands on *TFG* exon 7 joined to *NR4A3* exon 3, a junction this panel already carries. Four patent sequence records agree at the

seam (GenBank DI433544.1, DI438966.1, LG067227.1 and LG067228.1), corroborating a sequence rather than four patients, being one family from one group. As at *TCF12*, what the deposit supplies is the exon and not the distribution: no source states what

fraction of *TFG*-rearranged tumours break there, and *TFG* does not appear in the partner counts of the 58-case cohort every coverage figure here is denominated on, so this changes which junctions are reported and no percentage.

**Table 3. Predicted specificity per screened junction.** One row per junction, in Table 2's order. Rows are in ascending order of partner, then of donor exon number, then of NR4A3 acceptor exon number. Figures are for the design with the highest gap-level margin at that junction, which is the ranking the Methods define, and NOT for that junction's cleanest design — the two are often different molecules, and the cleanest ones are in Table 4. The gap-level margin is the count of bases inside the six-nucleotide catalytic gap that no wild-type parent carries at that position, taken on the shorter side of the junction; it runs from 1 to 3, the geometry's ceiling of half the gap rounded down, and it is a count of bases and not a score. Every figure in a row is that named design's own EXCEPT the last column, which is a median and a maximum over every design screened at the junction and is therefore a junction aggregate rather than a property of the molecule beside it: at EWSR1 e9::NR4A3 e3 the cell reads 1 (22) while the design the row names returns 9, and the two part company at 24 of the 38 rows. The margin column is therefore the best among the designs that RETURNED a screen at this depth: 7 of the panel's 190 default-depth submissions failed at the remote service, which is why a junction can show fewer than five designs screened here, and why Table 2 — which selects from the deeper re-screens — can name a design of HIGHER margin at the same junction. Table 2 can also name one of LOWER margin, for a different reason and not a disagreement: it ranks by parent liability, then pre-mRNA sites, then gene loci, and reaches gap-level margin only as a tie-break, so a cleaner design outranks the highest-margin one this table names. RefSeq lists one accession per annotated variant, so a near-match count is a count of transcript accession RECORDS and not of distinct genes, which is why the default-depth count is given collapsed to distinct gene loci beside it, and the size of that collapse is measured rather than left to the reader: the default-depth pair runs at a median of 2.3 records per locus over the 15 rows whose cell carries no censoring mark, and most at TCF12 e9::NR4A3 e3, whose 14 records are two loci. Across the deep screens' uncensored designs, taken over every stored near-match rather than the gap-paired subset, the collapse runs at a median of 5.52 records per locus and a maximum of 28.0; §5 gives the same figure at the DEFAULT ceiling, which is smaller and does not describe a column read at the deeper one. **Δ The deep columns are not a records-and-their-own-loci pair:** the deep locus column counts only the GAP-PAIRED subset of the records the near-match columns beside it count, so dividing one by the other is a ratio of two different hit sets and not a records-per-locus figure. **Δ The two censoring marks point in OPPOSITE directions and sit on adjacent columns:** “≥” is a LOWER bound, because the search was capped, and “≤” an UPPER bound, because the stored figure over-counts. 22 of the 38 rows carry “≥”, 20 carry “≤” and 20 carry both. A “≥” marks a right-censored count, and the two columns are censored by DIFFERENT caps, which is why an uncensored transcript count here can exceed the retained 15: the alignment screen itself returns at most 50 hits per query, so a transcript count at 50 is a lower bound, while the locus column is recounted from the 15 hits the screens RETAIN, so a locus count from a design with more than 15 hits is a lower bound too. All 38 junction screens are filtered by alignment orientation. XM\_ /XR\_ records are computationally predicted gene models rather than curated transcripts, and are counted separately for that reason. None of these numbers is a measurement of off-target activity.

<sup>1</sup> A near-match count is what the search returned on EITHER strand; a match on the strand opposite the target window cannot be hybridised by an antisense oligonucleotide and is not a liability. Across this corpus 44% of apparent gap-paired hits (738 of 1,677) are of that kind, which is why the two columns differ and why the raw count alone should not be read as load. This column counts only the 15 RETAINED hits. The gap-paired locus column is recounted from those hits wherever they are the complete list, and is exact there; a “≤” marks a truncated design, where the column instead carries the screen's own count over every ranked hit, computed under a locus assignment since corrected that split some genes across accessions and therefore over-counts. The two columns are not in conflict where a truncated design shows “≥0” sense-strand hits and a non-zero gap-paired locus count: the sense-strand hits are real and simply fall outside the stored window, which is precisely why such a design cannot be called clean.

<sup>2</sup> Counted over the gap-paired loci only, not over all of that design's near-match loci.

<sup>3</sup> The same design re-screened at a tenfold deeper alignment ceiling, with retention raised to match it so that no hit list is truncated. Because no list is truncated, the gap-paired locus column at this depth is recounted from the complete stored hits under the current locus assignment and is exact; it is not the screen's own stored figure, which was computed before that assignment was corrected and splits any gene whose description carries a comma across one accession per transcript variant. It is therefore the same quantity, counted the same way, as the locus figures in Table 2 and in the Results. The three columns are the counterparts of the default-depth columns to their left, given beside them rather than in place of them because the default depth is where the corpus-wide counts elsewhere in the paper were computed and the two must stay comparable. Read together they are the paper's censoring result at the level of a single row: a default-depth count is a lower bound whether or not it reached the 50-hit cap, and three junctions whose default cell reads zero in the gap-paired column carry gap-paired hits at ten times the depth. Three of the panel's 190 records failed at this ceiling; they are absent from the deep set rather than counted as zero in it. Every sequence in this table is an antisense 16-mer, tiled in the 5-6-5 locked-nucleic-acid (LNA)/DNA/LNA architecture on a phosphorothioate backbone that §6 specifies; the bases alone, ordered as unmodified DNA, are a different molecule. Research use only: not for administration to any person or animal (Declarations). The canonical machine-readable copy of every sequence is `fusion-junction-aso-sequences.csv`, which is what to order from rather than this PDF.

† No design screened at this junction clears the parent screen: every one of them pairs a wild-type parent gene through the whole catalytic gap, at the parent-duplex criterion applied throughout. This table ranks by gap-level margin, so the sequence in such a row is that junction's highest-margin design and nothing more; it is not a design any screen passes, and Table 2 gives the same junction no best-available reagent for that reason — its “designs clearing the parent screen” cell reads 0. Do not order the sequence in a marked row.

■ This design pairs a wild-type parent gene through the whole catalytic gap at the ten-base-pair criterion applied throughout, and the gene it pairs is named beside the length. The marker is on the DESIGN, where † is on the JUNCTION: a row can be unmarked by † and still carry ■, because this table prints each junction's highest-margin design rather than its cleanest. Do not order the sequence in a row marked ■ — pairing a parent through the whole gap is this paper's central negative (§4.5) and surrenders the only advantage the modality has. An unmarked row is not a clearance. That ten-base-pair cut is a criterion this work ADOPTS rather than measures, and the reading moves with it: at seven base pairs 175 of the 190 panel designs pair a parent through the whole gap against 87 at ten, 181 do so at some length, and only nine of the 38 seams have a design that clears against 35 at ten (§2.9). A cell below the cut is a reading at one cut and nothing wider. **Δ** At sixteen junctions Table 2 names a DIFFERENT DESIGN from this table, because the two tables rank on different keys. At seven of those sixteen one of the pair is condemned and the other is not, and at all seven of those seven the two share fifteen of their 16 contiguous bases — one register apart. At a further three junctions this table names a condemned design where Table 2 names none at all, which is not a disagreement but is also not a clearance. Check the junction AND the whole sequence against `fusion-junction-aso-sequences.csv` before ordering either.



**Table 3. Predicted specificity per screened junction · numbered notes are under the caption, on this table's first page — † no design at this junction clears the parent screen; do not order the sequence in a marked row — ▀ this design pairs a wild-type parent through the whole catalytic gap; do not order it · both markers are read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction            | designs screened | best gap-level margin | that design            | near-matches, either strand (transcripts → loci; ≥ is a lower bound) | of the retained hits, on the sense strand <sup>1</sup> (≥ is a lower bound) | loci with a gap-paired hit (≤ is an upper bound) | of those, predicted models only <sup>2</sup> | at the deeper ceiling: near-matches <sup>3</sup> | of those, on the sense strand <sup>3</sup> | loci with a gap-paired hit <sup>3</sup> | longest parent duplex through the gap (bp) | ≤1-mismatch matches across that junction's designs, median (max) |
|---------------------|------------------|-----------------------|------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------|-----------------------------------------|--------------------------------------------|------------------------------------------------------------------|
| EWSR1 e1::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATCCGTGGAC-3' | 0 → 0                                                                | 0                                                                           | 0                                                | 0                                            | 27                                               | 18                                         | 0                                       | 0                                          | 0 (0)                                                            |
| EWSR1 e4::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATCAGTGGGA-3' | 11 → 4                                                               | 11                                                                          | 3                                                | 2                                            | 90                                               | 67                                         | 8                                       | 7 ( <i>TFG</i> )                           | 4 (10)                                                           |
| EWSR1 e7::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATTCTGCTGC-3' | 32 → ≥9                                                              | ≥6                                                                          | ≤2                                               | 0                                            | 300                                              | 65                                         | 6                                       | 8 ( <i>FUS</i> )                           | 9 (21)                                                           |
| EWSR1 e9::NR4A3 e3  | 2                | 3                     | 5'-GGGCATATCACCAGGC-3' | 29 → ≥2                                                              | ≥6                                                                          | ≤2                                               | 0                                            | 165                                              | 81                                         | 7                                       | 7 ( <i>TCF12</i> )                         | 1 (22)                                                           |
| EWSR1 e10::NR4A3 e3 | 5                | 3                     | 5'-GGGCATATCTAGATCA-3' | 35 → ≥2                                                              | ≥15                                                                         | ≤4                                               | 0                                            | 138                                              | 137                                        | 5                                       | 7 ( <i>NR4A3</i> )                         | 4 (32)                                                           |
| EWSR1 e12::NR4A3 e3 | 5                | 3                     | 5'-GGGCATATCATCAAC-3'  | 9 → 4                                                                | 6                                                                           | 1                                                | 1                                            | 189                                              | 141                                        | 6                                       | 8 ( <i>TFG</i> )                           | 2 (22)                                                           |
| EWSR1 e13::NR4A3 e3 | 5                | 3                     | 5'-GGGCATATCTCCACGG-3' | 31 → ≥3                                                              | ≥12                                                                         | ≤2                                               | 0                                            | 63                                               | 47                                         | 2                                       | 8 ( <i>TCF12</i> )                         | 2 (25)                                                           |
| EWSR1 e15::NR4A3 e3 | 5                | 3                     | 5'-GGGCATATCCGGGGGC-3' | 36 → ≥2                                                              | ≥12                                                                         | ≤1                                               | 0                                            | 93                                               | 62                                         | 1                                       | 0                                          | 1 (10)                                                           |
| FUS e1::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCGTTTGAG-3' | ≥50 → ≥7                                                             | ≥13                                                                         | ≤1                                               | 0                                            | 129                                              | 116                                        | 2                                       | 8 ( <i>TCF12</i> )                         | 10 (38)                                                          |
| FUS e3::NR4A3 e3    | 4                | 3                     | 5'-GGGCATATTGTTCTGG-3' | 18 → ≥4                                                              | ≥1                                                                          | ≤2                                               | 0                                            | 148                                              | 34                                         | 4                                       | 8 ( <i>TCF12</i> )                         | 3 (23)                                                           |
| FUS e5::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCTCCACCT-3' | 32 → ≥8                                                              | ≥8                                                                          | ≤2                                               | 1                                            | 127                                              | 47                                         | 4                                       | 8 ( <i>TCF12</i> )                         | 2 (3)                                                            |
| FUS e7::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCACCAAAT-3' | 34 → ≥5                                                              | ≥8                                                                          | 0                                                | 0                                            | 141                                              | 107                                        | 4                                       | 7 ( <i>TCF12</i> )                         | 12 (39)                                                          |

**Table 3. Predicted specificity per screened junction · numbered notes are under the caption, on this table's first page — † no design at this junction clears the parent screen; do not order the sequence in a marked row — ■ this design pairs a wild-type parent through the whole catalytic gap; do not order it · both markers are read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction              | designs screened | best gap-level margin | that design            | near-matches, either strand (transcripts → loci; ≥ is a lower bound) | of the retained hits, on the sense strand <sup>1</sup> (≥ is a lower bound) | loci with a gap-paired hit (≤ is an upper bound) | of those, predicted models only <sup>2</sup> | at the deeper ceiling: near-matches <sup>3</sup> | of those, on the sense strand <sup>3</sup> | loci with a gap-paired hit <sup>3</sup> | longest parent duplex through the gap (bp) | ≤1-mismatch matches across that junction's designs, median (max) |
|-----------------------|------------------|-----------------------|------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------|-----------------------------------------|--------------------------------------------|------------------------------------------------------------------|
| FUS e8::NR4A3 e3      | 5                | 3                     | 5'-GGGCATATCGGAGTCA-3' | 4 → 3                                                                | 3                                                                           | 1                                                | 0                                            | 4                                                | 3                                          | 1                                       | 0                                          | 0 (1)                                                            |
| FUS e10::NR4A3 e3     | 5                | 3                     | 5'-GGGCATATCATCAAAC-3' | 9 → 4                                                                | 6                                                                           | 1                                                | 1                                            | 189                                              | 141                                        | 6                                       | 8 (TFG)                                    | 2 (22)                                                           |
| FUS e11::NR4A3 e3     | 5                | 3                     | 5'-GGGCATATCTCCTCGC-3' | 30 → ≥1                                                              | ≥15                                                                         | ≤1                                               | 0                                            | 60                                               | 30                                         | 1                                       | 9 (EWSR1)                                  | 0 (3)                                                            |
| FUS e13::NR4A3 e3     | 5                | 3                     | 5'-GGGCATATCCATGTGA-3' | 6 → 5                                                                | 2                                                                           | 2                                                | 1                                            | 20                                               | 8                                          | 2                                       | 7 (EWSR1)                                  | 1 (9)                                                            |
| TAF15 e1::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCCGACATG-3' | 5 → 2                                                                | 0                                                                           | 0                                                | 0                                            | 5                                                | 0                                          | 0                                       | 0                                          | 0 (0)                                                            |
| TAF15 e4::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCTGACTGA-3' | 34 → ≥3                                                              | ≥15                                                                         | ≤4                                               | 1                                            | 78                                               | 57                                         | 7                                       | 11 (NR4A3) ■                               | 10 (53)                                                          |
| TAF15 e6::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCTTGTGTG-3' | 11 → 7                                                               | 7                                                                           | 4                                                | 2                                            | 62                                               | 10                                         | 5                                       | 9 (TFG)                                    | 2 (2)                                                            |
| TAF15 e8::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCACCAAAA-3' | 41 → ≥4                                                              | ≥13                                                                         | ≤2                                               | 0                                            | 133                                              | 93                                         | 3                                       | 7 (TCF12)                                  | 10 (17)                                                          |
| TAF15 e9::NR4A3 e3    | 5                | 3                     | 5'-GGGCATATCAGCATCT-3' | 23 → ≥7                                                              | ≥0                                                                          | ≤2                                               | 0                                            | 68                                               | 48                                         | 5                                       | 6 (EWSR1)                                  | 1 (4)                                                            |
| TAF15 e11::NR4A3 e3   | 5                | 3                     | 5'-GGGCATATCATCAAAC-3' | 9 → 4                                                                | 6                                                                           | 1                                                | 1                                            | 189                                              | 141                                        | 6                                       | 8 (TFG)                                    | 2 (22)                                                           |
| TAF15 e12::NR4A3 e3   | 4                | 2                     | 5'-AGGGCATATCTCGCCG-3' | 42 → ≥4                                                              | ≥11                                                                         | ≤3                                               | 0                                            | 42                                               | 33                                         | 3                                       | 11 (NR4A3) ■                               | 3 (4)                                                            |
| TAF15 e14::NR4A3 e3 † | 5                | 3                     | 5'-GGGCATATCTCCTCCT-3' | ≥50 → ≥2                                                             | ≥11                                                                         | ≤2                                               | 0                                            | 174                                              | 95                                         | 6                                       | 10 (EWSR1) ■                               | 19 (40)                                                          |

**Table 3. Predicted specificity per screened junction · numbered notes are under the caption, on this table's first page — † no design at this junction clears the parent screen; do not order the sequence in a marked row — ▀ this design pairs a wild-type parent through the whole catalytic gap; do not order it · both markers are read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction             | designs screened | best gap-level margin | that design            | near-matches, either strand (transcripts → loci; ≥ is a lower bound) | of the retained hits, on the sense strand <sup>1</sup> (≥ is a lower bound) | loci with a gap-paired hit (≤ is an upper bound) | of those, predicted models only <sup>2</sup> | at the deeper ceiling: near-matches <sup>3</sup> | of those, on the sense strand <sup>3</sup> | loci with a gap-paired hit <sup>3</sup> | longest parent duplex through the gap (bp) | ≤1-mismatch matches across that junction's designs, median (max) |
|----------------------|------------------|-----------------------|------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------|-----------------------------------------|--------------------------------------------|------------------------------------------------------------------|
| TCF12 e3::NR4A3 e3 † | 5                | 3                     | 5'-GGGCATATCTGATCCA-3' | 15 → 4                                                               | 10                                                                          | 2                                                | 2                                            | 374                                              | 246                                        | 6                                       | 11 (NR4A3) ▀                               | 2 (35)                                                           |
| TCF12 e5::NR4A3 e3   | 5                | 3                     | 5'-GGGCATATCCATCAGA-3' | 26 → ≥1                                                              | ≥15                                                                         | ≤17                                              | 0                                            | 83                                               | 70                                         | 1                                       | 7 (EWSR1)                                  | 2 (32)                                                           |
| TCF12 e7::NR4A3 e3   | 4                | 2                     | 5'-GGCATATCAAGCGCTG-3' | 2 → 2                                                                | 0                                                                           | 0                                                | 0                                            | 2                                                | 0                                          | 0                                       | 11 (TCF12) ▀                               | 0 (1)                                                            |
| TCF12 e9::NR4A3 e3   | 4                | 3                     | 5'-GGGCATATCTTGCATA-3' | 14 → 2                                                               | 8                                                                           | 1                                                | 0                                            | 86                                               | 64                                         | 6                                       | 8 (NR4A3)                                  | 8 (23)                                                           |
| TCF12 e11::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATCTAGAATG-3' | 15 → 5                                                               | 2                                                                           | 1                                                | 1                                            | 46                                               | 17                                         | 4                                       | 7 (NR4A3)                                  | 2 (6)                                                            |
| TCF12 e13::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATCTGTGAGA-3' | 17 → ≥8                                                              | ≥6                                                                          | ≤1                                               | 1                                            | 170                                              | 80                                         | 4                                       | 11 (NR4A3) ▀                               | 1 (12)                                                           |
| TCF12 e17::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATCTCTATAA-3' | 8 → 3                                                                | 0                                                                           | 0                                                | 0                                            | 118                                              | 101                                        | 5                                       | 7 (TCF12)                                  | 1 (6)                                                            |
| TCF12 e19::NR4A3 e3  | 5                | 3                     | 5'-GGGCATATCTCTGACT-3' | 12 → 4                                                               | 8                                                                           | 1                                                | 1                                            | 102                                              | 79                                         | 3                                       | 8 (TCF12)                                  | 1 (2)                                                            |
| TFG e2::NR4A3 e3 †   | 5                | 3                     | 5'-GGGCATATCTTCATCT-3' | ≥50 → ≥1                                                             | ≥15                                                                         | ≤2                                               | 0                                            | 207                                              | 117                                        | 5                                       | 10 (TFG) ▀                                 | 35 (87)                                                          |
| TFG e3::NR4A3 e3     | 5                | 3                     | 5'-GGGCATATCAAATAAT-3' | 14 → 6                                                               | 9                                                                           | 3                                                | 0                                            | 217                                              | 161                                        | 6                                       | 8 (NR4A3)                                  | 9 (17)                                                           |
| TFG e4::NR4A3 e3     | 5                | 3                     | 5'-GGGCATATCATTTTCA-3' | ≥50 → ≥3                                                             | ≥15                                                                         | ≤5                                               | 2                                            | 318                                              | 238                                        | 15                                      | 11 (TFG) ▀                                 | 41 (100)                                                         |
| TFG e5::NR4A3 e3     | 5                | 3                     | 5'-GGGCATATCTGAAACC-3' | 41 → ≥6                                                              | ≥11                                                                         | ≤7                                               | 1                                            | 112                                              | 72                                         | 10                                      | 11 (NR4A3) ▀                               | 3 (21)                                                           |

**Table 3. Predicted specificity per screened junction · numbered notes are under the caption, on this table's first page — † no design at this junction clears the parent screen; do not order the sequence in a marked row — ▮ this design pairs a wild-type parent through the whole catalytic gap; do not order it · both markers are read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| junction         | designs screened | best gap-level margin | that design            | near-matches, either strand (transcripts → loci; ≥ is a lower bound) | of the retained hits, on the sense strand <sup>1</sup> (≥ is a lower bound) | loci with a gap-paired hit (≤ is an upper bound) | of those, predicted models only <sup>2</sup> | at the deeper ceiling: near-matches <sup>3</sup> | of those, on the sense strand <sup>3</sup> | loci with a gap-paired hit <sup>3</sup> | longest parent duplex through the gap (bp) | ≤1-mismatch matches across that junction's designs, median (max) |
|------------------|------------------|-----------------------|------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------|-----------------------------------------|--------------------------------------------|------------------------------------------------------------------|
| TFG e6::NR4A3 e3 | 5                | 3                     | 5'-GGGCATATCTTCAATC-3' | 37 → ≥3                                                              | ≥2                                                                          | 0                                                | 0                                            | 238                                              | 193                                        | 3                                       | 9 (TFG)                                    | 11 (21)                                                          |
| TFG e7::NR4A3 e3 | 5                | 3                     | 5'-GGGCATATCTGAATAC-3' | 27 → ≥3                                                              | ≥3                                                                          | ≤3                                               | 1                                            | 43                                               | 12                                         | 4                                       | 11 (NR4A3) ▮                               | 4 (26)                                                           |

## 2.4 · Strand orientation, and designs with no sense-strand near-match

All 38 in-frame junctions were screened with orientation filtered, covering 183 designs, and Table 3 gives the per-junction result. Of the 1,677 apparent gap-paired hits across the retained hit lists, 738 sit on the minus strand, or 44%. An antisense oligonucleotide cannot base-pair with those at all.

The proportion is not uniform. It runs from 0% at *TFG* exon 4, where no apparent risk is minus-strand, to 100% at both *EWSR1* exon 1 and *TCF12* exon 7, where every one is. That non-uniformity is what makes the filter worth applying rather than approximating. A uniform inflation would rescale every junction and leave the distances between them intact; this one does not, so an apparent count is not a proxy for the filtered one even between neighbours. *EWSR1* exons 7 and 13 return 55 and 57 apparent gap-paired hits (all but indistinguishable) and after filtering they stand at 6 and 53.

Under the stricter criterion — no sense-strand near-match anywhere, not merely across the catalytic gap as in §2.3 — nine designs at six junctions, a different nine from the nine multi-partner designs of §2.2, carry none among non-parent transcripts after filtering (Table 4), spanning four of the five partners — five of them do-not-order designs for a reason no near-match screen sees, marked ▮ in the list and explained after it. Three are at *EWSR1* exon 1 (5'-GGGCATATCCGTGGAC-3', 5'-GGCATATCCGTGGACG-3' ▮, 5'-GCATATCCGTGGACGC-3' ▮), one at *FUS* exon 8 (5'-AGGGCATATCGGAGTC-3'), one at *TAF15* exon 1 (5'-GGGCATATCCGACATG-3'), and four at *TCF12* — 5'-GGGCATATCTCTATAA-3' at exon 17, 5'-CAGGGCATATCTTGCA-3' ▮ at exon 9, and 5'-GGCATATCAAGCGCTG-3' ▮ and 5'-GCATATCAAGCGCTGC-3' ▮ at exon 7. Five of those nine carry ▮ here and `do_not_order` in the canonical file, because the mature-parent screen of §2.5 condemns them at eleven or twelve base pairs: this is a list of designs clean on the near-match screens, and it is not a list of designs to order. The exhaustive transcript scan agrees independently: each returns no exact and no single-mismatch match anywhere in 186,185 transcripts. The two screens fail in different ways, so their agreement is not a restatement. One is a heuristic alignment search over both strands; the other an exhaustive substitution scan over the sense orientation only. The pre-mRNA screen, over a compartment neither of those reaches, does not overturn them either: none of the nine carries a

pre-mRNA site of the strict class §2.5 defines (gap-paired, sense-strand), though §2.5 is silent on the wider sense-strand forty, of which 21 pair all of the gap but one or two positions.

The graded re-score agrees, with one instructive exception. Scoring every retained hit by the residual cleavage a gap-internal mismatch is predicted to permit, under both literature bounds, returns a residual load of zero for all nine. It returns zero for one further design too, at *FUS* exon 11, which is not counted as clean here. That design returns 21 near-matches, of which only 15 are retained, and all 15 are minus-strand. The graded score therefore sees nothing it can score, while the cleanliness criterion refuses the design because the strand of the six unretained hits is unknown. The graded model has no censoring guard, so it can award a zero the hit list does not support, and the stricter count is the one reported. A zero for the nine is arithmetic rather than an independent measurement: it follows from their having no sense-strand hit to score.

Every junction was then re-screened at a tenfold deeper alignment ceiling, with retention raised to match so that no hit list is truncated: 38 junctions and 187 design records. The result withdraws most of the set above. The 187 are the panel's 190 less three that failed at the remote service on this pass, two at *FUS* exon 5 and one at *TFG* exon 2. They had returned 23, 41 and 31 near-matches at the default depth, but a raw either-strand count decides nothing about sense-strand cleanliness (§5); what disqualifies them is that none carries a deep hit list at all, which §2.7 requires of a candidate, and no count below depends on them. Only three of the nine still carry no sense-strand near-match: 5'-AGGGCATATCGGAGTC-3' at *FUS* exon 8, 5'-GGGCATATCCGACATG-3' at *TAF15* exon 1 and 5'-GGCATATCAAGCGCTG-3' ▮ at *TCF12* exon 7 — the last of which the mature-parent screen condemns at eleven base pairs against wild-type *TCF12* and §2.7 excludes, so it is not to be ordered. Each returned the same count at both depths. The other six did not. The three *EWSR1* exon-1 designs had returned no near-match at all at the default ceiling and return 27, 29 and 84; 5'-GGGCATATCTCTATAA-3' at *TCF12* exon 17 goes from 8 to 118 on either strand, and 5'-CAGGGCATATCTTGCA-3' ▮ at *TCF12* exon 9 from 7 to 67; and 5'-GCATATCAAGCGCTGC-3' ▮ at *TCF12* exon 7, which had one near-match and none on the sense strand, returns 18 with two. Three of the six carry hits that span the catalytic gap and so are gap-paired rather than merely sense-strand matches: 64 for 5'-GCATATCCGTGGACGC-3' ▮, 14 for

5'-GGGCATATCTCTATAA-3' and 11 for 5'-CAGGGCATATCTTGCA-3' ▀. Three of the designs named in this paragraph carry ▀ because the mature-parent screen condemns them at a twelve-base-pair duplex — 5'-CAGGGCATATCTTGCA-3' against wild-type *NR4A3*, 5'-GCATATCAAGCGCTGC-3' against wild-type *TCF12* and 5'-GCATATCCGTGGACGC-3' against wild-type *EWSR1* — so none of them is to be ordered, and they are named here only for what the depth change does to their hit counts. A count of zero at the default ceiling was not a count of zero, which is the sharpest form of the bound §5 sets out. The three named above are clean on the near-match screens alone; composing them with the mature-parent screen changes which design stands at *TCF12* exon 7, as §2.7 sets out, so this list is not the candidate set and must not be ordered as one.

The deeper pass also decided what the default one could not. Seven of the 190 designs had failed at the remote service and carried no count at all — a different seven from the seven §5 reports as

withheld by retention alone, which do carry default-depth counts; all seven returned at the deeper ceiling, six of them dirty and one (5'-GGGCATATCAAGCGCT-3' at *TCF12* exon 7) with three near-matches and none on the sense strand. So the set of designs with a complete hit list and no sense-strand near-match is four at this depth rather than three: a design the shallower pass never screened joins the three that survived it. The deeper counts are reported as their own measurement and no figure quoted above is restated from them.

The orientation call is corroborated independently of any of this. Ten designs return perfect 16/16 BLAST matches while the sense-only exhaustive scan reports no exact match, and every one of those BLAST hits is on the minus strand. That last is read off the alignments rather than deduced from the disagreement, and the deduction would not have been available: the two screens do not search one corpus (§6, screens 1 and 2), so a hit to a record outside the pinned GRCh38.p14 set would reconcile the two results without any statement about orientation at all.

**Table 4. The 9 designs with no sense-strand near-match at the default search depth — six of them at that depth only, and five of them not to be ordered.** Six of these lose the property when the same junctions are re-screened at a tenfold deeper alignment ceiling, three of them having returned no near-match at all here; §2.4 reports that measurement and names the three that survive it. This table is the default-depth result, retained because it is the depth at which the corpus-wide counts elsewhere in the paper were computed. The gap-level margin is the count of bases inside the six-nucleotide catalytic gap that no wild-type parent carries at that position, taken on the shorter side of the junction; it runs from 1 to 3, the geometry's ceiling of half the gap rounded down, and it is a count of bases and not a score. Every design that QUALIFIES is listed, at each of the 6 junctions where one does; this is not one row per junction, and it is not every design at those junctions, which are tiled by five registers each. A design qualifies only if its retained hit list is not truncated — no more near-matches than the 15 the screens store — because the strand of an unstored hit cannot be recovered, so a truncated list cannot establish that nothing on the sense strand remains. The underlying search is itself capped, so these are the designs whose near-match lists are shortest, not the designs whose lists are known to be exhaustive.  $\Delta\Delta G^\circ 37$  is the margin by which the fusion duplex is favoured over the better of the two runs a parent pairs at the junction itself, for an unmodified DNA:RNA hybrid; it does not score the mature-parent duplexes of §2.5, of which 87 designs of the panel carry one, 61 of them against wild-type *NR4A3*, and 59 of THOSE 61 at one recurring site rather than anywhere else in a parent — the mature *NR4A3* exon-2/exon-3 seam every design's acceptor half reaches. Because the fusion duplex pairs both LNA wings and each parent duplex only one, it is a lower bound on the modified oligonucleotide's discrimination rather than an upper one. RefSeq lists one accession per annotated variant, so every near-match count in this table is a count of transcript accession RECORDS and not of distinct genes. Across the deep screens' uncensored designs, taken over every stored near-match rather than the gap-paired subset, the collapse runs at a median of 5.52 records per locus and a maximum of 28.0; §5 gives the same figure at the DEFAULT ceiling, which is smaller and does not describe a column read at the deeper one. **▲ The deep columns are not a records-and-their-own-loci pair:** the deep locus column counts only the GAP-PAIRED subset of the records the near-match columns beside it count, so dividing one by the other is a ratio of two different hit sets and not a records-per-locus figure. None of these numbers is a measurement of off-target activity, and none speaks to cleavage. **This table condemns nothing and clears nothing.** Its final column is a verdict from ONE screen, the near-match screen, and five of these rows carry the **■** of the mature-parent screen: a wild-type parent pairs their whole catalytic gap at the ten-base-pair criterion, which is this paper's central negative. **Those five verdicts are carried in this table, and mostly nowhere else.** Table 3 prints one row per junction, so one of them — 5'-GGCATATCAAGCGCTG-3' — is also printed in Table 3 and carries **■** there; the other four are marked in no other display item. Table 3 selects each junction's HIGHEST-MARGIN design, which at those seams is a different molecule: at EWSR1 e1::NR4A3 e3 it prints 5'-GGGCATATCCGTGGAC-3', reading 0, unmarked; at TCF12 e9::NR4A3 e3 it prints 5'-GGGCATATCTTGCATA-3', reading 8 (*NR4A3*), unmarked. Do not read an unmarked Table 3 row as a clearance of a design marked here. A design can survive every near-match screen here and still be one not to order. Three columns read one value in every row, and they are not one kind of fact. “of those, on the sense strand” (at the default depth) reads 0 in all 9 rows BY CONSTRUCTION: it is the property membership of this table is defined by. “exact /  $\leq 1$ -mismatch matches” reads 0 / 0, “residual cleavage load, both bounds<sup>4</sup>” reads 0 in all 9 rows, which is a measurement over these designs and not an unfilled cell; neither separates one row from another. Every sequence in this table is an antisense 16-mer, tiled in the 5-6-5 locked-nucleic-acid (LNA)/DNA/LNA architecture on a phosphorothioate backbone that §6 specifies; the bases alone, ordered as unmodified DNA, are a different molecule. Research use only: not for administration to any person or animal (Declarations). The canonical machine-readable copy of every sequence is `fusion-junction-aso-sequences.csv`, which is what to order from rather than this PDF.

<sup>4</sup> A DISCRIMINATION-WEIGHTED COUNT OF NEAR-MATCH SITES, and the unit is sites: each of the design's near-matches enters at one, reduced by the modelled loss of RNase-H1 cleavage at however many mismatches fall in its catalytic gap, and the weights are summed — so a 0 means no near-match survives either weighting, and the column is not a rate, a concentration or a quantity of cleaved RNA. Computed under the optimistic five-fold and the pessimistic no-discrimination bound on RNase-H1 single-mismatch discrimination. A single value means the two bounds agree; where two are printed, the width is TRUNCATION of the saved hit list and not statistical uncertainty. No RNase-H1 cleavage is measured anywhere in this work.

<sup>5</sup> The four conventional antisense design rules audited are GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, and no CpG dinucleotide.

<sup>6</sup> Whether the design still carries no sense-strand near-match once its junction is re-screened at the tenfold deeper ceiling. The verdict is computed from the three deep columns beside it, not asserted, so this table cannot come to disagree with §2.4 about which designs survive. It is a verdict on that screen alone and not on the parent screen of note <sup>7</sup>. The six that do not are the reason this table's default-depth zeros must not be read on their own.

<sup>7</sup> The longest contiguous duplex a wild-type parent gene forms through this design's whole catalytic gap, with the gene that forms it. **■** marks ten base pairs or more, the criterion applied throughout: **do not order a design marked ■** — pairing a parent through the whole gap is this paper's central negative and surrenders the only advantage the modality has. An unmarked row is not a clearance, only a reading at that one cut. That ten-base-pair cut is a criterion this work ADOPTS rather than measures, and the reading moves with it: at seven base pairs 175 of the 190 panel designs pair a parent through the whole gap against 87 at ten, 181 do so at some length, and only nine of the 38 seams have a design that clears against 35 at ten (§2.9). A cell below the cut is a reading at one cut and nothing wider.



**Table 4. The 9 designs with no sense-strand near-match at the default search depth — six of them at that depth only, and five of them not to be ordered · numbered notes are under the caption, on this table's first page — ▀ this design pairs a wild-type parent through the whole catalytic gap; do not order it · the marker is read at the ten-base-pair criterion, which this work adopts rather than measures; an unmarked row is not a clearance, only a reading at that one cut**

| design                 | junction            | GC (%) | gap-level margin | $\Delta\Delta G^{\circ}37$ (kcal/mol) | near-matches, either strand | of those, on the sense strand | exact / $\leq 1$ -mismatch matches | residual cleavage load, both bounds <sup>4</sup> | conventional rules failed <sup>5</sup> | at the deeper ceiling: near-matches | of those, on the sense strand | loci with a gap-paired hit | longest wild-type parent duplex through the gap (bp) <sup>7</sup> | survives the near-match screen <sup>6</sup> |
|------------------------|---------------------|--------|------------------|---------------------------------------|-----------------------------|-------------------------------|------------------------------------|--------------------------------------------------|----------------------------------------|-------------------------------------|-------------------------------|----------------------------|-------------------------------------------------------------------|---------------------------------------------|
| 5'-GCATATCCGTGGACGC-3' | EWSR1 e1::NR4A3 e3  | 62.5   | 1                | 7.981                                 | 0                           | 0                             | 0 / 0                              | 0                                                | GC outside 40–60%, contains a CpG      | 84                                  | 83                            | 1                          | 12 ( <i>EWSR1</i> )<br>▀                                          | no                                          |
| 5'-GGCATATCCGTGGACG-3' | EWSR1 e1::NR4A3 e3  | 62.5   | 2                | 10.085                                | 0                           | 0                             | 0 / 0                              | 0                                                | GC outside 40–60%, contains a CpG      | 29                                  | 22                            | 0                          | 11 ( <i>EWSR1</i> )<br>▀                                          | no                                          |
| 5'-GGGCATATCCGTGGAC-3' | EWSR1 e1::NR4A3 e3  | 62.5   | 3                | 12.189                                | 0                           | 0                             | 0 / 0                              | 0                                                | GC outside 40–60%, contains a CpG      | 27                                  | 18                            | 0                          | 0                                                                 | no                                          |
| 5'-AGGGCATATCGGAGTC-3' | FUS e8::NR4A3 e3    | 56.2   | 2                | 10.895                                | 3                           | 0                             | 0 / 0                              | 0                                                | contains a CpG                         | 3                                   | 0                             | 0                          | 0                                                                 | yes                                         |
| 5'-GGGCATATCCGACATG-3' | TAF15 e1::NR4A3 e3  | 56.2   | 3                | 11.894                                | 5                           | 0                             | 0 / 0                              | 0                                                | contains a CpG                         | 5                                   | 0                             | 0                          | 0                                                                 | yes                                         |
| 5'-GGGCATATCTCTATAA-3' | TCF12 e17::NR4A3 e3 | 37.5   | 3                | 8.556                                 | 8                           | 0                             | 0 / 0                              | 0                                                | GC outside 40–60%                      | 118                                 | 101                           | 5                          | 7 ( <i>TCF12</i> )                                                | no                                          |
| 5'-GCATATCAAGCGCTGC-3' | TCF12 e7::NR4A3 e3  | 56.2   | 1                | 7.980                                 | 1                           | 0                             | 0 / 0                              | 0                                                | contains a CpG                         | 18                                  | 2                             | 0                          | 12 ( <i>TCF12</i> ) ▀                                             | no                                          |
| 5'-GGCATATCAAGCGCTG-3' | TCF12 e7::NR4A3 e3  | 56.2   | 2                | 10.085                                | 2                           | 0                             | 0 / 0                              | 0                                                | contains a CpG                         | 2                                   | 0                             | 0                          | 11 ( <i>TCF12</i> ) ▀                                             | yes                                         |
| 5'-CAGGGCATATCTTGCA-3' | TCF12 e9::NR4A3 e3  | 50.0   | 1                | 9.325                                 | 7                           | 0                             | 0 / 0                              | 0                                                | none                                   | 67                                  | 18                            | 4                          | 12 ( <i>NR4A3</i> )<br>▀                                          | no                                          |

## 2.5 · The parents: liability in pre-mRNA and in mature transcript

RNase-H1 is active in the nucleus and gapmers engage pre-mRNA, so a screen over mature transcripts cannot see intronic or intron–exon-spanning sites. That omission is not neutral in its direction. A junction gapmer's two halves are both exonic, and in a parent pre-mRNA an exon is followed by an intron rather than by the next exon. Parent pre-mRNA is therefore precisely where a design's donor half sits beside sequence no mature screen has compared it against. A mature-only screen returns a low count partly by construction.

That argument cuts one way and the paper does not pretend otherwise. The fusion junction is itself made by splicing: in an unspliced fusion transcript the donor exon is followed by intron, exactly as in a parent, so the target these designs are built for exists only in spliced transcript, while the parent pre-mRNA sites counted here are additionally available on nascent transcript. The screens do not resolve that asymmetry between target and liability, and nothing here measures how much of either species a gapmer meets.

Of the 190 designs, 53 have a near-match somewhere in parent pre-mRNA. Nineteen carry one that meets all three conditions that would make it dangerous: it is on the sense strand, it pairs the catalytic

gap in full, and it touches intronic sequence. That third condition is what makes such a site invisible to both transcript screens, rather than a re-count of something already reported.

The two liability classes are not disjoint designs, and the arithmetic has to be done as a union rather than a sum. Thirteen of those 19 are already among the 87 designs the mature-parent screen below returns (screen 4, which finds a parent duplex of at least ten base pairs through the whole gap), so the pre-mRNA compartment adds six designs that screen misses entirely, and the two screens together condemn 93 of 190 rather than the  $87 + 19 = 106$  a sum would give. That union is arithmetic and not a claim that the two liabilities are equivalent: an intron–exon-spanning site exists only between transcription of that region and its splicing, while a mature-transcript site is present for as long as the transcript is, and nothing here measures either abundance. The 93 is a count of designs condemned by at least one screen, not a graded liability. What the compartment adds is therefore a liability class invisible to every mature screen, not a second population of comparable size: the six are the number that matters for a laboratory choosing among designs that already passed the mature-parent screen.

The step from 53 to 19 runs through two filters and only the second is a threshold. Thirteen of the 53 carry their pre-mRNA sites on the minus strand alone, which an antisense oligonucleotide cannot pair at all, leaving forty with a sense-strand parent pre-mRNA site. The step from forty to 19 is then a threshold rather than a measurement, and the class it removes is the one the Methods (§6) decline to dismiss: the 19 are those pairing the catalytic gap in full, and the remaining 21 pair all of it but one or two positions. Of their 28 sites, 26 are a single gap mismatch short; five of the 28 are in *NR4A3* itself. Under the bounds this work adopts, a single mismatch inside the gap does not abolish cleavage, so those 21 are not a null result. They are excluded because a graded count over this compartment would need a discrimination model the literature does not supply for a parent duplex. The same condition governs the mature-parent screen below, which considers only windows pairing the whole gap. The headline counts of this section should be read as the fully-paired class, not as the whole parent liability: the 19 is drawn from the forty and the forty from the 53, while the 21 the 19 leaves out is counted separately rather than being a tally it comes from. The mature-parent counts below carry the same exclusion, and their own wider tally is the 181 designs a parent can pair the whole gap for at any length (§2.9).

Those 19 designs fall into two classes that do not mix, and only one is mechanistically interesting. Nine are intron–exon-spanning, and all nine read the same two sites in *NR4A3*: each begins in the last six or seven nucleotides of intron 2 (an intron 2,208 nucleotides long, so the site sits at its 3' end and not near its start) and spans the boundary into exon 3. That follows from the design problem. A junction gapmer's acceptor half is the 5' end of *NR4A3* exon 3, and the wild-type *NR4A3* transcript reaches that same exon across its own splice junction. So a design whose donor half also matches the 3' end of intron 2, within the mismatch budget, pairs across the real splice site. That is a route to wild-type *NR4A3* engagement which does not pass through the fusion at all, in the compartment where RNase-H1 is active. It is the discrimination question this paper is about.

**And that site carries a second liability no screen here grades.** The sequence those nine designs cover is not incidental: the last

twelve nucleotides of *NR4A3* intron 2 are CTGTCCCTGCAG, a pyrimidine-rich tract closing on the AG that terminates the intron, so the site is the wild-type 3' splice acceptor itself. An oligonucleotide occupying a splice acceptor can alter the splicing of the transcript it sits on by occupancy alone, which is the steric mechanism §3 names as unevaluated here, and that route requires no catalytic gap, no ten-base-pair duplex and no RNase-H1. Every screen in this paper grades hybridisation-dependent cleavage liability (§5), so none of them sees it, and the one- to five-fold single-mismatch discrimination bounds this work imports do not bound it. What that would do to wild-type *NR4A3* is not predicted here; it is named because the screens are silent on it rather than reassuring about it. The other ten are wholly intronic and every one is in *TCF12*, which contributes 365,096 of the 517,157 intronic nucleotides searched. That is 71% of the search space accounting for 100% of the class: volume alone predicts about seven of the ten, so it accounts for most of the concentration and the remainder should not be read as anything about *TCF12*.

The liability tracks the tiling register, of which the gap-level margin is a function. At margin 1, 12 of 76 designs carry a gap-paired sense-strand pre-mRNA site of the strict class just defined — the 19, not the 53 or the 40; at margin 2, 7 of 76; at margin 3, none of 38. Eight of the nine designs reading those two *NR4A3* boundary sites are at the shortest donor-side register, which needs the fewest intronic bases to match. None of the nine designs with no sense-strand near-match on either transcript screen carries one.

The second of the two liability compartments (mature parent transcript rather than pre-mRNA) is the larger. Each of the first three screens misses it for its own reason. The alignment screen excludes parent records by design and filters at  $\geq 14/16$  identity. The exhaustive transcript scan admits only one mismatch. The pre-mRNA screen searches unspliced sequence and so cannot reach a mature exon–exon junction. A parent duplex of 11 or 12 contiguous base pairs that pairs the whole catalytic gap is therefore invisible to all three, while satisfying the duplex criterion adopted here — which §5 and §6 record as stated rather than measured for this architecture.

Screen 4 compares every design's target window to every window of all six mature parents. 87 of 190 designs have a duplex of at least ten base pairs, and 61 of those 87 are against wild-type *NR4A3*, the transcript this modality must spare. Six transcripts are searched, so a duplex could in principle fall on a gene with no relation to the design's own fusion, and that reading has to be separated from the one the title makes: 85 of the 87 pair one of the design's own two parent genes — *NR4A3* or its own donor — and two do not, one *FUS* design paired by *TAF15* and one *TAF15* design paired by *EWSR1*, both inside the FET paralogue family. The liability is therefore self-targeting almost throughout, which is what makes it a property of the modality rather than an ordinary off-target count. Sub-threshold readings do not follow that pattern: the two lead reagents of §4.1 run to eight and nine base pairs against *TFG*, a gene neither fusion involves. The count is a floor at the threshold chosen, and seven base pairs is the shortest end of the same cited range (§6): at seven the same screen returns 175 of 190 rather than 87, and 9 of the 38 junctions retain a design clearing it rather than 35. Loosening the cut enlarges the negative on both of those readings and does not strengthen it, because the nulls below move with it. Re-run at seven, the exon-terminus chimera ensemble reaches 91.4% against the 92.1% that 175 of 190 is, and the comparison attributed to *NR4A3*

reverses outright, 73 of 190 or 38.4% observed against 46.6% for the null. At seven this screen is measuring what any chimera of two real transcripts does.

Ten is adopted, and reporting two cuts is still a choice of two cuts. Whether 87 of 190 is a property of these designs or a property of where the criterion was put is a question about the whole range, so the whole range is measured — the observed arm, every null ensemble, and the junction-level reading, at every cut the instrument can reach. The shortest counted run is the catalytic gap itself, six base pairs, which is the only cut with an enzymological referent, since a counted run of any length presents exactly those six as RNA:DNA hybrid and the rest as locked-wing pairs; thirteen is the longest run any arm returns.

| cut (bp) | liable designs | %    | Wilson 95% | strongest null %                     | scramble null % | observed – strongest | strongest null vs the observed interval | junctions with a clearing design | designs still liable at the five published-breakpoint junctions | designs NR4A3 pairs specifically |
|----------|----------------|------|------------|--------------------------------------|-----------------|----------------------|-----------------------------------------|----------------------------------|-----------------------------------------------------------------|----------------------------------|
| 6        | 181            | 95.3 | 91.2–97.5  | 98.2 (exon-terminus, novel acceptor) | 91.4            | –2.9                 | outside, above                          | 6 of 38                          | 25 of 25                                                        | 158                              |
| 7        | 175            | 92.1 | 87.4–95.2  | 91.4 (exon-terminus)                 | 74.3            | +0.7                 | inside                                  | 9 of 38                          | 25 of 25                                                        | 111                              |
| 8        | 143            | 75.3 | 68.7–80.9  | 76.2 (exon-terminus)                 | 43.8            | –1.0                 | inside                                  | 23 of 38                         | 21 of 25                                                        | 84                               |
| 9        | 98             | 51.6 | 44.5–58.6  | 56.8 (exon-terminus, novel acceptor) | 18.4            | –5.2                 | inside                                  | 31 of 38                         | 14 of 25                                                        | 63                               |
| 10       | 87             | 45.8 | 38.9–52.9  | 40.6 (exon-terminus)                 | 6.2             | +5.2                 | inside                                  | 35 of 38                         | 11 of 25                                                        | 62                               |
| 11       | 84             | 44.2 | 37.3–51.3  | 34.7 (exon-terminus)                 | 1.8             | +9.5                 | outside, below                          | 37 of 38                         | 11 of 25                                                        | 62                               |
| 12       | 35             | 18.4 | 13.6–24.5  | 14.5 (exon-terminus)                 | 0.4             | +3.9                 | inside                                  | 38 of 38                         | 5 of 25                                                         | 24                               |
| 13       | 6              | 3.2  | 1.5–6.7    | 4.7 (exon-terminus)                  | 0.1             | –1.6                 | inside                                  | 38 of 38                         | 1 of 25                                                         | 5                                |

The strongest null at every cut is one of the two exon-terminus chimera arms (the column names which, because they change places at six and nine) and the comparison against it does not behave as a real effect behaves. **The excess is not monotone in the cut and changes sign four times:** the observed rate is below the strongest null at six, eight, nine and thirteen, and above it at seven, ten, eleven and twelve. The strongest null lies inside the observed rate's own nominal interval at every cut but two — at six it lies above it, and at eleven, which is not the cut reported here, below it. **An earlier version of this section said that ten is the cut at which the observed rate stands clear of every null. The ladder was written to check that sentence and it does not survive: at ten the exon-terminus null is inside the interval, as the closing paragraphs of this section already said in another form.** Ten is the cut this work adopts and states every count at, on the criterion's stated provenance (§6) rather than on separation, and no cut in the reachable range resolves an excess over a chimera of two real exon termini. The central negative is therefore not weakened by the ladder; it is the same negative, holding across the criterion's whole range instead of at one point in it.

Three further readings come off the same table and none of them is available from a single cut. The junction-level and design-level counts move in opposite directions, so a looser criterion enlarges the class and empties the panel at once: at six, 181 of 190 designs are liable and only 6 of 38 junctions retain a design that clears. At the

**The cut ladder.** Every cut the instrument can reach, from the catalytic gap's own six base pairs to the thirteen that is the longest run any arm returns: the observed arm, its Wilson interval, the strongest null ensemble at that cut and the scramble null beside it, the signed excess of observed over strongest null, the junction-level reading, and how many designs pair wild-type NR4A3 specifically. This is the paper's central negative and is set here, in the argument that reads it, rather than with the numbered tables: the excess over the strongest null changes sign four times down the column, so no cut in the range is a boundary the data picks out. Derived, like every table here, from the released artefacts; the adopted cut of ten is one row of it and is adopted, not measured.

five junctions any patient is reported to carry, every design is liable at six and at seven, and the count falls gradually rather than stepping — 21 of those 25 at eight, 14 at nine, 11 at ten — so the deliverable is more cut-dependent than the corpus is. And the count of designs wild-type NR4A3 pairs specifically (asked of NR4A3 alone rather than attributed to whichever parent returned the longest run) is 62 at ten against the 61 attributed there, and 158 at six against 77: the attributed figure this paper reports is a floor on the NR4A3-specific one, and the two separate as the cut loosens. Those 61 are not 61 distinct sites: 59 of them are the same one, the mature exon-2/exon-3 seam every design's acceptor half reaches, which is the mature-transcript counterpart of the pre-mRNA concentration above. One further design pairs NR4A3 at eleven base pairs but another parent at twelve, so it is attributed elsewhere. The count falls steeply with the gap-level margin: 50 of 76 designs at margin 1, 29 of 76 at margin 2, and 8 of 38 at margin 3. That is what the margin's definition predicts, since at margin 1 a parent needs one lucky base to pair the whole gap and at margin 3 it needs three. Five of the nine designs of §2.4 carry such a duplex at 11 or 12 base pairs, including 5'-CAGGGCATATCTTGCA-3' against wild-type NR4A3 itself — a do-not-order design, ■ in Table 4. The margin is therefore a predictor of parent engagement rather than a guarantee against it, because it counts bases unique to the fusion at the junction without asking whether a parent carries them elsewhere.

Eighty-seven of 190 is 45.8%, with a nominal binomial Wilson 95% interval of 38.9–52.9% — nominal because it treats the 190 records as independent draws, which the close of this section states they are not. Clustering on the 38 junctions the five registers of each are tiled from, with the within-junction correlation estimated by one-way analysis of variance, gives a design effect of 1.42, an effective sample of 134 and an interval of 37.6–54.2%, which is the one to read. That interval is a Wilson interval evaluated at the effective sample, 133.57 before rounding, and the correction it carries is for register clustering only: it does not reach the other non-independence the close of this section names, because the nine molecules recorded at more than one junction fall in different clusters rather than in one. Collapsing the 190 records onto the 176 distinct molecules instead gives 82 of 176, 46.6%, nominal Wilson 39.4–54.0%. Both intervals sit on a complete enumeration rather than on a sample (the 190 are every 5-6-5 register whose seam falls strictly inside the catalytic gap, at all 38 junctions) so what they describe is a differently tiled panel of the same junctions, not sampling error about a larger population of designs. A count of that kind means little without a null, so the same screen was run over arbitrary 16-mers. Only the query changes: the same six mature parents, the same forward orientation, the same ten-base-pair threshold. Scrambling each design's own target window, which preserves its base composition and is the scrambled-gapmer control §4.4 asks a laboratory to make, gives 6.2% (5.9–6.4%); a dinucleotide-preserving shuffle gives 10.0% (9.7–10.3%); 16-mers drawn from uniform bases give 6.9% (6.7–7.2%), and from the panel's pooled base composition 7.2% (6.9–7.4%). Every parenthesis on a null rate in this section is a Monte-Carlo interval on that ensemble's mean, narrowing with the number of draws rather than with evidence, and it is not the spread against which the observed rate should be read; the exon-terminus paragraph below sets the two against each other. The uniform-base figure does not stand alone; a calculation on the same uniform bases agrees with it: the gap must pair, at  $4^{-6}$ , and the run must then extend four further nucleotides across the two wings, at  $1/64$ , which over the 19,921 parent windows searched predicts 7.3%. That last step is a Poisson conversion and treats those windows as independent trials, which they are not: they tile 20,011 nucleotides, so consecutive windows share 15 of their 16 bases and a site that lands in one lands in its neighbours too. Clumping of that kind makes a Poisson figure sit high, and the sampled uniform arm 0.4 points below it at 6.9% is what bounds the assumption here. The observed rate is 6.6 times the sampled uniform-base arm, and the NR4A3 sub-analysis the modality actually turns on appears to separate further still: 32.1% of designs carry their qualifying duplex against wild-type NR4A3, against 1.8% of scrambles — the count attributed to NR4A3, not the NR4A3-specific count of the ladder's last column. That sub-analysis's own nominal interval is 25.9–39.0%, and the clustering correction applied above does not widen it: its design effect is 0.82 rather than 1.42. That is not a reading about the registers. Permuting the liability labels across the 38 junctions gives a design effect of 1.00 on average, a standard deviation of 0.21 and a 95% range of 0.62 to 1.45, so 0.82 is not distinguishable from no clustering at all, while the aggregate 1.42 sits at the top of that same range rather than outside it, so the clustering the aggregate carries is marginal rather than established. A design effect below one would narrow the interval, and the nominal one is reported instead. Scrambles are the weakest null run here; against the random-offset chimera null of the next paragraph, the same sub-analysis gives 32.1% against 9.3%.

Neither comparison survives the exon-terminus null two paragraphs below, which returns 28.8% (28.3–29.2%) on it: the liability attributed to NR4A3 is no better resolved as specific to the reported breakpoints than the aggregate is.

A second null asks whether that excess is a fact about reported breakpoints or merely about the design rule. Joining a random window of a real donor parent to a random window of real NR4A3, split at a junction offset the panel's registers use, reproduces everything the rule specifies, namely donor sequence 5', NR4A3 sequence 3' and the junction inside the catalytic gap, while destroying only the fact that the two pieces meet where a tumour joins them. Those chimeras reach 23.8% (23.3–24.2%).

That arm, however, destroys more than the breakpoint, and the excess over it should not be read as a share attributable to where the disease joins the two genes. A real design's halves do not sit at arbitrary interior positions: the donor half ends at an exon 3' terminus and the NR4A3 half begins at an exon 5' terminus, because that is what a splice junction is. Drawing both halves at real exon termini of the same two transcripts — a syntactically valid exon-exon junction, almost always at an exon pair no patient is reported to carry — gives 40.6% (40.1–41.1%), against 45.8% observed, and 40.5% (40.0–41.0%) when the NR4A3 exon-3 acceptor that every junction tiled here uses is excluded from the draw altogether. Almost always, not never: the arm draws both termini freely, so 7.7% of its 38,000 draws reproduce a panel design's target window exactly and 1.4% land on one of the five junctions a patient is reported to carry. Excluding those draws moves the arm to 40.0% and widens the excess from 5.2 to 5.8 points — still inside the observed rate's interval, and in the direction that runs against this section's conclusion rather than for it. The parenthetical ranges on every null rate here are Monte-Carlo intervals on the ensemble mean over 38,000 draws, not confidence intervals comparable to the observed Wilson interval: they narrow with more drawing rather than with more evidence, and the spread of what 190 designs drawn from this arm would give is about 3.6 percentage points rather than the one printed — and that 3.6 is itself nominal, since at the design effect of 1.42 above the comparable spread on the observed side is 4.3 points, against which the 5.2-point excess at ten is about 1.2 standard deviations. The observed rate's own interval contains both. The liability is therefore a property of joining two exon termini of these two transcripts, and this panel does not resolve a residual specific to the reported breakpoints.

Two tighter readings agree with that, and one of them needs no null model at all. Holding the gene pair, the NR4A3 exon-3 acceptor and the tiling register fixed, and varying only which donor exon 3' terminus the donor half ends at (the reported one excluded, so every draw is this paper's own design rule at an exon pair the disease is not reported to use) returns 44.5% against the 45.8% observed. That is an excess of 1.3 percentage points — 0.4 of the standard deviation of the count the null predicts, before any clustering correction, and a junction-cluster bootstrap interval of  $-7.3$  to  $+10.0$  points. Within the panel itself the same answer comes off the counts directly: the five junctions a patient is reported to carry are liable at 44.0%, 11 of 25, against 46.1%, 76 of 165, at the 33 in-frame junctions nobody is reported to carry — a difference of  $-2.1$  points at a junction-label permutation  $p$  of 1.00, and  $p \geq 0.42$  at every cut of the ladder above. Both readings cluster on the junction rather than on the record. What



the screen resolves is the gene pair and the splice geometry; it resolves nothing about where the disease joins them.

The two termini do not contribute equally, and the asymmetry is the informative part. Requiring only the donor half to end at a real exon terminus leaves the rate at 22.5% (22.1–23.0%), close to the arbitrary-offset draw and far from the 40.6% both termini give; the whole of the difference appears when the *NR4A3* half is required to begin at a real acceptor. What the screen is detecting tracks the acceptor boundary of wild-type *NR4A3* — the transcript the modality exists to spare — rather than the donor consensus. Two further arms locate it no more finely: holding the six gap bases and scrambling the wings gives 9.1%, and the mirror gives 8.8%, because a run reaching ten base pairs needs the real gap and the real flanks together. None of these rates is a significance test and none is offered as one, and the two tests that are reported above relabel or resample the 38 junctions rather than the 190 records. The 190 records are 176 distinct molecules tiled at overlapping registers across 38 junctions, so they are not independent draws, and a test treating them as 190 would be wrong about its own denominator.

Nor is any of it a transcriptome-wide rate. Every figure in this section, observed and null alike, is a rate over 19,921 windows of the 20,011 nucleotides these six mature parents hold — 0.0028% of the 718,571,139-nucleotide, 186,185-transcript span the exhaustive scan measured (§6, screen 2) — and the criterion does not survive the enlargement: over that span the same uniform-base arithmetic predicts about  $2.7 \times 10^3$  qualifying sites for any 16-mer whatever, real design, scramble and chimera alike, so every sequence would meet it. What screen 4 counts is therefore a self-liability and not an off-target rate (the 85 of 87 above) and for that quantity a six-transcript search is already exhaustive rather than merely narrow.

## 2.6 · The *NR4A3* exon-2 acceptors, and the un-rearranged allele

Every junction above joins a donor exon to *NR4A3* exon 3, because the graded exon-pair atlas that emits them (§2.1) drops exon-2 acceptors as non-coding: *NR4A3* exon 2 lies upstream of the start codon, so a junction at that acceptor cannot be built from the 38-junction panel at all. A frame-based exclusion is correct for a fusion protein and does not transfer to an RNase-H mechanism, which cleaves a transcript whether or not its reading frame survives, and the acceptors it excludes are not hypothetical: the *EWSR1* type 2 transcript joins *EWSR1* exon 7 to *NR4A3* exon 2,<sup>24</sup> sequenced as one of the five cases of a whole-transcriptome cohort<sup>25</sup> and again in an independent patient.<sup>26</sup> At the *TAF15* seam, functional work has engineered both the intron-2 cryptic-exon acceptor and the exon-3 one, calling the *TAF15* exon 6 to *NR4A3* intron 2 form the less common of those two.<sup>27</sup>

Those acceptors are now designed and screened to the panel's depth, at four seams with a published exon-resolved breakpoint, each tiled by the same five registers and graded by the same five screens, and every count below is at the tenfold deeper alignment ceiling — with two exceptions at one seam, both limits of the instrument rather than properties of the design. Neither the pre-mRNA screen nor the mature-parent screen carries *PGR* in its parent set: both read the same six genes, and the committed cache holds no *PGR* unspliced sequence, so that seam's zero in either compartment is an absent reading of its own donor rather than a clean one. A scan outside the screens closes the mature half, returning no window of wild-type

*PGR* mature transcript within two mismatches of any of that seam's five designs (*pgr-parent-engagement-noncoding-acceptor.json*); the unspliced half stays unread, and closing it needs a networked re-fetch of the cache rather than a re-analysis. Their best available designs are 5'-CAGTGGGCTTCTGCTG-3' at *EWSR1* exon 7, the type 2 transcript, at gap-level margin 2 and 51 gap-paired near-matches over 7 loci; 5'-AGTGGGCTCTCCACGG-3' at *EWSR1* exon 13, at margin 3 and 25 over 6; 5'-AGTGGGCTCTTGTGTG-3' at *TAF15* exon 6, at margin 3 and 128 over 6; and 5'-AGTGGGCTCTCCATT-3' at *PGR* exon 2, a sixth fusion partner outside the five modelled here, at a seam reported in a single patient,<sup>28</sup> at margin 3 and 51 over 14 (Table 5). None of the four is clean. Each of these seams is tiled by five junction-spanning registers and no register at any of them is clean either, so the least loaded of the four is the least dirty of its own seam's five rather than a different kind of result. They are reported beside the panel and never pooled into it, because the grade that excludes their junctions from the 38 is unchanged. *PGR* carries a further caveat: its design is screened against the non-canonical-acceptor table rather than derived through the panel's own transcript models. Its transcript accession is given with the other six in §6, so the design is reproducible even though the panel does not model it.

**Where a design's acceptor half is *NR4A3* sequence that is not exonic in the mature transcript (the 5' untranslated exon 2, or the cryptic exon within intron 2) the patient's own un-rearranged *NR4A3* allele carries that same sequence too, sitting behind an intron instead of behind a partner's donor exon. Some designs therefore pair their whole catalytic gap against that un-rearranged allele, and the mature-parent screen, which reads spliced transcript only, passes the two it can read and never runs on the third, whose cryptic-exon acceptor it cannot address.** This is the result most consequential for anyone ordering these oligonucleotides. In a fusion transcript that acceptor sequence follows the partner's donor exon; in the un-rearranged allele it follows *NR4A3* intronic sequence, so the question is whether the design's donor half matches that intron closely enough for the whole catalytic gap to pair. For three designs it does: at *EWSR1* exon 13 joined to *NR4A3* exon 2, 5'-CAGTGGGCTCTCCACG-3' and 5'-GCAGTGGGCTCTCCAC-3', each pairing across the wild-type intron-1/exon-2 boundary at two mismatches with neither inside the gap; and at *TAF15* exon 6 joined to the intron-2 cryptic exon, 5'-TGATGAGGGCCTTGTG-3', likewise gap-paired at two mismatches. All three must not be ordered or used, for that reason, and are excluded from every best-design field above. Both seams keep a reagent, 5'-AGTGGGCTCTCCACGG-3' and 5'-ATGAGGGCCTTGTGTG-3', the second's catalytic gap carrying three *TAF15*-derived bases the *NR4A3* locus does not have and returning no wild-type site at all on the two screens that reach this seam. Two of the three condemned designs, and the kept one, sit at the *EWSR1* exon-13 seam as consecutive registers differing only by a single-base slide: the kept reagent, 5'-AGTGGGCTCTCCACGG-3', is shifted one base 3' of 5'-CAGTGGGCTCTCCACG-3' ■ and two of 5'-GCAGTGGGCTCTCCAC-3' ■. The third condemned design, 5'-TGATGAGGGCCTTGTG-3' ■, is at the *TAF15* exon-6 seam and is not a register of that *EWSR1* set — **but it stands in exactly the same relation to the reagent kept beside it:** it is shifted two bases 5' of 5'-ATGAGGGCCTTGTGTG-3', and the fourteen bases the slide leaves overlapping are identical. Neither condemned

*EWSR1* exon-13 register may be ordered in place of the kept *EWSR1* reagent, the condemned *TAF15* exon-6 design may not be ordered in place of the kept *TAF15* one, and in both cases a slide of one or two bases is all that separates a reagent from a design this paper condemns. Every seam in this section joins its donor to *NR4A3* exon 2 or to the intron-2 cryptic exon, and none of them to *NR4A3* exon 3: the *EWSR1* exon-13 to *NR4A3* exon-3 reagent of §4.1 is a different molecule at a different seam. Of the two kept reagents, 5'-ATGAGGGCCTTGTTGTG-3' is the one not certifiable under the criterion §4.5 states, and §3 carries it only under that qualification rather than as a cleared reagent: its acceptor is a cryptic exon, which three of the five screens cannot address at all, so what it holds is a quiet reading on two instruments rather than a clearance. What is lost is not a seam but the assumption that designs tiled across one seam are interchangeable.

Two things about that finding matter more than the three sequences. The first is how they were reached. The mature-parent exclusion had passed two of them, and every other design at their seam; at the third's seam it never ran at all, on that design or on any other there. That exclusion is a screen over spliced cDNA and structurally unable to see intronic sequence; a parent screen that returns nothing at such a seam is the silence of an instrument that cannot look at the compartment in question. The same three were returned independently by an exhaustive scan of the *NR4A3* unspliced sequence, by the pre-mRNA screen and by the genome scan, on a fixed known-positive control that fired on exactly the one design it was required to fire on. The second is what decides it, because the obvious answer is wrong. It is not the gap-level margin

this paper ranks by: the two condemned *EWSR1* designs carry 2 and 1 donor bases inside the catalytic gap, while a design at the same seam with a margin of 1 and five donor bases in its gap does not cleave the un-rearranged allele at all. What decides is how much donor sequence the gap holds, the rest being acceptor sequence the wild-type allele already carries verbatim — a property of the donor rather than of the acceptor. The sequence bears it out: *EWSR1* exon 13 ends CACTCCGTGGAG against the last twelve nucleotides of *NR4A3* intron 1, CCTTGCCTGTAG, matching at 7 of 12 positions with a shared terminal AG, whereas *TAF15* exon 6 ends ACCACACACAAG and matches at 4, mismatching in every register — which is why *TAF15* exon 6 joined to *NR4A3* exon 2 returns no such design while *EWSR1* exon 13 joined to *NR4A3* exon 2 returns two. Both are named here by donor and acceptor together, because *TAF15* exon 6 is also the donor of the intron-2 cryptic-exon seam above, which is a different seam and does return one. A design must therefore be checked against the acceptor gene's unspliced sequence whenever its acceptor half is not exonic in the mature transcript, and no spliced-transcript screen substitutes for it.

The *TAF15* intron-2 acceptor and a second cryptic-exon seam, *EWSR1* exon 10 to the same acceptor, are designed but not comparably screened: three of the five instruments cannot address them at all, resolving a junction by exon index where a cryptic exon 5' of the *NR4A3* start codon has none, so their counts are absent rather than low and must not be read beside the panel's. The pre-mRNA and genome arms do reach them, and at the *TAF15* seam decide between its five designs rather than ranking them.



**Table 5. Every seam the coverage ladder qualifies, with the ladder's bounds and §4's two contrast arms beside them, what each costs on each screen and what each buys in coverage.** The rows are in the order §4 decides them: the two lead reagents, the rungs of the coverage ladder above them, the bounds above those, the remaining junction with a published exon-resolved breakpoint and a reagent through all five deep screens, the three NR4A3 exon-2 acceptor seams the ladder carries no entry for, reported beside the panel, and the two contrast arms. The gap-level margin is the count of bases inside the six-nucleotide catalytic gap that no wild-type parent carries at that position, taken on the shorter side of the junction; it runs from 1 to 3, the geometry's ceiling of half the gap rounded down, and it is a count of bases and not a score. Membership is the coverage ladder's and not this table's: every junction its best-supported buildable panel qualifies — a published exon-resolved breakpoint, and all five specificity screens run over that junction's designs, each condition read from the table that owns it — has a row here whether or not §4 names its reagent, and the generator refuses to build if a qualifying junction has no row.  $\Delta$  Every screen runs over every qualifying junction, but two of them are gene-scoped: of the nine junctions that qualify, eight had all five read that junction's OWN parent genes, and PGR e2::NR4A3 e2 had three of five — the parent-scoped screens' gene set does not carry that donor, so those readings are unmeasured rather than clean. A row can therefore qualify and still buy no coverage, which is a statement about the denominator and not about the reagent. The screen condition counts screens that RAN over a junction's designs and not compartments measured for every gene a row names: the pre-mRNA screen behind the exon-2 acceptor rows carried unspliced sequence for six parent genes — *EWSR1*, *FUS*, *NR4A3*, *TAF15*, *TCF12* and *TFG* — so a seam whose donor is outside that set has an absent reading of its own donor's introns in that compartment rather than a clean one. *PGR* is that case here, for the reason §2.6 gives, and the row is in this table on the same footing as the others. The reagent column is editorial, but which of the two membership classes a row falls in is the ladder's decision as well, and it is legible in the last two columns: a junction the ladder NAMES IN ONE OF ITS ENTRIES carries that entry's cumulative figure and increment, and a junction that qualifies but appears in no entry carries “adds nothing” and the reason instead. That is why one NR4A3 exon-2 acceptor seam — EWSR1 e7::NR4A3 e2 — is a rung here while the other three are reported beside the panel: the ladder prices the type 2 transcript's breakpoint on the same single series, and such a rung can still print an increment of zero, because that series never named the transcript — an unnamed count rather than a measured absence. Being a rung of the ladder is not being pooled into the panel, which is the separate statement below and is unchanged. Cumulative coverage is the coverage of the reagent set through that row, so the two leads are one rung and carry one figure between them; it is discounted by the breakpoint distribution of a single series and is not a partner figure, and its parenthetical range is composed from each breakpoint fraction's own Wilson bound rather than from the point estimate. That range is a composed endpoint and carries no nominal coverage level: it is not a confidence interval, and §4.1 gives the reason the four quantities cannot be composed to one. Every rung and every bound prints the increment it adds over the row above it, so no figure reads as bought by the row it sits on. Each coverage figure and each increment is rounded to one decimal independently, from the unrounded fraction rather than from each other, so a row's figure plus the increment printed on the row below it need not reproduce that row's figure in the last place. A bound row is what coverage would be if every remaining breakpoint of that partner were covered, which nothing measures. A bound that names no reagent still has a row, and the *EWSR1* one is the larger of the two steps between the last buildable rung and the table's top figure: the three breakpoints it prices are ones the retrieved record does not resolve to an exon, so no sequence, geometry or screen result exists for them and every such cell is empty. If those breakpoints are private rather than recurrent, no stocked panel reaches them at any size. A row that adds nothing prints why, because the two reasons differ: the partner is absent from the 58-case cohort behind the denominator, or the partner is present and that exon pair carries no count in the measured within-partner distribution. The exon-2 acceptor rows are from the non-canonical-acceptor table and are never pooled into the panel, since the grade that excludes their junctions from the 38 is unchanged. A contrast arm carries no coverage figure and must not borrow its junction's, which is already counted a row above. The three controls §4 requires have no row and can have none: the fusion-negative isogenic comparator is a cell line rather than a reagent; the positive control is specified as a class rather than a sequence, a gapmer against an abundant housekeeping transcript; and the scrambled control is a draw from a stated shuffling procedure rather than one oligonucleotide. None of the three therefore has a sequence, a geometry or a screen result for these columns. Gap-paired near-matches are at the tenfold deeper alignment ceiling where every hit list is complete, and the parent duplex is the longest contiguous run containing the whole catalytic gap that any wild-type parent forms, whatever its length. None of these numbers is a measurement of off-target activity, and no row is a claim of efficacy. **The duplex column is not filtered at the criterion.** A reading of ten base pairs or more is what Tables 3 and 4 mark  $\blacksquare$  and do not order, and no row here reaches it — the highest cell is 9 bp — so this table carries no  $\blacksquare$ ; that is a statement about these 12 rows and not a clearance. That ten-base-pair cut is a criterion this work ADOPTS rather than measures, and the reading moves with it: at seven base pairs 175 of the 190 panel designs pair a parent through the whole gap against 87 at ten, 181 do so at some length, and only nine of the 38 seams have a design that clears against 35 at ten (§2.9). A cell below the cut is a reading at one cut and nothing wider. Every cell that carries a reading carries the verdict the SAME number takes at the loose end of that range — that number is the row's `mature_parent_duplex_through_gap_bp` in `fusion-junction-aso-sequences.csv`, so the comparison is reproducible from the canonical file and not only here: ten of the eleven rows that print a duplex fall inside the do-not-order class there, and two of them are the lead reagents §4 names — which is the reading Box 1 states of those two and which no marker on this table used to carry. The sequences in this table are antisense gapmers at more than one geometry, and each row's own geometry is the one stated beside it: the architecture gives the locked-nucleic-acid (LNA) wing, DNA gap and LNA wing in nucleotides, so 5-6-5 is a 16-mer, 5-8-5 an 18-mer and 5-10-5 a 20-mer, each on a phosphorothioate backbone that §6 specifies; the bases alone, ordered as unmodified DNA, are a different molecule. Research use only: not for administration to any person or animal (Declarations). The canonical machine-readable copy of every sequence is `fusion-junction-aso-sequences.csv`, which is what to order from rather than this PDF.

| reagent                                                                                                                                                     | junction                                         | sequence                           | geometry | gap-level margin | sense-strand gap-paired near-matches → loci at the deeper ceiling | longest mature-parent duplex through the gap, and the reading at seven base pairs | cumulative coverage of published junctions — not a coverage measurement | basis                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------|----------|------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------|
| <b>MEASURED RUNGS — the cumulative-coverage cell is the coverage of the reagent set through this row, discounted by one series' breakpoint distribution</b> |                                                  |                                    |          |                  |                                                                   |                                                                                   |                                                                         |                                                        |
| lead reagent                                                                                                                                                | EWSR1 e12::NR4A3 e3                              | 5'-GGGCATATCATCAAAC-3'             | 5-6-5    | 3                | 123 → 6                                                           | 8 bp ( <i>TFG</i> ); liable at seven                                              | 68.4% (39.9–82.8)                                                       | single series, cumulative                              |
| lead reagent                                                                                                                                                | TAF15 e6::NR4A3 e3                               | 5'-GGGCATATCTTGTGTG-3'             | 5-6-5    | 3                | 8 → 5                                                             | 9 bp ( <i>TFG</i> ); liable at seven                                              | 68.4% (39.9–82.8)                                                       | single series, cumulative                              |
| coverage rung                                                                                                                                               | EWSR1 e13::NR4A3 e3                              | 5'-GGGCATATCTCCACGG-3'             | 5-6-5    | 3                | 24 → 2                                                            | 8 bp ( <i>TCF12</i> ); liable at seven                                            | 79.0% (50.3–89.2) (+10.6)                                               | single series, cumulative                              |
| coverage rung                                                                                                                                               | EWSR1 e7::NR4A3 e2                               | 5'-CAGTGGGCTTCTGCTG-3'             | 5-6-5    | 2                | 51 → 7                                                            | 8 bp ( <i>TAF15</i> ); liable at seven                                            | 79.0% (50.3–89.2) (+0.0)                                                | single series, cumulative                              |
| <b>ARITHMETIC BOUNDS — the cell is what coverage WOULD be if every remaining breakpoint of that partner were covered, which nothing here measures</b>       |                                                  |                                    |          |                  |                                                                   |                                                                                   |                                                                         |                                                        |
| coverage bound (no reagent named)                                                                                                                           | BOUND — every remaining EWSR1 breakpoint covered | — (3 further reagents, none named) | —        | —                | —                                                                 | —                                                                                 | 94.8% (+15.9)                                                           | arithmetic bound                                       |
| coverage bound                                                                                                                                              | TCF12 e5::NR4A3 e3                               | 5'-GGGCATATCCATCAGA-3'             | 5-6-5    | 3                | 17 → 1                                                            | 7 bp ( <i>EWSR1</i> ); liable at seven                                            | 98.3% (+3.4)                                                            | arithmetic bound                                       |
| <b>QUALIFYING SEAMS THE LADDER PRICES AT NOTHING — the cell reads “adds nothing” and the basis beside it gives which of the two reasons applies</b>         |                                                  |                                    |          |                  |                                                                   |                                                                                   |                                                                         |                                                        |
| published seam in the panel                                                                                                                                 | TFG e7::NR4A3 e3                                 | 5'-GGCATATCTGAATACT-3'             | 5-6-5    | 2                | 24 → 6                                                            | 9 bp ( <i>TAF15</i> ); liable at seven                                            | adds nothing                                                            | partner absent from the cohort behind the denominator  |
| beside the panel                                                                                                                                            | EWSR1 e13::NR4A3 e2                              | 5'-AGTGGGCTCTCCACGG-3'             | 5-6-5    | 3                | 25 → 6                                                            | 8 bp ( <i>EWSR1</i> ); liable at seven                                            | adds nothing                                                            | partner in the cohort, this exon pair uncouncted in it |
| beside the panel                                                                                                                                            | TAF15 e6::NR4A3 e2                               | 5'-AGTGGGCTCTTGTGTG-3'             | 5-6-5    | 3                | 128 → 6                                                           | 9 bp ( <i>NR4A3</i> ); liable at seven                                            | adds nothing                                                            | partner in the cohort, this exon pair uncouncted in it |
| beside the panel                                                                                                                                            | PGR e2::NR4A3 e2                                 | 5'-AGTGGGCTCTTCCATT-3'             | 5-6-5    | 3                | 51 → 14                                                           | 9 bp ( <i>NR4A3</i> ); liable at seven                                            | adds nothing                                                            | partner absent from the cohort behind the denominator  |

**Table 5. Every seam the coverage ladder qualifies, with the ladder's bounds and §4's two contrast arms beside them, what each costs on each screen and what each buys in coverage**

| reagent                                                                                                                                            | junction               | sequence                 | geometry | gap-level margin | sense-strand gap-paired near-matches → loci at the deeper ceiling | longest mature-parent duplex through the gap, and the reading at seven base pairs | cumulative coverage of published junctions — not a coverage measurement | basis              |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------|----------|------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------|
| <b>CONTRAST ARMS — not coverage rows; the cell is empty because their junction's coverage is counted a row above and must not be counted twice</b> |                        |                          |          |                  |                                                                   |                                                                                   |                                                                         |                    |
| gap-length control                                                                                                                                 | EWSR1<br>e12::NR4A3 e3 | 5'-AGGGCATATCATCAAACC-3' | 5-8-5    | 4                | 3 → 1                                                             | 0 bp; clear at seven                                                              | —                                                                       | not a coverage row |
| margin contrast arm                                                                                                                                | EWSR1<br>e12::NR4A3 e3 | 5'-GCATATCATCAAACCA-3'   | 5-6-5    | 1                | 34 → 6                                                            | 8 bp ( <i>FUS</i> ); liable at seven                                              | —                                                                       | not a coverage row |

## 2.7 · The surviving candidates, and a genome-wide check

Composing the mature-parent screen of §2.5 with the deeper re-screen of §2.4 leaves three of the four designs that re-screen returns clean as candidates in the whole panel, and they are not equally secure. The un-rearranged-allele exclusions of §2.6 remove none of them, because the seams they condemn cannot be built from the 38-junction panel at all. 5'-AGGGCATATCGGAGTC-3' at *FUS* exon 8 and 5'-GGGCATATCCGACATG-3' at *TAF15* exon 1 carry no sense-strand near-match at ten times the default search depth, no single-mismatch match on the exhaustive transcript scan, no pre-mRNA site and no mature-parent duplex. Neither depends on the ten-base-pair threshold, because no window of any parent pairs their catalytic gap at any length: their longest run is zero rather than merely short. 5'-GGGCATATCAAGCGCT-3' at *TCF12* exon 7 passes the same screens, but not in the same way. Its longest parent run is eight base pairs against wild-type *NR4A3*, which is below the threshold rather than absent, so it is a candidate at ten and at no cut of eight or below — including the seven-base-pair end of the range §5 bounds the criterion by, where it joins the liability class. The fourth design with a clean deep screen, 5'-GGCATATCAAGCGCTG-3' at the same junction, is excluded by an eleven-base-pair duplex with wild-type *TCF12* — its own donor parent, not the acceptor. The *TCF12* exon-7 member therefore differs between the near-match set of §2.4 and the candidate set here, and the two sequences are consecutive registers of one seam differing by a single-base slide: 5'-GGCATATCAAGCGCTG-3' is the design §2.4 lists and this screen excludes, and 5'-GGGCATATCAAGCGCT-3' is the candidate, first screened only at the deeper ceiling. Neither may be ordered for the other. That is the honest size of the candidate set by screen, and it is not a set to make: no junction among them has a published patient breakpoint, which is why §4.4 uses all three as mechanism controls rather than candidates — the exon-resolved *TAF15* breakpoints reported in EMC are exon 6, no *TCF12* breakpoint is reported at exon 7, and for *FUS* none has been published at all. Selecting within each junction rather than across the panel changes what is available, not what is clean: Table 2 applies the same criteria at all 38, where 35 have a design that clears the parent screen and *TAF15* exon 14, *TCF12* exon 3 and *TFG* exon 2 have none. All five junctions of the panel with a published exon-resolved breakpoint have one: four of them at the top gap-level margin, with longest parent runs of eight, eight, nine and seven base pairs, and *TFG* exon 7 at

a margin of two with a longest parent run of nine. Those are readings at ten base pairs, and they are the readings that move most: at the seven-base-pair end of the range §5 bounds this criterion by, no design at any of the five clears the screen, and at eight, two of the five do.

Both parent-liability classes of §2.5 (the pre-mRNA sites and the mature-transcript duplexes) were bounded the same way: exhaustive over six parent transcripts and silent about every other gene. The genome scan, screen 5, removes that bound for the pre-mRNA class alone. It runs at two mismatches, and a contiguous run of eleven or twelve base pairs inside a 16-mer leaves five or four positions unpaired respectively, so the mature-transcript duplexes stay bounded by the six transcripts searched — the same reason §2.5 gives for their being invisible to the alignment screens.

A raw genome-wide count is not a result at this threshold. Chance alone predicts of order  $10^3$  near-matches per 16-mer over a genome for any 16-mer whatever, so the informative readings are stratified. Exact 16/16 matches are the class where chance expectation is of order one: 1.37 expected per design against 236 observed over the 176 designs, or 1.34 each, which is at chance on the mean and on no individual design: 79 of the 176 return none where that expectation predicts about 45, and twelve return five or more where it predicts about two, to a maximum of 20. The corpus sits at chance; the designs are dispersed around it, and a reagent should be read from its own row rather than from the mean. Load relative to that expectation separates designs where a total cannot — the median design sits at 0.97 of its expectation and 14 of 176 exceed twice it. And the repeat split, which a soft-masked reference supplies at no extra cost, shows 52.5% of hits fully repeat-masked. The baseline that figure needs is not the whole assembly's 51.4%, because a hit can only arise in a window free of ambiguous bases and 151,138,112 windows were dropped for carrying one; a masked base is never an N, so every masked base survives into the scanned sequence and the masked share of what was scanned lies between 51.4% and 54.1%. The upper end of that band is a bound and not a value: a maximal run of  $k$  ambiguous bases costs  $k + 15$  windows, so the dropped-window count over-states the bases removed, and dividing by it over-states the share. The two figures are also not in the same units — 52.5% is a share of hit *windows* fully masked, and the band a share of *bases* — and a window is fully masked only when all sixteen of its bases are, which is rarer than a base being masked. The same-units baseline, the share of scanned 16-mer windows

that are fully masked, is not computed here. So the comparison supports no direction at all: 52.5% falls inside the band, and the two quantities it would be read against are counted in different units. What can be said is that the hits are not repeat-enriched.

The decisive reading is a lookup rather than a count: does any design have a gap-paired, strand-agreeing site in *NR4A3*, in a parent gene, or in an *NR4A* paralogue anywhere in the genome? Twenty of 176 do. No candidate above is among them, and the two secure at any parent-duplex threshold — 5'-AGGGCATATCGGAGTC-3' at *FUS* exon 8 and 5'-GGGCATATCCGACATG-3' at *TAF15* exon 1, neither of them a reagent of §4.1 — carry a low load: 0.33 and 0.24 of expectation at  $\leq 2$  mismatches against a corpus median of 0.97, and 0.06 and 0.04 for gap-paired sites, each pair given in that order, the *FUS* design first. These ratios are taken either-orientation against an either-orientation null, unlike the sense-filtered transcript columns beside them: strand agreement is applied to the named-target lookup above but not to the load ratio, and for the *EWSR1* exon-12 reagent 58% of the sites the *gap-paired* ratio counts lie on a strand an antisense oligonucleotide cannot pair — 156 of its 371 gap-paired sites at two mismatches or fewer are hybridisable. The stratum has to be named, because the other axis the loads above are given on returns a different figure: over the  $\leq 2$ -mismatch set, 437 of 1,062 sites are hybridisable and the non-hybridisable share is 59%. The two candidates' ranks on those axes, in that same order, 26th and 13th of 176 and 5th and 1st, are reported because the ordering is what the scan supports; the null assumes independent uniform bases and resolves more-than-chance from at-chance and nothing finer (§5), so the ratios should not be read as calibrated distances below chance. The third candidate is not quiet on this axis and its reading is given for the same reason: 5'-GGGCATATCAAGCGCT-3' sits at 1.46 of expectation at  $\leq 2$  mismatches, 144th of 176 rather than near the clean end, and at 7.0 times expectation at  $\leq 1$  mismatch. That is the strongest statement this work can make about them, and it is a statement about predicted hybridisation and not about cleavage.

## 2.8 · Expression of the off-target loci

No screen above establishes that a design's off-target gene is transcribed in the organs a systemic dose reaches, and that discount applies to every count in this paper. Read against reference expression data (Table 6), off-target count and off-target exposure run against each other at three of the four junctions covered: the

design with the heavier gap-paired load has the less-expressed loci. Exposure is graded at two cuts, 1 and 10 transcripts per million (TPM), against the three tissues a systemically dosed phosphorothioate gapmer is taken to reach — liver, kidney cortex and kidney medulla (§6). The fourth, *TCF12* exon 5, runs with its load rather than against it: its reagent returns the second lightest load of the four, 17 hits resolving to one locus, and that locus, *PIK3CG*, sits below the lower cut in all three exposure tissues. The single locus is the reagent's, not the junction's: Table 6 lists twelve at that seam, because its rows are a union over every tiling register read at a junction and can outnumber the per-reagent counts below, which are each best design's own. Its record column counts gap-paired hits summed over every design read at that seam, one per hit, not the number of transcript variants the locus is annotated with — a distinction the two extremes make plain, since *NRP1*'s five records are one accession returned by five designs while *HNRNPA2B1*'s hundred are fifty accessions returned by two. The *EWSR1* exon 12 reagent's six loci carry 123 of the 649 gap-paired hits returned across the four junctions of Table 6, and none of the four measurable ones reaches the upper cut in liver or either kidney compartment; *ANKS1B* supplies 67 of them and sits below the lower cut in all three, peaking instead in brain at 24.9 transcripts per million (TPM) in the Genotype-Tissue Expression project's (GTEx) cervical spinal cord, a whole-body reading carried in the released artefact rather than in either of Table 6's two compartments, and *CHST5*, the smallest of the six by hit count, peaks in gut on the same reading. The *EWSR1* exon 13 to *NR4A3* exon 3 reagent's two loci are both transcribed at the upper cut in those same compartments. The *TAF15* exon 6 to *NR4A3* exon 3 reagent's five loci separate the other way: *NRP1* reaches 6.6 to 17.8 TPM across all three exposure tissues and is the only one all five of that junction's tiling registers return, on five gap-paired hits to a single accession. It is at or above the upper cut in two of those three, and robustness to register orders the loci differently again — though not independently of the hit count, since a locus returned by more registers accrues more hits by construction, and the two still do not order the loci together: *NRP1* leads on register robustness and sits mid-range on hits, its five records being one accession returned once per register; the tumour-compartment proxy orders them a third way. What these readings can and cannot decide between two reagents is stated where that choice is made (§4.1).

**Table 6. Where the off-target loci at the junctions with a published breakpoint are expressed.** Every gene locus returned by the deeper screens that were READ at four of the five junctions of the 38-junction panel with a published exon-resolved EMC breakpoint, over the tiling registers read at each, against reference expression data. The four *NR4A3* exon-2 acceptor seams of §2.6 also carry a published exon-resolved breakpoint and carry no expression reading here, so this is four of nine such seams in the paper and not four of five.  $\triangle$  That is not every screen at every junction: at the *EWSR1* exon-12 lead seam only the multi-partner reagent's own screen is read, so its rows carry a denominator of one, which note <sup>8</sup> states per row. The two compartments answer different questions and are never combined: a systemically dosed phosphorothioate gapmer is taken to distribute predominantly to liver and kidney — a premise taken from the chemistry, for which no measurement or citation was retrieved here — so liver, kidney - cortex and kidney - medulla are read as the exposure compartment, while the soft-tissue column is the normal tissue of the compartment EMC arises in and stands in for a tumour no reference atlas contains. That column is a MAXIMUM over six named proxies — Muscle - Skeletal, Adipose - Subcutaneous, Nerve - Tibial, Cells - Cultured fibroblasts, Artery - Tibial and Skin - Sun Exposed (Lower leg) — one of which is a cultured cell line rather than a tissue, and the tissue printed beside each value is the one the maximum was taken at. Values are median transcripts per million (TPM) from version 8 of the Genotype-Tissue Expression project (GTEx) across each tissue's donors, in every expression column including the soft-tissue maximum. The two cuts behind the last column are stated for legibility and are not thresholds of concern: below 1 TPM in all three exposure tissues reads as below detection, at or above 10 TPM in any of them as the level at which an off-target hypothesis would have to be tested. Every raw median is released so another cut can be applied without re-running. Gap-paired hit records are the gap-paired near-matches the deeper screens returned at that locus, one per accession per design, added up over every design tiled across the junction; the column totals 649, which is the gap-paired hit count over the four junctions of this table and not over the whole 38-junction panel. It is a count of what the search returned and not of how many accessions RefSeq lists for the gene, so it is not annotation depth and not a property of the locus on its own: a locus that every register returns at one accession is counted once per register, and a locus returned at several accessions is counted once per accession per register — *HNRNPA2B1*'s hundred records over two registers are fifty accessions each, which §2.8 works through. The column beside it gives how many of those records are CURATED RefSeq transcripts rather than the computationally predicted XM\_ /XR\_ gene models Table 3 also counts separately: 263 of the 649 records in this table are curated, and the locus contributing most of them is *G3BP2*, at 48 of its own 56. Both columns count records the search returned rather than properties of the locus, so a locus is neither cleaner for having few curated records nor dirtier for having many; what the split says is how much of a large cell rests on predicted annotation. Tiling registers is how many of the designs tiled across that junction return the locus, which is robustness to where the window is placed; the two columns therefore move together rather than being independent axes, and neither is ranked on, neither is expression and neither is affinity. A locus with no reading carries the reason rather than a zero, because an absent reading is not a reading of absence. Every hit behind this table sits at 14 of 16 identity, the loosest the screen admits, so nothing here distinguishes these loci from one another on affinity. None of these numbers is a measurement of cleavage, and no expression figure is a predicted cleavage event.

<sup>8</sup> The denominator is how many designs at that seam THIS TABLE READS, and not how many junction-spanning registers the seam admits — 5 at every junction of this panel (Table 7). At the *EWSR1* exon-12 lead seam the multi-partner reagent's own screen is the only one read, so those rows carry a denominator of one; at the other seams no design is selected and every screened register is read, because a ranking is not a reagent and the union across registers is what the panel has to cover.

◆ A locus returned by the design Table 2 names as the best available at that seam, which is the molecule Table 5 prices and §4 names. The unmarked rows are returned by other registers tiled across the same junction and not by that reagent. The marker identifies and does not rank: every locus keeps its row, the union is still what this table reports, and a reagent's own loci are neither cleaner nor dirtier for being its own.

**Table 6. Where the off-target loci at the junctions with a published breakpoint are expressed · numbered notes are under the caption, on this table's first page — ♦ A locus returned by the design Table 2 names as the best available at that seam; the marker identifies and does not rank**

| junction            | gene locus            | gap-paired hit records | of those, curated rather than predicted | tiling registers returning it <sup>a</sup> | Liver | Kidney - Cortex | Kidney - Medulla | soft-tissue proxy maximum            | exposure-organ reading           |
|---------------------|-----------------------|------------------------|-----------------------------------------|--------------------------------------------|-------|-----------------|------------------|--------------------------------------|----------------------------------|
| EWSR1 e12::NR4A3 e3 | <i>ANKS1B</i> ♦       | 67                     | 32                                      | 1 of 1                                     | 0.03  | 0.46            | 0.28             | 3.6 (Artery - Tibial)                | below the lower cut in all three |
| EWSR1 e12::NR4A3 e3 | <i>ZNF667</i> ♦       | 37                     | 3                                       | 1 of 1                                     | 0.31  | 1.63            | 2.58             | 6.2 (Nerve - Tibial)                 | detectable, below the upper cut  |
| EWSR1 e12::NR4A3 e3 | <i>GMCL1</i> ♦        | 9                      | 1                                       | 1 of 1                                     | 4.52  | 4.98            | 6.72             | 18.3 (Artery - Tibial)               | detectable, below the upper cut  |
| EWSR1 e12::NR4A3 e3 | <i>LOC105374140</i> ♦ | 5                      | 0                                       | 1 of 1                                     | —     | —               | —                | —                                    | no gene model — not measurable   |
| EWSR1 e12::NR4A3 e3 | <i>LOC105370997</i> ♦ | 4                      | 4                                       | 1 of 1                                     | —     | —               | —                | —                                    | no gene model — not measurable   |
| EWSR1 e12::NR4A3 e3 | <i>CHST5</i> ♦        | 1                      | 1                                       | 1 of 1                                     | 0.07  | 0.35            | 0.78             | 0.8 (Nerve - Tibial)                 | below the lower cut in all three |
| TAF15 e6::NR4A3 e3  | <i>G3BP2</i>          | 56                     | 48                                      | 2 of 5                                     | 11.92 | 17.03           | 23.12            | 77.0 (Cells - Cultured fibroblasts)  | at or above the upper cut        |
| TAF15 e6::NR4A3 e3  | <i>LINC02030</i>      | 22                     | 22                                      | 2 of 5                                     | 0.00  | 0.00            | 0.02             | 0.3 (Skin - Sun Exposed (Lower leg)) | below the lower cut in all three |
| TAF15 e6::NR4A3 e3  | <i>MIR9-2HG</i>       | 18                     | 18                                      | 2 of 5                                     | —     | —               | —                | —                                    | no reading taken                 |
| TAF15 e6::NR4A3 e3  | <i>LAMA4</i>          | 13                     | 3                                       | 1 of 5                                     | 1.28  | 3.13            | 6.06             | 268.6 (Cells - Cultured fibroblasts) | detectable, below the upper cut  |
| TAF15 e6::NR4A3 e3  | <i>ZFPM2</i> ♦        | 12                     | 0                                       | 3 of 5                                     | 0.64  | 0.55            | 0.40             | 9.6 (Artery - Tibial)                | below the lower cut in all three |
| TAF15 e6::NR4A3 e3  | <i>GNAL</i>           | 10                     | 10                                      | 2 of 5                                     | 0.66  | 1.74            | 1.81             | 8.6 (Artery - Tibial)                | detectable, below the upper cut  |
| TAF15 e6::NR4A3 e3  | <i>NRP1</i> ♦         | 5                      | 0                                       | 5 of 5                                     | 6.62  | 16.87           | 17.81            | 104.7 (Cells - Cultured fibroblasts) | at or above the upper cut        |
| TAF15 e6::NR4A3 e3  | <i>SLC17A3</i>        | 4                      | 4                                       | 2 of 5                                     | 9.14  | 33.61           | 8.09             | <0.1 (Adipose - Subcutaneous)        | at or above the upper cut        |
| TAF15 e6::NR4A3 e3  | <i>CA5B</i> ♦         | 3                      | 3                                       | 3 of 5                                     | 0.66  | 1.57            | 1.96             | 11.7 (Artery - Tibial)               | detectable, below the upper cut  |
| TAF15 e6::NR4A3 e3  | <i>CA5BP1-CA5B</i> ♦  | 3                      | 3                                       | 3 of 5                                     | —     | —               | —                | —                                    | no reading taken                 |
| TAF15 e6::NR4A3 e3  | <i>EEFSEC</i>         | 3                      | 2                                       | 1 of 5                                     | 16.15 | 14.37           | 15.57            | 39.1 (Nerve - Tibial)                | at or above the upper cut        |
| TAF15 e6::NR4A3 e3  | <i>ANKRD26P3</i>      | 1                      | 1                                       | 1 of 5                                     | 0.00  | 0.00            | 0.00             | <0.1 (Muscle - Skeletal)             | below the lower cut in all three |
| TAF15 e6::NR4A3 e3  | <i>GBP4</i>           | 1                      | 1                                       | 1 of 5                                     | 3.12  | 5.59            | 10.08            | 18.2 (Adipose - Subcutaneous)        | at or above the upper cut        |



**Table 6. Where the off-target loci at the junctions with a published breakpoint are expressed · numbered notes are under the caption, on this table's first page — ♦ A locus returned by the design Table 2 names as the best available at that seam; the marker identifies and does not rank**

| junction            | gene locus            | gap-paired hit records | of those, curated rather than predicted | tiling registers returning it <sup>a</sup> | Liver  | Kidney - Cortex | Kidney - Medulla | soft-tissue proxy maximum            | exposure-organ reading           |
|---------------------|-----------------------|------------------------|-----------------------------------------|--------------------------------------------|--------|-----------------|------------------|--------------------------------------|----------------------------------|
| TAF15 e6::NR4A3 e3  | <i>LOC105376349</i>   | 1                      | 0                                       | 1 of 5                                     | —      | —               | —                | —                                    | no gene model — not measurable   |
| TAF15 e6::NR4A3 e3  | <i>LOC124907518</i> ♦ | 1                      | 0                                       | 1 of 5                                     | —      | —               | —                | —                                    | no gene model — not measurable   |
| TAF15 e6::NR4A3 e3  | <i>NRXN3-AS1</i>      | 1                      | 0                                       | 1 of 5                                     | —      | —               | —                | —                                    | no reading taken                 |
| TAF15 e6::NR4A3 e3  | <i>ST3GAL1</i>        | 1                      | 0                                       | 1 of 5                                     | 28.58  | 16.36           | 8.06             | 27.2 (Muscle - Skeletal)             | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>FBNP1</i> ♦        | 42                     | 22                                      | 2 of 5                                     | 7.08   | 12.12           | 14.06            | 56.5 (Nerve - Tibial)                | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>EHMT2</i>          | 34                     | 22                                      | 2 of 5                                     | 6.89   | 13.23           | 16.70            | 53.0 (Nerve - Tibial)                | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>ZNF215</i>         | 27                     | 10                                      | 1 of 5                                     | 0.13   | 0.67            | 1.27             | 5.2 (Cells - Cultured fibroblasts)   | detectable, below the upper cut  |
| EWSR1 e13::NR4A3 e3 | <i>ESYT2</i>          | 26                     | 4                                       | 2 of 5                                     | 10.42  | 14.26           | 28.28            | 139.9 (Artery - Tibial)              | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>LRP5L</i>          | 8                      | 0                                       | 2 of 5                                     | 7.83   | 6.32            | 11.16            | 8.0 (Nerve - Tibial)                 | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>CDC42SE1</i> ♦     | 6                      | 2                                       | 2 of 5                                     | 22.74  | 44.73           | 78.97            | 187.6 (Nerve - Tibial)               | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>THEMIS</i>         | 6                      | 0                                       | 1 of 5                                     | 0.13   | 0.08            | 0.18             | 0.3 (Adipose - Subcutaneous)         | below the lower cut in all three |
| EWSR1 e13::NR4A3 e3 | <i>LOC105374651</i>   | 4                      | 0                                       | 1 of 5                                     | —      | —               | —                | —                                    | no gene model — not measurable   |
| EWSR1 e13::NR4A3 e3 | <i>ZC3H4</i>          | 4                      | 0                                       | 1 of 5                                     | 6.41   | 5.89            | 9.71             | 26.2 (Nerve - Tibial)                | detectable, below the upper cut  |
| EWSR1 e13::NR4A3 e3 | <i>ERBIN</i>          | 2                      | 0                                       | 1 of 5                                     | 9.80   | 9.38            | 14.68            | 43.8 (Cells - Cultured fibroblasts)  | at or above the upper cut        |
| EWSR1 e13::NR4A3 e3 | <i>ZNF236</i>         | 2                      | 0                                       | 1 of 5                                     | 1.68   | 2.29            | 2.90             | 8.7 (Nerve - Tibial)                 | detectable, below the upper cut  |
| TCF12 e5::NR4A3 e3  | <i>HNRNPA2B1</i>      | 100                    | 28                                      | 2 of 5                                     | 184.12 | 247.24          | 457.30           | 656.6 (Nerve - Tibial)               | at or above the upper cut        |
| TCF12 e5::NR4A3 e3  | <i>PIK3CG</i> ♦       | 51                     | 9                                       | 3 of 5                                     | 0.14   | 0.17            | 0.37             | 1.2 (Adipose - Subcutaneous)         | below the lower cut in all three |
| TCF12 e5::NR4A3 e3  | <i>MROH2A</i>         | 28                     | 2                                       | 1 of 5                                     | 1.04   | 0.42            | 0.07             | 0.3 (Skin - Sun Exposed (Lower leg)) | detectable, below the upper cut  |
| TCF12 e5::NR4A3 e3  | <i>LINC02940</i>      | 9                      | 0                                       | 1 of 5                                     | —      | —               | —                | —                                    | no reading taken                 |

**Table 6. Where the off-target loci at the junctions with a published breakpoint are expressed · numbered notes are under the caption, on this table's first page — ♦ A locus returned by the design Table 2 names as the best available at that seam; the marker identifies and does not rank**

| junction           | gene locus          | gap-paired hit records | of those, curated rather than predicted | tiling registers returning it <sup>a</sup> | Liver | Kidney - Cortex | Kidney - Medulla | soft-tissue proxy maximum          | exposure-organ reading           |
|--------------------|---------------------|------------------------|-----------------------------------------|--------------------------------------------|-------|-----------------|------------------|------------------------------------|----------------------------------|
| TCF12 e5::NR4A3 e3 | <i>EXOC2</i>        | 6                      | 0                                       | 1 of 5                                     | 5.27  | 6.54            | 8.87             | 22.5 (Nerve - Tibial)              | detectable, below the upper cut  |
| TCF12 e5::NR4A3 e3 | <i>KCNG3</i>        | 4                      | 0                                       | 2 of 5                                     | 0.00  | 0.09            | 0.06             | 1.3 (Nerve - Tibial)               | below the lower cut in all three |
| TCF12 e5::NR4A3 e3 | <i>PLAC8</i>        | 4                      | 4                                       | 2 of 5                                     | 1.43  | 0.32            | 0.30             | 0.7 (Adipose - Subcutaneous)       | detectable, below the upper cut  |
| TCF12 e5::NR4A3 e3 | <i>EFCAB11</i>      | 2                      | 2                                       | 1 of 5                                     | 0.37  | 0.92            | 1.20             | 2.1 (Cells - Cultured fibroblasts) | detectable, below the upper cut  |
| TCF12 e5::NR4A3 e3 | <i>LOC107984281</i> | 2                      | 2                                       | 2 of 5                                     | —     | —               | —                | —                                  | no gene model — not measurable   |
| TCF12 e5::NR4A3 e3 | <i>LOC107985219</i> | 2                      | 0                                       | 2 of 5                                     | —     | —               | —                | —                                  | no gene model — not measurable   |
| TCF12 e5::NR4A3 e3 | <i>LOC107987169</i> | 1                      | 0                                       | 1 of 5                                     | —     | —               | —                | —                                  | no gene model — not measurable   |
| TCF12 e5::NR4A3 e3 | <i>LOC124905457</i> | 1                      | 0                                       | 1 of 5                                     | —     | —               | —                | —                                  | no gene model — not measurable   |

## 2.9 · Gap length trades junction specificity against parent-duplex competence

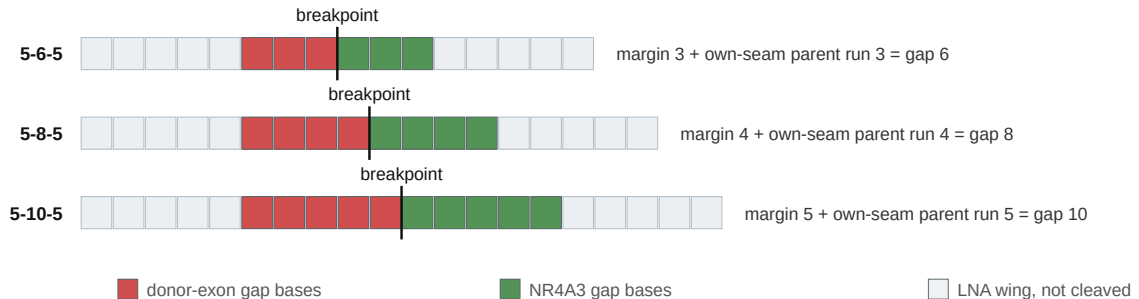
The panel above is one geometry. Tiling the same junctions at 5-8-5 and 5-10-5, wing fixed at five nucleotides, resolves what a longer catalytic gap buys and what it costs (Table 7, Figure 3).

## A longer catalytic gap buys junction-unique bases and concedes gap DNA at the design's own seam, one for one

This ordinate is the design's own seam arithmetic, not the mature-parent search Tables 2, 3, 5 and 7 print under "longest parent duplex through the gap". That search FALLS as the gap lengthens (181 of 190 at 5-6-5; 130 of 266 at 5-8-5; 87 of 342 at 5-10-5) and is FLAT at the ten-base-pair criterion (87 of 190 at 5-6-5; 88 of 266 at 5-8-5; 87 of 342 at 5-10-5). Only the quantity drawn here rises.

### A · the catalytic gap is tiled by the two counts

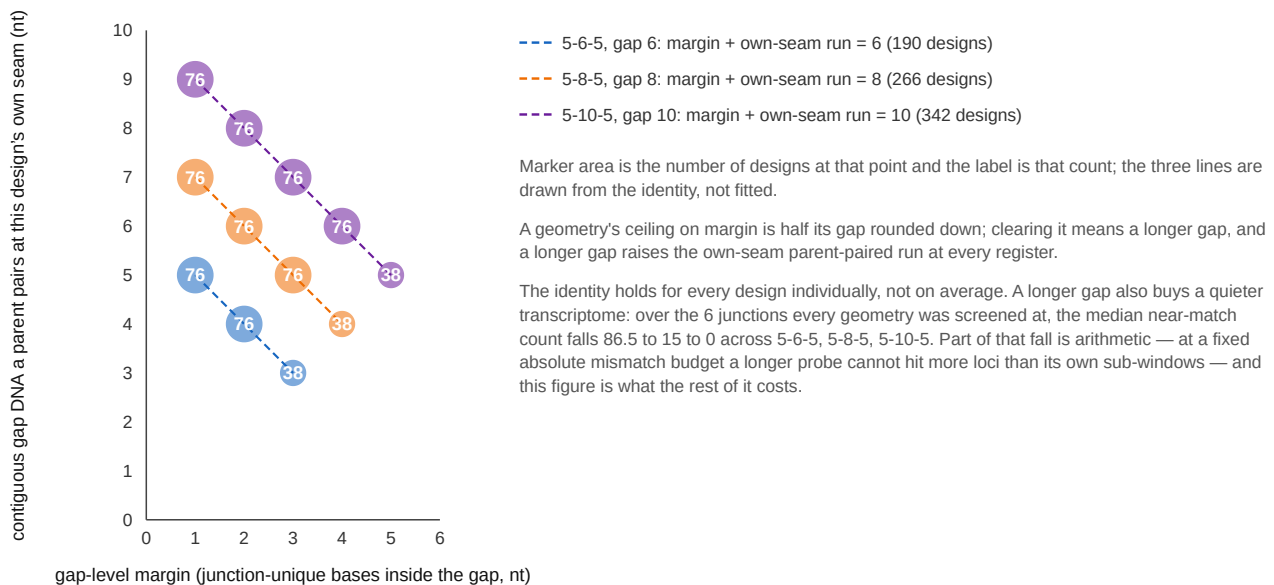
The best-margin design at EWSR1 e12 :: NR4A3 e3, at each geometry. Wings fixed at five nucleotides, so only the gap changes. The register drawn divides its gap evenly at every geometry, which is the highest margin the geometry allows and not the common case: 38 of 190 at 5-6-5; 38 of 266 at 5-8-5; 38 of 342 at 5-10-5. Panel B counts every register there is.



The same three molecules' mature-parent duplex through the whole gap — the column the tables print — reads 8 bp (TFG) at 5-6-5, none at 5-8-5, none at 5-10-5. It is a search over six wild-type parent transcripts, not this arithmetic, and on this molecule the two disagree.

### B · every design in all three geometries, with no scatter to fit

798 designs over 38 junctions. Both axes are drawn at 32 px per nucleotide, so the identity lines really are at 45 degrees.



**Figure 3. The margin a longer catalytic gap wins is the gap DNA a parent pairs at the design's own seam.** (A) The best-margin design at *EWSR1* exon 12 joined to *NR4A3* exon 3, drawn at 5-6-5, 5-8-5 and 5-10-5 with the wings held at five nucleotides. Every base inside the catalytic gap comes from the donor exon or from the acceptor exon, so the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side tile the gap and sum to it. (B) Every fusion-specific design in all three geometries, 798 over 38 junctions, plotted as gap-level margin (abscissa) against the contiguous run of gap DNA one of the design's OWN two parents pairs at its seam (ordinate). That is not the quantity the markers and the do-not-order verdicts are set on. Those come from a SEARCH over all six mature parent transcripts, and the two separate: for the molecule drawn in (A) this figure draws 3 nucleotides where Tables 2, 3, 5 and 7 print an 8 bp duplex against wild-type TFG. The two also move in opposite directions across the geometries — the searched duplex FALLS (181 of 190, then 130 of 266, then 87 of 342 pair a parent at any length) and is flat at the ten-base-pair criterion (87, 88, 87), while the quantity drawn here RISES (76 of 190, 228 of 266, 342 of 342). A longer gap concedes seam-level parent DNA; it does not concede more of the liability this paper condemns. In (A) red marks the gap bases contributed by the donor exon, green those contributed by *NR4A3*, and pale grey the LNA wings, which are not cleaved; the shorter of the two coloured runs is the margin and the longer is what a wild-type parent can pair, so which colour is which depends on the register — the junction drawn here divides its gap evenly in all three geometries, and 76 of the 190 designs at 5-6-5 have the longer run on the donor side; in (B) the three geometries are blue (5-6-5), orange (5-8-5) and purple (5-10-5). Both panels repeat these keys beside the drawing. Marker area is the number of designs at that point and the label is that count; the three lines are drawn from the identity, not fitted, and it holds for each design individually rather than on average. Within one geometry the two move inversely along a line of slope  $-1$ , and both axes are drawn at the same scale (32 px per nucleotide) so that slope is true as plotted rather than only in the units; a geometry's ceiling on margin is half its gap rounded down, and clearing it means a longer gap and a higher parent-paired run at every register (§2.9, Table 7).

**Table 7. Gap length against junction specificity, at one junction and across the design space.** The same junctions tiled and screened at three gapmer geometries, wing held at five nucleotides so that only the catalytic gap changes. Inside the gap the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side are complements: they sum to the gap, which the generating module asserts for every design rather than assuming. Within one geometry the gap is fixed, so each nucleotide of gap-level margin is one FEWER nucleotide of contiguous wild-type-parent duplex; what raises the parent-paired run at every register is a longer gap, which is the only way past a geometry's ceiling of half its gap rounded down. The two directions are reported separately and never combined into a score. Near-match counts fall partly for a reason the instrument guarantees rather than measures: at a fixed budget of two mismatches every locus a longer design can reach is also reached by each of its own shorter sub-windows, so the set can only shrink, and two mismatches is a fractionally stricter test at 20 nucleotides than at 16. Only the size of the fall and which designs reach zero are measurements. The three blocks carry different denominators and are not comparable across blocks: the junction block is one molecule, the matched-junction block is the six junctions every geometry was screened at, and the corpus block is each geometry's whole design space, which is not screened at the same junctions. The exhaustive GRCh38 genome scan is unavailable at 18 and 20 nucleotides by construction, so no row reports it. Two of the corpus rows carry a ten-base-pair criterion and they are not the same measurement. "...and that duplex reaches ten base pairs" is the mature-parent screen, a search over every window of all six parent transcripts, and it is the row \$2.5's 87 of 190 and \$2.9's 87 / 88 / 87 are read from. "At the design's own seam" is arithmetic on the junction itself: because the wing is five throughout, a parent's hybrid at that seam is five base pairs plus its share of the gap, so pairing five nucleotides of contiguous gap DNA and reaching a ten-base-pair seam hybrid are the same condition and are reported as one row. The three near-match rows count what the alignment screen returned on the sense strand, which is what Tables 2 and 5 print under the same name; none of them is a count of cleavage events.  $\Delta G^{\circ}37$  values are for an unmodified DNA:RNA hybrid; the wing is five at every geometry, so LNA affinity enters each parent duplex identically and cannot explain a difference between the columns. None of these numbers is a measurement of cleavage. The sequences in this table are antisense gapmers at more than one geometry, and each row's own geometry is the one stated beside it: the architecture gives the locked-nucleic-acid (LNA) wing, DNA gap and LNA wing in nucleotides, so 5-6-5 is a 16-mer, 5-8-5 an 18-mer and 5-10-5 a 20-mer, each on a phosphorothioate backbone that \$6 specifies; the bases alone, ordered as unmodified DNA, are a different molecule. Research use only: not for administration to any person or animal (Declarations). The canonical machine-readable copy of every sequence is `fusion-junction-aso-sequences.csv`, which is what to order from rather than this PDF.

| Table 7. Gap length against junction specificity, at one junction and across the design space |                        |                          |                            |
|-----------------------------------------------------------------------------------------------|------------------------|--------------------------|----------------------------|
|                                                                                               | 5-6-5 (16-mer)         | 5-8-5 (18-mer)           | 5-10-5 (20-mer)            |
| <b>At the <i>EWSR1</i> e12 / <i>TAF15</i> e11 / <i>FUS</i> e10 junction</b>                   |                        |                          |                            |
| design                                                                                        | 5'-GGGCATATCATCAAAC-3' | 5'-AGGGCATATCATCAAACC-3' | 5'-CAGGGCATATCATCAAACCA-3' |
| gap-level margin                                                                              | 3                      | 4                        | 5                          |
| sense-strand gap-paired near-matches                                                          | 123                    | 3                        | 0                          |
| gene loci carrying one                                                                        | 6                      | 1                        | 0                          |
| near-matches ( $\leq 2$ mismatches, deeper ceiling)                                           | 189                    | 50                       | 20                         |
| $\leq 1$ -mismatch matches over 186,185 transcripts                                           | 1                      | 0                        | 0                          |
| mature-parent duplex through the whole gap (bp)                                               | 8 ( <i>TFG</i> )       | 0                        | 0                          |
| contiguous DNA a wild-type parent pairs (nt)                                                  | 3                      | 4                        | 5                          |
| most stable parent $\Delta G^{\circ}37$ (kcal/mol)                                            | -7.77                  | -8.66                    | -10.25                     |
| <b>Over the six junctions screened at every geometry</b>                                      |                        |                          |                            |
| designs screened                                                                              | 30                     | 42                       | 54                         |
| median near-matches                                                                           | 86.5                   | 15                       | 0                          |
| median sense-strand gap-paired near-matches                                                   | 21                     | 0                        | 0                          |
| designs carrying none                                                                         | 8 of 30                | 28 of 42                 | 54 of 54                   |
| most gap-paired loci on any one design                                                        | 7                      | 2                        | 0                          |
| designs with no near-match at all                                                             | 0 of 30                | 7 of 42                  | 39 of 54                   |
| <b>Over each geometry's whole design space</b>                                                |                        |                          |                            |
| junction-spanning registers per junction                                                      | 5                      | 7                        | 9                          |
| fusion-specific designs                                                                       | 190                    | 266                      | 342                        |
| best gap-level margin available                                                               | 3                      | 4                        | 5                          |
| a mature parent can pair the whole gap                                                        | 181 of 190             | 130 of 266               | 87 of 342                  |
| ...and that duplex reaches ten base pairs, the criterion applied throughout                   | 87 of 190              | 88 of 266                | 87 of 342                  |
| at the design's own seam, the parent pairs $\geq 5$ nt of contiguous gap DNA                  | 76 of 190              | 228 of 266               | 342 of 342                 |
| designs pairing the gap in parent pre-mRNA                                                    | 19 of 190              | 11 of 266                | 9 of 342                   |
| median most stable parent $\Delta G^{\circ}37$ (kcal/mol)                                     | -8.66                  | -11.60                   | -14.58                     |

What a longer gap buys and what it costs are the two halves of one gap, so the trade is exact rather than approximate — exact as an identity between base counts, which is the whole of what it is. It fixes how many nucleotides fall on each side of the seam, and settles nothing about the chemistry those nucleotides carry or about what the enzyme does with them. Inside the gap, the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side are complements: they sum to the gap. That holds for every design in all three panels rather than on average, and it fixes

the direction of both trades. Within one geometry the gap is fixed, so the two move inversely: at 5-6-5 the 38 designs at margin 3 concede three nucleotides of contiguous parent-paired gap DNA, the 76 at margin 2 concede four and the 76 at margin 1 concede five. What the identity forbids is raising the margin past a geometry's ceiling, which is half its gap rounded down; that takes a longer gap, and a longer gap raises the parent-paired run at every register, the best-margin design running from margin 3 conceding 3 nucleotides at 5-6-5 to margin 5 conceding 5 at 5-10-5. Lengthening the gap therefore

makes RNase-H1 more competent against the fusion and against the parent together, and no register choice within a geometry escapes that, though it does trade the two against each other.

Both directions are large. The best available gap-level margin rises from 3 to 4 to 5, and the junction-spanning registers per junction from five to seven to nine. At the *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 junction, the design carrying that margin sheds its gap-paired sense-strand matches completely: 123 across the gap at six gene loci become 3 at one locus and then none, though 20 near-matches remain (Table 7), so it is that risk class and not the load of Box 1 that goes to zero. Over the six junctions screened at every geometry, designs carrying no such risk rise from 8 of 30 to 28 of 42 to 54 of 54, and the most risk loci on any one design falls from seven to two to none. Part of that fall is guaranteed by the instrument rather than measured: at a fixed budget of two mismatches, every locus a longer design can reach is also reached by each of its own shorter sub-windows, so the reachable set can only shrink as the design lengthens, and two mismatches is a fractionally stricter test at 20 nucleotides than at 16. Only the size of the fall, and which designs reach zero, are measurements. The parent-side quantities below carry no such qualification, being computed from the junction rather than from a search.

Against that, the contiguous DNA a wild-type parent pairs at the same junction rises from 3 to 4 to 5 nucleotides, and the most stable parent duplex becomes more stable,  $-7.77$  to  $-8.66$  to  $-10.25$  kcal/mol. The corpus shows the same trade. Designs whose parent pairs at least five nucleotides of contiguous gap DNA, the shorter of the two reported minima for RNase-H1, rise from 76 of 190 to 228 of 266 to 342 of 342 — those minima being stated for a DNA gap internal to a fully paired duplex, where a parent-paired run of the same length terminates the duplex at the seam, so meeting the count is necessary and not sufficient — and the median most stable parent duplex becomes more stable,  $-8.66$  to  $-11.60$  to  $-14.58$  kcal/mol. At 5-10-5 that count is every design, and necessarily so, since the larger half of a gap of ten cannot be under five: every design at that geometry concedes a parent-paired run reaching the shorter of the two minima. At 5-6-5, 114 of 190 designs keep the parent below it. Which minimum is read decides the contrast, and that is why both are carried: at the six-nucleotide minimum the same series reads 0 of 190, 152 of 266 and 304 of 342, no 5-6-5 design reaching it at all, since a margin of at least one leaves the parent at most five of that geometry's six gap nucleotides.

A third consequence of a longer gap is counted nowhere in this paper, and it runs against the longer geometries. Every screen here counts a site only where all gap positions pair, so a single mismatch inside the gap deletes that site from every count — but a mismatch does not delete the DNA:RNA hybrid, and how much competent hybrid survives one depends on the gap's length. In a six-nucleotide gap a mismatch at the third position leaves contiguous runs of two and three, below both of the minima §6 cites; in a ten-nucleotide gap a mismatch at the fifth leaves runs of five and four, the longer of which reaches the shorter minimum. Lengthening the DNA gap therefore widens the RNase-H1-competent window, and the population of imperfectly matched sites the enzyme could still cleave grows entirely outside what these screens count. That a gap mismatch reduces rather than abolishes cleavage is the premise this work adopts (§6); how that tolerance scales with gap length is bounded by no source retrieved for this paper, and the direction of

the omission is against 5-8-5 and 5-10-5. It bears on a liability §5 and §6 raise and tie only to affinity and wing content — sequence-dependent hepatotoxicity — which §5 rightly says this work cannot attribute to a mechanism. That the mechanism the phrase points at is RNase-H1-dependent cleavage of unintended transcripts is adopted here as a premise, no source having been retrieved for it; on that premise the axis this work comes nearest to grading is the one its screens cannot read. The liability is in any case geometry-dependent in a way the panel does not hold fixed: the wings stay at five locked residues while the oligonucleotide lengthens, so the locked share of the molecule falls from ten bases in sixteen at 5-6-5 to ten in eighteen and ten in twenty — 62.5%, 55.6% and 50%. The longer geometries trade affinity down while trading the catalytic window up, and this paper grades neither axis.

Two liabilities the transcript screens do not reach appear to move the favourable way, and neither reading survives one fixed criterion. A mature parent can pair the whole gap for 181 of 190 designs at 5-6-5, 130 of 266 at 5-8-5 and 87 of 342 at 5-10-5, but the whole gap is a six-nucleotide coincidence in the first and a ten-nucleotide one in the last, so the three counts are not the same test. Held to the ten-base-pair criterion applied everywhere else here, the count of liable designs does not fall: 87 of 190, 88 of 266 and 87 of 342, the two criteria coinciding at a gap of ten because the gap alone is then already a ten-base-pair hybrid. As a share those are 45.8%, 33.1% and 25.4%, and the share falls because a longer oligonucleotide has more junction-spanning registers per seam — five, seven and nine.

Neither framing is the whole reading, and the credit side is a reading at one cut. The liable count is flat at 87, 88 and 87, and what the extra registers add is choices at a junction rather than a lower liability per design: at the ten-base-pair criterion the parent-clean designs available at each junction rise with them, from 2.7 to 4.7 to 6.7 on average — five, seven and nine registers per junction less 2.29, 2.32 and 2.29 liable ones, so the rise of about two per step is the rise in registers rather than a fall in liability. That series is not cut-independent, and the cut it is read at flatters the shortest geometry. Re-derived at the seven-base-pair end of the range §5 bounds this criterion by, it reads 0.4, 3.6 and 6.7 — and at 5-6-5, 29 of the 38 junctions then have no parent-clean design at all, against none at 5-8-5 and none at 5-10-5. What a longer gap cannot do is raise the margin without conceding parent-paired gap DNA, and that is the identity above rather than a matter of which denominator is chosen. Designs pairing the gap in parent pre-mRNA fall from 19 of 190 to 9 of 342, but that arm is a search at a fixed two-mismatch budget and inherits the nesting bound above rather than the parent-side quantities' freedom from it. Nor is the effect confined to the longest geometry: 5'-CAGGGCATATCAAGCGCT-3' at *TCF12* exon 7 returns no near-match at all, where the two 16-mers surviving at that junction return three and two.

## 2.10 · Duplex thermodynamics and conventional design rules

Scored as free energies (nearest-neighbour values for an unmodified DNA:RNA hybrid, with no locked or phosphorothioate parameters applied anywhere in this work, §6) — every one of the 190 designs favours the fusion duplex over the better of the two runs a parent can pair at the junction itself, by 4.8 to 13.1 kcal/mol with a median of 9.6. The comparison is to the seam and not to the transcriptome: the mature-parent duplexes of §2.5, which for 87 of the 190 reach ten base pairs or more against a parent (and for 59 of them that duplex is not elsewhere at all, but runs past the seam into the wild-type *NR4A3*

exon-2/exon-3 junction the acceptor half already meets) are not scored here. Every design favours the fusion because a parent pairs roughly half the oligonucleotide, and half a duplex is much the weaker one. That separates two things a base count conflates. The *difference* in binding free energy is not marginal here. That is not the same as saying binding discriminates: what would decide that is whether a parent seam is occupied at the exposure concentration, which depends on the modified duplex's absolute stability and is not computed anywhere in this paper (§6). Two things therefore remain unresolved rather than one — occupancy at the binding step, and discrimination at the level of *catalysis*, where RNase-H1 requires a DNA:RNA segment within the gap and where a mismatch in it reduces rather than abolishes cleavage by a factor the literature bounds at one- to five-fold. The thermodynamic result narrows neither; it orders them.

The two rankings agree in direction. Grouping designs by the gap-level margin the Methods (§6) define, the mean of that free-energy margin (written  $\Delta\Delta G^{\circ}37$  and computed as  $\Delta G^{\circ}37$  of the better parent run minus  $\Delta G^{\circ}37$  of the full fusion duplex, so that it is positive where the fusion duplex is the more stable of the two and larger is better) rises monotonically with it, from 8.3 kcal/mol at margin 1 to 9.9 at margin 2 and 10.7 at margin 3. That agreement is arithmetic rather than corroboration: the design's own seam hybrid (the run either parent shares with it at the junction itself, which is not the screened parent duplex of §2.5 and is not searched for anywhere else in the transcriptome) is exactly 11 minus the gap-level margin for all 190 designs (11 being one wing's five bases plus the gap's six, so the longer of a design's two seam runs is a whole wing plus whatever of the gap the margin leaves) so the free energy is largely ordering that same length in kilocalories. Largely, because the identity is on the longer run while the free energy is taken against the more stable one, and for 6 of the 190 that is the shorter side: all six sit at margin 2, where the two runs are nine and seven bases and the seven-base run is the more stable of the pair. That run is taken by construction and never extended outward, so it is a minimum: where a parent's next base beyond the seam happens to match, the real hybrid is longer and more stable than the one scored, and the margin reported for that design is correspondingly generous. The §2.5 screens do not recover those cases, since they retain only sites pairing the whole gap. What it adds is the size of the difference; it is not purely a restatement of the margin, because composition reverses the order in 19.9% of cross-margin design pairs and the margin-3 range sits inside the margin-1 range, and the same caution applies to the margin's agreement with the parent screens of §2.5.

Conventional design rules select differently, and against three of the designs this paper calls candidates — though not against the two lead reagents of §4.1, which fail no rule (a narrower clearance than it reads, for the reason two paragraphs below); the *EWSR1* exon-13 reagent §4.1 adds for coverage fails two. Of the 190 designs, 106 satisfy all four rules; the rules bind at different rates, with every design free of a G-quadruplex motif but 13 carrying a homopolymer run of four or more, 43 a CpG dinucleotide and 58 falling outside the 40–60% GC window. The failures overlap, so they do not sum to the 84 designs that fail at least one. The 13 with a homopolymer run are neither base-neutral nor spread across the molecule: five carry a guanine run, five a thymine run and three an adenine run, none a cytosine run, and all 13 come from three junctions — *EWSR1* exon 15, *TFG* exon 4 and *TAF15* exon 8. Every one of those runs sits at the 3' end, six of them wholly inside the 3' locked wing and seven

straddling the gap and that wing; none lies in the 5' wing and none is confined to the gap. The quadruplex column passes everything because of what it asks: four separate runs of two or more guanines, which a 16-mer of this composition has little room to carry. The guanine feature that is present is a different one. A run of three or more guanines is carried by 118 of the 190, at the 5' end in 38 of them — one register per junction, including both lead reagents of §4.1 — and four registers of the *EWSR1* exon-15 junction carry a run of five, which the homopolymer rule catches and the quadruplex rule does not. A clean quadruplex column is therefore not a statement about this panel's guanine content, and the two should not be read as one.

Where that 5' run sits is a fact about the chemistry and not only about the sequence. Both lead reagents of §4.1 begin 5'-GGG, and at 5-6-5 with a five-nucleotide wing those three guanines are positions 1 to 3 of the 5' locked wing: a contiguous locked G-tract, not a DNA one. This work adopts as a premise that a contiguous guanine run inside a locked wing carries synthesis, aggregation and melting-temperature consequences distinct from the ones a homopolymer rule written for unmodified DNA is aimed at; no source retrieved for this paper bounds them, and nothing here measures them. What the audit can be held to is what the audit reads: each of the four rules is computed from the base string alone and none of them takes the sugar or the backbone as an input, and the 5-6-5, 5-8-5 and 5-10-5 geometries are the 2'-O-methoxyethyl convention rather than a locked one (§6). Passing all four is therefore a DNA/MOE rule set returning clean on a locked molecule, and it is not a chemistry clearance.

The disagreement is sharpest where it matters most. Of the nine designs with no sense-strand near-match (Table 4), exactly one satisfies all four rules. Seven contain a CpG dinucleotide; four fall outside the 40–60% GC window — the three *EWSR1* exon-1 designs above it at 62.5% and 5'-GGGCATATCTCTATAA-3' below it at 37.5%. The CpG rule is applied position-blind, and this architecture argues in two directions at once. The canonical immunostimulatory motif is a CpG in sequence context rather than a bare dinucleotide, and only the six-nucleotide gap of a 5-6-5 gapmer carries an unmodified sugar, the wings carrying a 2' modification at every position — but that gap is phosphorothioate DNA rather than unmodified DNA, and a phosphorothioate backbone is the feature that makes an oligodeoxynucleotide a competent ligand for this receptor at all. One consideration lowers the prior and the other raises it; neither is measured here. Whether a CpG inside a locked wing is recognised at all is not established here and no source is cited for it, so the audit is left position-blind and the reading below is stated as the direction the architecture argues for rather than as a result. Only 5'-CAGGGCATATCTTGCA-3' at *TCF12* exon 9 passes every rule — and it is a do-not-order design, ■ in Table 4, pairing wild-type *NR4A3* itself through the whole catalytic gap at twelve base pairs, which is this section's disagreement at its sharpest: the one molecule conventional triage clears outright is the one this paper's central screen condemns. The multi-partner lead reagent 5'-GGGCATATCATCAAC-3', which is not among the nine, also passes all four. The cleanest designs this work found are, with one exception, molecules conventional triage would flag, in six of the seven cases for a CpG that reaches into the catalytic gap, which is the stretch the argument above says the rule would bite on if it bites anywhere. Removing such a motif by base substitution would change one of the bases the gap-level margin is computed on; 5-



methylation of the gap cytosine would remove it at no cost to the margin at all, and §6 specifies the gap cytosines as unmethylated precisely so that this audit reads as written rather than as a question about a vendor's default. Both are reported rather than composed into a single score.

### 3 • Discussion

Designability is not the constraint. Junction-spanning designs exist at every in-frame NR4A3 fusion junction, though at three of them every design pairs a wild-type parent through the catalytic gap — three at ten base pairs, 29 at seven, and 32 at any length, which is the same cut dependence §2.5 states and the reason none of these counts is quoted without it. On the one existence statistic tested, no partner is uniformly clean or uniformly dirty: with all 38 junctions screened, four of the five partners have a junction whose cleanest design carries no sense-strand near-match across the gap at the deeper ceiling, and all five do at the default one. The exon a fusion breaks at therefore bears on whether a design is clean across the gap in that sense, and not the gene alone — though no test of a partner effect was run, and *TFG* sits higher on the genome-wide load on an axis confounded with base composition (§2.3), and the count of such junctions is itself a function of search depth, as §5 sets out.

Clean designs are much scarcer than the default search depth implies, and two independent findings converge on that. One is search depth, measured in §2.4 and bounded corpus-wide in §5. The other is invisible to depth at any setting: five of the nine designs form an eleven- or twelve-base-pair duplex with a mature wild-type parent that pairs the whole catalytic gap, one of them with *NR4A3* itself, where no screen filtering on global identity can see it. Three designs survive every screen applied here, two of them at any parent-duplex threshold (§2.7).

The limiting step is discrimination between the fusion and its parents, and it is not resolved here. Both cited bounds are measured against a single substitution in an otherwise fully paired duplex. Neither transfers to a parent that leaves half the oligonucleotide unpaired and the catalytic gap only partly so: they bound the near-match case, and no retrieved measurement bounds the parent case. The two parent compartments of §2.5 sharpen that rather than softening it. For nine designs of the panel (a different nine from the nine of Table 4 discussed above), the route to wild-type *NR4A3* is not a gap-level discrimination problem at all. Those nine pair the catalytic gap in full across the wild-type intron-2/exon-3 boundary, at two mismatches that both fall in the locked-nucleic-acid (LNA) wing, and the compartment such a duplex would form in is the nuclear one RNase-H1 occupies. That the site exists is not that the duplex forms: pairing the whole gap forces both mismatches opposite locked residues, where destabilisation is largest and where no penalty is modelled anywhere in this work (§6), so these counts are an upper bound on hybridisation. None of Table 4's nine carries such a site of the strict class §2.5 defines, and that section is silent on whether any sits in the wider sense-strand forty. In mature parent sequence 87 designs pair a wild-type parent (any of the six, not *NR4A3* alone, which 61 of them pair) over a contiguous duplex of at least ten base pairs. The general point is that a fusion-junction design's most plausible wild-type liability is its own parent, reached either across a splice junction or in the mature transcript, and both are invisible to a screen that ranks candidates by global identity. A third compartment is invisible to all of them: at a seam whose

acceptor half is not exonic in the mature transcript, the patient's own un-rearranged *NR4A3* allele carries the same sequence behind an intron, and three designs the mature-parent screen clears or cannot read — the pre-mRNA screen returned all three, as §2.6 reports — pair their whole catalytic gap there. The four reports of parental sparing cited here were all made on molecules already synthesised, and three of the four went further than cells: an shRNA to the *FGFR3* side of *FGFR3::TACC3* improved survival in glioma-bearing mice,<sup>13</sup> a *PML::RARα*-specific siRNA prevented disease in NOD/SCID mice,<sup>15</sup> and liposomal siRNAs against *TMPRSS2::ERG* treated mice bearing orthotopic and subcutaneous xenografts.<sup>16</sup> Only the fourth<sup>14</sup> describes no in vivo model in its retrieved record. That raises rather than lowers the bar the comparison above has to clear, and the point stands unchanged: every one of those readouts required the molecule to exist first, and the comparison above is available before anything is synthesised. Whether other groups apply an equivalent comparison before synthesising is not established here: no survey of published design pipelines was performed, and the argument above is about what particular instruments can see rather than about what the field does. The null of §2.5 is what makes that a finding rather than a restatement of the design rule: arbitrary sequence meets this screen at 6.2%, and a chimera keeping the whole rule while joining the two parents at random offsets meets it at 23.8%, against 45.8% for designs at real breakpoints. That last comparison overstates what is specific to the disease, and the stricter null says so: a chimera drawn at real exon termini of the same two transcripts, almost always at junctions no patient is reported to carry, meets the screen at 40.6% — so most of the liability is what joining two exon termini of these genes gives, and the excess at the reported breakpoints is not resolved at this panel's size (§2.5).

All of that presumes that sparing wild-type *NR4A3* is worth the specificity cost, and that premise deserves examination rather than assumption. The published evidence cuts both ways and neither way is decisive. On the permissive side, *NR4A3* has two close paralogues and the family is functionally redundant where it has been tested: *NR4A1* and *NR4A3* are described as functionally redundant suppressors of acute myeloid leukaemia, and the three receptors are highly homologous.<sup>29,30</sup> A conditional double knockout of *Nr4a1* and *Nr4a3* disturbs haematopoietic stem-cell homeostasis, and even then the cells retain regenerative and differentiation capacity;<sup>29</sup> no single-knockout arm is reported there, so the double deletion is shown to be sufficient and is not shown either way on necessity. On the restrictive side, a separate study makes the loss of *NR4A3* consequential rather than silent when paralogue reserve is reduced: mice hypoallelic across the two genes develop a myelodysplastic or myeloproliferative neoplasm.<sup>31</sup> That same source restates, from an earlier report of its own authors rather than as a finding of its own, that abrogation of both genes leads to rapid postnatal leukaemia; it is cited here as a restatement and the earlier report was not retrieved. The family is also not uniform in direction — within atherosclerosis, *NR4A1* and *NR4A2* attenuate lesion formation while *NR4A3* aggravates it<sup>32</sup> — so paralogue redundancy cannot be assumed to be substitution.

Three limits on that reading matter more than the reading itself. Every source cited here is haematopoietic or vascular, and none addresses the tissue an EMC arises in; two of the four are reviews rather than primary reports; and the perturbation described throughout is germline or conditional gene deletion, which is a different and more complete perturbation than partial, reversible,

dose-limited knockdown by an oligonucleotide. The honest position is therefore that wild-type *NR4A3* knockdown has an unquantified cost that is probably not zero and probably not catastrophic, and that the case for junction selectivity does not rest on it. It rests on the donor side instead, on the premise that knocking down the FET donors *EWSR1* and *TAF15* costs more than knocking down the two non-FET partners of §2.3, *TCF12* and *TFG*. That premise is adopted here and is not established here: this paper cites no essentiality measurement for any of the four donor genes, and nothing below rests on the contrast holding. What does not depend on it is the failure mode, which is the same whichever donor a design sits at: a reagent cleaving a parent transcript is failing at the one thing that distinguishes this modality from knocking down *NR4A3* directly, which requires no junction at all. A design that cannot spare the parents has no advantage left to trade.

Free-energy calculation does not narrow the interval either. Every design shows a large free-energy *difference* against the parent runs it meets at the seam, but that difference is not occupancy: no modified-duplex stability was computed for any parent seam, so the binding step is unresolved alongside the catalytic one rather than settled ahead of it (§6). Two things could narrow that interval, and no further sequence analysis is either of them: a measurement, or a physics-based estimate of cleavage geometry on the RNase-H1-heteroduplex complex, for which experimental structures exist. Neither is attempted here. Gap length is not a third, for the arithmetic reason §2.9 gives: a longer gap buys a markedly quieter transcriptome while at the same time pairing more of one wild-type parent through the gap. The two are not one effect: the transcriptome quiets because at a fixed two-mismatch budget a longer design can reach no locus its own shorter sub-windows do not, so the reachable set only shrinks, while the parent duplex grows out of the gap identity itself. It is the same limit reached from the other side rather than a way around it. The field's own answer to poor single-base discrimination has been positional chemical modification of the gap rather than length,<sup>33</sup> and that is the design direction this result points to, now for a demonstrated reason rather than by analogy. A steric-block mechanism, which does not require gap-level discrimination, is a second alternative this work does not evaluate. It is also an ungraded liability and not only an opportunity: the wild-type *NR4A3* 3' splice acceptor that §2.5's pre-mRNA class concentrates at can be occupied with no catalytic gap paired, and what that would do to wild-type *NR4A3* is outside every screen reported here (§5).

Delivery remains unsolved for a tumour, and separates into three routes with different requirements. A characterised EMC-enriched surface antigen is a prerequisite of the systemic receptor-targeted route only; local and inhaled administration require none. No such antigen could be named when the question was put to the disease's own tissue: of the twelve candidate surface antigens for which both a lineage reading against comparator sarcomas and a measured EMC tumour-versus-normal-organ contrast exist (the latter from four EMC and 27 normal-organ libraries across six organs in Gene Expression Omnibus (GEO) deposit GSE28866) three cleared both measured axes, and none cleared those two axes and a wider prior on normal-tissue distribution together. *CD44* and *CSPG4* cleared the two measured axes and were then refused by that prior; *RET* cleared the same two and carries no reading against the prior, so it is ungraded rather than excluded; the other nine did not clear both measured axes. That bounds what was examined rather than establishing that no antigen exists: 86 of the 100 genes on the committed surface

panel carry no row in that deposit and are unmeasured rather than excluded, every reading is transcript-level, and none of it speaks to protein, surface localisation, antigen density or internalisation. Every reading in this paragraph is released as `aso-delivery-antigen.json`. For the inhaled route, EMC's distant spread is lung-dominant, at 35–45% of patients and a median of approximately 28 months to metastasis.<sup>6</sup> Inhaled oligonucleotides have reached human dosing in non-oncology indications. An inhaled antisense oligonucleotide has been dosed in healthy volunteers in phase 1,<sup>34</sup> though that was a splice-switching oligonucleotide rather than an RNase-H1-active gapmer, so it establishes the route and not the mechanism used here. An inhaled siRNA has reached phase 2b–3 in patients.<sup>35</sup> Both target airway epithelium or alveolar parenchyma, which is the compartment inhalation naturally reaches. A hypocellular, matrix-rich sarcoma nodule sitting within that parenchyma is not. Inhaled delivery to lung tumours is an active preclinical field, with 68 records in the retrieval corpus behind this section, but only two of those carry clinical-stage language and neither is a trial. The route is therefore established in humans and not for this target.

**The testable surface is narrower than the literature makes it look, and a reader planning an experiment should know that before ordering anything.** A junction-spanning gapmer needs a junction to span, so the test article has to carry one. The EMC line a reader would reach for first, H-EMC-SS (RRID:CVCL\_1238), is the only one this work could establish as available from a cell repository, another reported line's distributor being unreadable and its availability therefore unanswered rather than answered, and no *NR4A3* fusion is detectable in it on the public record: a filtered fusion caller that ran against it — DepMap's filtered fusion call set, release 24Q4, model ACH-001519 — returned two calls, `AL158209.1--NEBL` and `VIM--RPS25`, neither naming *NR4A3* nor any FET gene; its *NR4A3* expression is 0.941 on a  $\log_2(\text{TPM} + 1)$  scale, near the floor in absolute terms while sitting at the 83rd percentile of the 1,673-line panel, whose median is 0.214, so it is weak corroboration and is graded as such, expression alone being able neither to establish nor to exclude a fusion; the reference registry that records a gene fusion for other EMC models records none for this one; and no retrieved source reports a positive junction in it. The operative consequence is narrow and is the only one this paper draws: no reagent named here can be assumed testable in that line on current evidence, and its junction would have to be established by RNA sequencing first, as §4.4 requires of any test article. A filtered caller's silence bounds detection at a sensitivity nothing here quantifies, and an absent reading is not a reading of absence. This is not a statement that the line is misidentified, nor a statement about what the line is instead. It carries a short tandem repeat profile concordant across three independent sources at every locus but one (D13S317, recorded as a single allele by one source and as two by the other two), and no problematic-line flag. Fusion-negative EMC tumours are themselves a recognised minority category: of eight extraskeletal myxoid chondrosarcomas in one series the fusion transcript was found in six, and two carried neither a fusion transcript nor a genomic rearrangement of either partner,<sup>36</sup> against a later report stating that more than 90% of these tumours carry an *NR4A3* rearrangement.<sup>37</sup> Absence of the fusion is therefore not by itself a reclassification. The observation is also not new: it is in print in one figure legend and is carried as a caution field in the reference registry. No retrieved source examines it as a subject, and it is not discoverable by anyone searching on model validity. Both

halves are released: the fusion-caller, expression and registry readings in `emc-atr-vulnerability.json`, which owns them, and the reading of that figure legend together with the reagent consequence in `emc-model-junction-evidence.json`.

What remains is five test articles. Each of the five has a reagent at its junction, but one of those reagents — the cryptic-exon one below — is not certifiable under the criterion §4.5 states, so the reagents matched to a test article are four that all five screens can be run on and a fifth carried under that qualification — a different set from the four of the coverage ladder in §4.1, overlapping it in two members. Addressable by all five is not cleared by all five: within the 38-junction panel the designs that clear every screen are the three of §2.7, and the two of those four that §4.1 names carry the off-target loads §4.3 reports. Three are the engineered constructs of the functional study cited above,<sup>27</sup> E-N, T-N\* and T-N, whose exon spans that paper states verbatim; two of the three, E-N and T-N\*, carry the same two junctions the reagents of §4.1 span — *EWSR1* exon 12 and *TAF15* exon 6, each joined to *NR4A3* exon 3 — so both of those reagents have a stated test article; the third construct, T-N, carries the intron-2 cryptic-exon seam of §2.6, whose reagent cannot be certified under the criterion §4.5 states, three of the five screens being unable to address a cryptic-exon acceptor at all. The other two are the patient-derived, identity-clean models reported with two EMC tumours,<sup>37</sup> USZ20-EMC1 (RRID:CVCL\_C6MX) and USZ22-EMC2 (RRID:CVCL\_C6MY), whose fusions are reported as *EWSR1* exon 13 and *TAF15* exon 6 joined to *NR4A3* exon 2 rather than exon 3; reagents exist at both acceptors, so each line has one under either reading of that exon label.

The two sources of a test article are not interchangeable and their limits run in opposite directions. Rebuilding the constructs is the faster of the two and the only one whose critical path contains no laboratory that has to answer an email, since the junction is specified by construction. It is not unconditioned: the published recipient background is a catalogue item supplied under a material transfer agreement, and every plasmid carrying the published retroviral backbone that could be read here is distributed to academic institutions and non-profits only, a restriction that binds before any price does. What it cannot buy is biological relevance: a complementary DNA over-expressed in a heterologous background is not the disease, so such an experiment could speak to junction-selective knockdown of the intended transcript and not to activity at endogenous expression from an endogenous locus. The published recipient background is in no cell-line registry either, so a rebuild would sit in a different background from the original — an isogenic mismatch to declare rather than gloss. The patient-derived models are the only source of a fusion-positive EMC cell, and are available on request from the originating laboratory with no repository deposit; what a transfer requires is stated nowhere that could be read, which is an absent statement rather than an absence of conditions, and the cells are slow once received, at reported doubling times of five to six days as sarco-spheres passaged every two to three weeks, which constrains any exposure window. One further reported line — the one whose distributor could not be read, above — cannot serve as a test article at all on current evidence, because its fusion partner and exons are unstated anywhere readable and it would have to be sequenced first. One constraint sits above all of them and no reagent choice moves it: every one of these sources ends at someone culturing cells, so the rate-limiting step is a laboratory rather than a

line, a construct or an oligonucleotide. §4 opens on what that means for this work.

One deliverable does not wait on that. No named reagent reaches every patient, and the released design and screening pipeline is the paper's second output: §4.5 states what it takes as input, what it does with it and what its result is worth, which is a candidate rather than a validated reagent.

#### 4 · Reagents, controls and the falsification experiment

**This work has no laboratory.** Nothing specified below has been performed, and the rate-limiting step for all of it is a laboratory rather than a line, a construct or an oligonucleotide (§3). What follows is a specification for someone else to run, not a report of a result.

This section is the paper's output for a laboratory. It names six things: the oligonucleotides to make, the arm that separates the two ways a weak result could arise, the predicted off-target load each carries, the controls without which the readout does not mean what it appears to mean, the number that would falsify, at its top margin, the ranking every candidate here is ordered by, and — for the patients no named reagent reaches — the released procedure by which a candidate can be designed for a breakpoint outside this panel. Nothing in it is a claim of efficacy. No sequence named below has been synthesised or tested.

##### 4.1 · The reagents to synthesise, and their population coverage

The experimental design that would resolve the central uncertainty — an isogenic fusion-positive/fusion-negative comparison — has been published in an analogous disease, though with a single-dose knockdown readout rather than the matched dose-response and wild-type-parent measurement §4.4 requires. Fusion-specific antisense oligonucleotides against *NAB2::STAT6* in solitary fibrous tumour, evaluated against CRISPR-engineered isogenic fusion-positive and fusion-negative cells, reduced fusion expression by 58% and proliferation by 22% in vitro.<sup>38</sup>

**Research use only.** Both sequences named in the next paragraph, and every sequence in Tables 2 and 5, are research reagents for laboratory investigation only. None is a medicine or a candidate drug, none has been synthesised or tested, and none may be administered to any human being or animal or supplied to anyone for that purpose. Ordering any of them from a commercial synthesis service is possible for anyone; doing so does not make it a treatment, and nothing in this manuscript should be read as licensing use in a patient. Do not order an oligonucleotide by copying it out of this PDF — use the canonical `fusion-junction-aso-sequences.csv` or `.fasta` (§6) — and do not order at all until the breakpoint of the cell line or patient sample has been established at nucleotide resolution by RNA sequencing (§4.4).

Applied here, the reagents to synthesise are the best available at the two most frequently reported junctions with a published exon-resolved breakpoint (Table 2): 5'-GGGCATATCATCAAC-3' at *EWSR1* exon 12 and 5'-GGGCATATCTTGTGTG-3' at *TAF15* exon 6. Both hold the top gap-level margin of 3, and neither pairs a parent through the catalytic gap at the ten-base-pair threshold, though both sit close to it: the *EWSR1* reagent's longest parent run is eight base pairs and the *TAF15* reagent's nine, both against wild-type *TFG*.



Where the criterion is set therefore decides how these two reagents read, and the reading is not favourable across the range \$5 bounds it by: at a cut of eight both would fall inside the class Box 1 marks do-not-order, at nine the *TAF15* reagent alone would, and only at ten does neither. The panel moves with them — 143 of the 190 designs pair a wild-type parent through the whole gap at a cut of eight, 98 at nine and 87 at ten. One of the two carries a second liability no transcript screen shows, and it belongs beside the sequence rather than two subsections later: the *EWSR1* reagent has a sense-strand near-match in wild-type *TAF15* precursor RNA at two mismatches, one of them inside the catalytic gap, spanning an intron–exon boundary (§4.3); the *TAF15* reagent has no sense-strand pre-mRNA site at all. The first also tests the multi-partner prediction, against a synthetic target only.

How much of the disease those two junctions represent is a junction figure, not a partner figure, and the two differ: 46 *EWSR1* and 9 *TAF15* among 58 molecularly confirmed cases is 94.8% between the partners,<sup>9</sup> while each reagent spans one exon pair. Discounted by the breakpoint distribution of an 18-case series<sup>22</sup> the two are 68.4%, roughly two thirds; Table 5 gives that figure, the rungs above it and the reagent at each.

What that figure is, and what it is not, has to be said in terms, because the arithmetic is easy to read as a result. **It is not a coverage measurement.** No patient was screened with either sequence, and a sequence that matches a reported junction is not thereby active against a tumour carrying it, so the number prices which published junctions the two reagents address and nothing about what they would do in a patient. It is also not a pooled proportion: the two cohorts are combined multiplicatively, a partner prevalence times a conditional within-partner fraction, which is a composition no pooling rule for proportions applies to. Four bounds sit on it, and they run in different directions. The interval is wide for the denominators rather than the estimate: taking each breakpoint fraction to its own Wilson bound spans 39.9% to 82.8%, the *EWSR1* arm resting on 15 tumours and the *TAF15* arm on three. **That interval propagates the breakpoint fractions only.** The partner shares are held at their point estimates, and their own Wilson intervals are 67.2–87.7% for *EWSR1* and 8.4–26.9% for *TAF15*, so an interval that varied all four quantities would be wider than the one reported. Wider, but not reachable by composing them: taking all four quantities to their upper bounds at once returns 101.4% of cases covered. The reason the composition cannot be repaired rather than widened is that the two partner shares are cells of one multinomial, whose Wilson upper bounds sum to 114.7% on their own — a note on how the interval is built, not a second attempt at the coverage figure. What is reported is therefore a composed-endpoint range carrying no nominal coverage level, and not a confidence interval. The third decimal is not resolved by the data behind it: the *TAF15* arm is three tumours out of three, so one tumour breaking elsewhere moves the total by about 5.2 percentage points, which is why the plain-language reading is roughly two thirds and why nothing here should be compared at one-tenth of a point. And the denominator has a denominator: every figure in this paragraph is a fraction of *molecularly confirmed* EMC, while what fraction of EMC reaches molecular confirmation at all is stated by no source retrieved here, so the population these percentages describe is a tested population and not the disease. It assumes too that the breakpoint distribution within *EWSR1*-rearranged tumours is the same in the 58-case cohort as in the 18-case one, 21 years apart, which nothing here tests and which

no published series is large enough to settle. Breakpoint sequencing of archival material would sharpen it; no further analysis of sequence will.

Two things sit outside the table. Coverage rises only by adding reagents, because no oligonucleotide can serve two breakpoints of the same partner: *EWSR1* exon 12 ending AATGGTTTGATG against exon 13's CACTCCGTGGAG, which agree over a single terminal base. Taken across every pair of in-frame junctions of one partner in this panel, the longest shared 3' donor run is five nucleotides, at *TFG* exons 2 and 6, and three within *EWSR1*. The cross-partner coverage of §2.2 is the exception that shows the rule, needing the ten identical donor bases the FET paralogues share. And the *NR4A3* exon-2 acceptor rows of §2.6 sit in Table 5 without entering the panel's own counts.

Two further reagents extend the set, both at the top gap-level margin of 3: 5'-GGGCATATCTCCACGG-3' at *EWSR1* exon 13 to *NR4A3* exon 3, and 5'-GGGCATATCCATCAGA-3' at *TCF12* exon 5 to *NR4A3* exon 3, whose junction is resolved to the nucleotide by the deposited chimeric cDNA of §2.3. The second comes with no distribution: one *TCF12*-rearranged tumour has ever been sequenced at this junction, no breakpoint series retrieved here contains a *TCF12* tumour at all, and the break-apart assay the later cohorts used locates no seam within the *NR4A3* locus, so recurrence there is untested rather than refuted. Table 5's top figure, 98.3%, is an upper bound rather than a reachable target for two reasons and not one: that *TCF12* arm is priced at its ceiling, worth 3.4 percentage points, and the figure also assumes every remaining *EWSR1* breakpoint covered, which is the larger of the two steps at 15.9 points and needs three further reagents the retrieved record does not resolve to an exon (Table 5). One fact about the screened set is invisible in the ladder, which prices rungs rather than counting seams: every junction with a published exon-resolved breakpoint in the retrieved record now carries a screened design, the five in this panel (§2.7) and the four *NR4A3* exon-2 acceptor seams of §2.6, of which Table 5 carries three under its own “beside the panel” label and the fourth as a coverage rung, reported beside it (§2.6). Eight of those nine designs are taken through all five screens. The ninth, at the *PGR* seam, is graded on three of them, and two screens miss it rather than one: the mature-parent screen and the pre-mRNA screen read the same six-gene set of parent sequence, which does not carry that donor, so neither of that junction's parent compartments is read by the screens, for the reason §2.6 gives. The mature half is closed outside them by the scan §2.6 names, which returns no wild-type *PGR* window within two mismatches of any of that seam's designs; the unspliced half is unmeasured rather than clean. That is a statement about how far the screening reached and not a coverage figure, and it displaces nothing above it: how many patients a reagent set reaches is the ladder's question, priced on the single series behind it. The Supplementary Information (SI) §S6 carries the rest of the ladder's bookkeeping — the count of those seams, a second basis that prices them on the pooled breakpoint record, which SI §S6 states is two of seven retrieved series and not the whole one, and the four that move the figure by nothing at all.

The *EWSR1* exon-13 to *NR4A3* exon-3 reagent should not be recommended on its transcriptome count, because the two axes that separate it from the exon-12 reagent point in opposite directions and only one of them bears on where an effect would land. On count it is the lighter of the two, 24 gap-paired near-matches at 2 loci against

123 at 6. On exposure it is the heavier: both of its loci are transcribed at the upper cut in the organs a systemic phosphorothioate gapmer distributes to, where none of the exon-12 reagent's measurable loci is (§2.8, Table 6). For a laboratory choosing between them, the exposure reading is the one that speaks to the question a count cannot: a locus matched but not transcribed in the organs a systemic dose reaches has no route to an effect, whereas the count says only how many gap-paired windows the screen returned there. Neither axis is a risk ranking, and this comparison does not make the exon-12 reagent the safer molecule. Every hit behind both is a 14 of 16 match, no screen here predicts cleavage at any of them because all five grade hybridisation only (§5), and an expressed gene is necessary and not sufficient for an effect. The reason to make the *EWSR1* exon-13 to *NR4A3* exon-3 reagent is coverage, and it is unaffected by either axis.

Two risks attach, in this order. The first is architectural, and the Methods (§6) disclose it. A six-nucleotide gap supports noteworthy but incomplete RNase-H1 activity where seven to ten are reported as optimal,<sup>39</sup> so weak knockdown is at least as likely to be the gap as the sequence. That risk is now addressable by a named second reagent rather than by a caveat.

#### 4.2 · A second geometry as a gap-length control

5'-AGGGCATATCATCAACC-3' is the 5-8-5 design at the same *EWSR1* exon 12 junction. It spans the same three partners' breakpoints and sits inside the reported activity optimum. It holds a gap-level margin of 4 where the 16-mer holds 3, and carries 3 sense-strand near-matches across the gap at one gene locus, against the 16-mer's 123 at six (§2.9, Table 7).

Synthesised alongside the 16-mer, at one extra oligonucleotide and one extra well per condition, it separates the two explanations a weak result would otherwise confound. A 5-8-5 arm that knocks down where the 5-6-5 arm does not attributes the failure to gap length rather than to sequence, though only if occupancy is held equal: the longer arm is also a longer perfect duplex against the fusion, so a positive result is consistent with better cleavage competence or simply with the fusion duplex being occupied where the 16-mer's was not, and this pair does not separate those. What it does not buy is parental sparing, since the same two nucleotides lengthen each parent's contiguous duplex from 3 to 4 nucleotides of gap DNA and its whole contiguous hybrid at that seam from 8 to 9 base pairs, and drive its free energy from  $-7.77$  to  $-8.66$  kcal/mol. Its longest mature-parent duplex through the whole gap does fall from 8 base pairs to none, but 8 sits below the ten-base-pair threshold applied throughout, so neither design counted as a mature-parent liability at this seam and that fall removes nothing the screens had counted at ten. That reading has to be stated with its other half, because the cut is used in both directions in this paper and must not be used in whichever direction enlarges the negative: at the seven-base-pair end of the range §5 bounds the criterion by, the same fall removes a duplex that does count, and it is the same dual reading Box 1 and §4.1 apply to the two lead reagents' own runs of eight and nine. Both arms therefore need the fusion-negative comparator below.

#### 4.3 · The predicted off-target load of the two lead reagents

The second of the two risks §4.1 names is transcriptome load, and it differs sharply between the two leads. The *EWSR1* exon-12 reagent carries the heavier load of the two: 123 gap-paired sense-strand near-

matches at the deeper ceiling, recounting to six gene loci, all at the screen's loosest admitted identity and none on a parent transcript (§5). It is not the heaviest in the panel — fourteen of the 187 design records re-screened at that ceiling carry more, to a maximum of 240 at *TCF12* exon 3. Table 3 prints one row per junction rather than one per design record, so those per-record counts are in the released screens rather than in a table. The *TAF15* reagent carries 8 such near-matches at five loci.

The parent compartments qualify that, in a way the transcript screens cannot show. The *EWSR1* reagent carries a sense-strand intron–exon-spanning near-match in wild-type *TAF15* pre-mRNA at two mismatches, one of them inside the catalytic gap, returned independently by the pre-mRNA screen and the genome scan. It falls outside the headline parent counts, because those require the gap to be paired in full; it sits inside §2.5's wider tallies, among the 53 with a pre-mRNA near-match, the forty on the sense strand and the 21 that pair all of the gap but one or two positions. By the bounds adopted above a single gap mismatch does not abolish cleavage. It is the multi-partner result's own cost rather than an incidental hit: the ten donor bases shared across *EWSR1*, *TAF15* and *FUS* that let one oligonucleotide span three junctions are the bases that place it against wild-type *TAF15*. The *TAF15* exon-6 to *NR4A3* exon-3 reagent carries no sense-strand pre-mRNA site at all, which is a second respect in which the two separate on something other than count.

That load should travel with the reagent. It is a liability to disclose and to control for rather than a disqualification, because on the genome scan the same design sits at chance in both directions that matter: 0.69 times the expected number of near-matches at two mismatches, against a corpus median of 0.97 on that axis, and 0.62 times the expected number of gap-paired ones, against a corpus median of 0.82 on that one. The two medians are different quantities and only the first is 0.97, so each ratio is read against its own (§2.7). The null behind those ratios assumes independent uniform bases and separates more-than-chance from at-chance and nothing finer (§5), so the scan neither aggravates the load nor exonerates it. Expression separates the two reagents the other way (§2.8, Table 6): none of the *EWSR1* reagent's measurable loci is expressed at the upper cut in the organs a systemic dose reaches, while the *TAF15* reagent's five include *NRP1*, which is. That does not reverse the ranking, since no screen here establishes that a two-mismatch duplex engages any of them, and it is not a statement about safety.

#### 4.4 · Controls and the decision threshold

The three designs that survive every screen are mechanism controls rather than candidates: 5'-AGGGCATATCGGAGTC-3' at *FUS* exon 8, 5'-GGGCATATCCGACATG-3' at *TAF15* exon 1 and 5'-GGGCATATCAAGCGCT-3' at *TCF12* exon 7, tiered as §2.7 describes. None sits at a junction a patient is reported to carry (§2.7), which is what makes them controls rather than candidates. 5'-GGGCATATCTCTATAA-3' at *TCF12* exon 17, which the default-depth screen returns as clean (Table 4), is not among them. At ten times the default search depth it carries 101 sense-strand near-matches, 14 of them spanning the catalytic gap.

What a knockdown experiment with these reagents can transfer to depends on how it is set up, and the first requirement is upstream of the assay. The breakpoint of the cell line or patient sample must be established at nucleotide resolution by RNA sequencing before any

oligonucleotide is ordered: every design here is specific to the exon pair or pairs it was tiled at — nine span more than one (§2.2) — and none is valid for an unverified junction. Routine diagnosis does not supply it: break-apart *NR4A3* fluorescence in situ hybridisation is the preferred single assay because it detects any rearrangement "irrespective of partner",<sup>6</sup> so on its own it does not locate the seam, and the 58-case cohort every coverage figure here is denominated on was identified that way, with RNA exome sequencing applied to three of its cases.<sup>9</sup> That cohort reports a partner for 57 of its 58 tumours, so partner assignment there rests on more than the break-apart assay; what no part of that workflow supplies is the exon pair.

Three assay controls are required, and a knockdown assay alone distinguishes none of them:

- a positive control gapmer against an abundant housekeeping transcript in the same cells, to separate failed delivery from a test reagent that reached its target and did not cleave it. It must be of the same 5-6-5  $\beta$ -D-oxy-LNA phosphorothioate geometry as the test articles, so that only the target sequence differs: a catalogue control of another chemistry or another gap length — a 5-10-5 2'-O-methoxyethyl gapmer, say — establishes neither that a 5-6-5 was delivered on comparable terms, since uptake and endosomal release track chemistry class and phosphorothioate content, nor that a six-nucleotide gap can direct cleavage in these cells, which §4.1 names as this panel's first risk. It says nothing about discrimination, which is fusion-versus-parent selectivity and is what the fusion-negative isogenic comparator below is for. §4.2's gap-length arm is not that control: it separates gap length from sequence as explanations of weak knockdown, and §4.2 states that it buys no parental sparing, both arms needing the comparator too;
- a scrambled gapmer of the same 5-6-5  $\beta$ -D-oxy-LNA phosphorothioate chemistry and geometry as the test article, and composition-matched to it — dinucleotide-preserving, so that GC fraction, total guanine content and the 5' run are held as well as base composition, since a scramble differing in those differs from the test article in melting temperature, protein binding and aggregation as well as in sequence. What it separates is the backbone-class component of toxicity and not the whole of it: §6 records that the relevant liability of high-affinity gapmers is sequence-dependent, and a scramble cannot by construction control for a sequence-dependent effect. The scramble actually ordered must itself be put through the mature-parent screen before it is made, because on this paper's own null 6.2% of scrambles pair a parent's whole catalytic gap at the ten-base-pair criterion and 1.8% do so against wild-type *NR4A3* (§2.5), which is the one transcript a control must not engage. That 6.2% is a mean over 190 windows, and the per-window rates should not be read as window properties: each rests on 200 draws, so its standard error is 1.7 percentage points, and the observed spread of 2.0% to 15.0% is close to what 190 windows of 200 draws from a single common rate would produce. Nine molecules are tiled at two or three junctions with an identical target window, and their repeated estimates disagree by up to 8.5 percentage points, which is the size of that noise. There is real between-window variation underneath it — the dispersion ratio is 1.61 on 189 degrees of freedom and the between-window standard deviation is about 1.3 percentage points — but it is smaller than the range suggests, so the screen is required for every scramble

rather than for the windows that happen to have drawn high. Two things decide whether that screen answers the question. The sequence to screen is the *target window*, the reverse complement of the oligonucleotide ordered, because screen 4 searches the sense strand only (§6): screening the antisense sequence directly searches the wrong strand and returns a false pass. And the rejection rule is the paper's own criterion — redraw the scramble where a wild-type parent pairs its whole catalytic gap over ten base pairs or more, wild-type *NR4A3* above all, since a control that engages the transcript the modality exists to spare is not a control. What a passed scramble is then clean at has to be said with it: it is clean at ten and not below it. Read at seven, the same null returns 74.3% of scrambles pairing a wild-type parent's whole catalytic gap and 23.9% doing so against wild-type *NR4A3*, so passing this rule certifies the criterion applied throughout this paper and does not certify that the control spares the parents;

- a fusion-negative isogenic comparator, since wild-type *NR4A3* may be too weakly expressed in an EMC line for the selectivity readout to be defined at all.

The test articles can be ordered from this paper. The experiment cannot, and the gap should be read before it is budgeted rather than discovered at the bench. Two of the three controls are named as classes rather than as molecules. No target transcript and no catalogue item is named for the positive control, which must carry the test articles' own 5-6-5  $\beta$ -D-oxy-LNA phosphorothioate geometry and is therefore a custom synthesis rather than a shelf reagent; and the scramble is a sequence a laboratory has to draw against its own test article's dinucleotide composition and then put through the mature-parent screen before ordering, so it does not exist until that is done. The third is not a molecule at all, and this paper names no supplier of one: a fusion-negative isogenic pair has to be engineered, and only the construct route of §3 supplies one, in the recipient background those constructs are expressed in. Neither patient-derived model of §3 has a fusion-negative counterpart reported or sourced anywhere here, so that route runs without this control unless one is built by excising the fusion from lines whose reported growth rate §3 gives — a project rather than a control. No assay is prescribed either — not the quantification platform, not the wild-type amplicon placement, whose two options are set out below and bias in opposite directions, and not the normalising reference, which the next paragraph constrains without naming. What this section fixes is the reagents, the arms, the reporting requirements and the threshold; what it leaves to the laboratory is the assay and the two control molecules.

Two limits of that set are invisible within it. The normalising reference transcript must be neither the positive control's target nor either transcript measured, or the positive-control well becomes uninterpretable and every other well is rescaled against a perturbed reference. And none of the three assay controls separates RNase-H1-dependent cleavage from steric block, which §3 names as unevaluated, so a knockdown seen with this set is consistent with both while the ranking under test models cleavage alone.

Where the wild-type measurement sits decides the answer, because the fusion carries *NR4A3* exons 3 to 8 and a wild-type assay must not read sequence the fusion also carries. The two available placements bias in opposite directions. An amplicon upstream of the acceptor cannot detect the fusion, but it lies on the 5' cleavage



fragment rather than the 3' one, and the two fragments are cleared by different routes at rates nothing here measures, so it reads wild-type knockdown through the 5' fragment's own persistence and biases apparent selectivity by an amount and in a direction that are not fixed in advance; an amplicon spanning the wild-type exon-2/exon-3 boundary reads knockdown directly, but at the *NR4A3* exon-2 acceptors of §2.6 the fusion itself carries that boundary. Both are defensible, no assay is prescribed here, and which placement was used should be reported with the result.

The decision threshold should be fixed before the experiment, in a form that can be registered. Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose-response in the same wells, so that a larger number is a more selective reagent. The cut is defined for that quantity and no other, and what that leaves out has to be said, because it is the half of the argument §3 rests the modality on. Wild-type *NR4A3* is the acceptor parent; no donor parent is read by this ratio, and neither is any other gene. So a reagent can clear the cut while pairing a parent transcript through its whole catalytic gap — the case §2.7 excludes a fourth *TCF12* design for, at eleven base pairs against its own donor — and each reagent of §4.1 carries its longest mature-parent duplex against a gene this ratio does not read (Table 5). The remedy is not a second cut. No retrieved measurement bounds the parent case (§3), so a threshold on a donor ratio would be a number with nothing behind it, where this one at least has a stated convention. It is a reporting requirement instead: wild-type transcript for the reagent's own donor parent and for the parent carrying its longest duplex, measured in the same wells on the same dose-response, reported beside the ratio and against no cut — the requirement the margin-contrast arm below already carries, stated for every arm rather than for that one. A design whose donor knockdown tracks its fusion knockdown has lost what §3 says the modality exists to buy, whatever the registered number reads.

What that leaves the registrable test able to settle should be stated without hedging, because it is narrower than the existence of a pre-registrable threshold makes it look. The quantity above is defined on wild-type *NR4A3*, and wild-type *NR4A3* is the one parent §3 states the case for junction selectivity does *not* rest on. A result on this cut can therefore falsify the ranking and cannot falsify the rationale. What would falsify the rationale is the corresponding ratio on a donor parent — wild-type *EWSR1* or wild-type *TAF15* half-maximal knockdown concentration over the fusion's, on the same matched dose-response — and this section states no cut for it, because no retrieved measurement bounds the parent case (§3). The donor measurement is required to be reported and is not required to clear anything. That asymmetry is a limit of the test rather than a property of the reagents, and it is why the reporting requirement above is not optional. Nor does any arm of this experiment read a phenotype: every quantity fixed here is a transcript measurement, so a reagent that clears the cut has been shown to discriminate and has been shown nothing about the growth or survival of an EMC cell. Whether such a cell requires the fusion at all is the assumption §1 records as unshown, and no experiment in this paper tests it.

A ratio of residual transcript at a single dose is not commensurate with it and must not be compared against the same cut: that ratio is bounded above by one divided by the fusion knockdown's complement, so at the 58% knockdown reported for the analogous published experiment<sup>38</sup> it cannot exceed  $1/(1 - 0.58) \approx 2.4$  however

selective the reagent is, and a cut of 5 would return falsification as arithmetic rather than as biology.

The replicate count follows from the variance rather than being asserted. Selectivity here compounds four normalised measurements, so the replicate standard deviation of its natural logarithm is the quantity to estimate in a pilot, and that standard deviation — taken across independent biological replicates of the same matched dose-response, in the same wells, at the wild-type placement declared with it — is the one thing the pilot has to return. Neither the pilot's replicate count nor its test article is fixed here, for the same reason no assay is: both are properties of a platform this section leaves open, and the replicate count is the quantity the pilot exists to decide rather than one it can be given. What is fixed is the scale, because a standard deviation taken on the ratio rather than on its logarithm is not the quantity the figures below are stated for.

At a standard deviation of 0.35 on that scale — natural log throughout, a 1.42-fold replicate spread — six independent biological replicates give about 80% power to falsify a true selectivity of 3, and three give about 30%, computed from a noncentral t. The bound that decides falsification is the upper limit of a two-sided 95% interval, equivalently a one-sided 97.5% bound, so the test's one-sided type-I rate is 2.5% rather than 5%; the figures above are for that bound and change if a one-sided 95% bound is used instead.

Above a standard deviation of about 0.65 no observed ratio at or above one can place that upper bound below 5 at three replicates, so the test can fail only where the reagent is anti-selective. Such a test is *void*: it cannot fail, which is a different outcome from one that fails to falsify, and voidness is a property of the assay's replicate variance and not of the design. The number of replicates should therefore be set from the pilot estimate, with three as a floor and not a target. A pilot standard deviation is itself an estimate with large sampling error at small replicate counts, so the void gate is applied to the upper confidence bound on the pilot's standard deviation rather than to its point estimate, and the pilot reports that bound with its degrees of freedom. That makes the pilot a gate as well as a sizing step, and its rule follows from the figures above without needing a further one: a pilot returning a standard deviation at or above the void figure has shown that the floor cannot falsify at this cut, so the decision at that point is a larger replicate count or no falsification test at all, and never three. The ranking is falsified only where the upper bound of the 95% confidence interval lies below the cut, never where a point estimate does. The ratio is reportable only where the wild-type *NR4A3* knockdown is itself resolved above a pre-stated limit of quantification — the quantity that can approach zero, which is the change in wild-type transcript and not its vehicle-well abundance; a wild-type change indistinguishable from none returns an unbounded selectivity that would otherwise pass by default, and the honest output there is a one-sided lower bound, which cannot falsify. The cut is 5.0, taken as a convention from the approximately five-fold near-match figure of §6 rather than measured for this comparison, on which §3 records that no retrieved measurement bounds the parent case. It sits at the optimistic end of the one- to five-fold span reported above, so a reading inside that span is consistent both with the ranking failing and with the discrimination this chemistry is already reported to give; the margin contrast below is what separates those two, and a single-arm result does not.

One property of that cut belongs to no single experiment. The rule is stated per reagent and the claim it tests is not: the ranking is one claim, §4.5 releases a procedure that generates reagents against it without limit, and any one reagent falsifying is read as falsifying the ranking. The family over which an error rate would have to be controlled is therefore open-ended, which is why no correction is imposed — a multiplicity adjustment needs a family size fixed in advance, and this one is fixed by how many laboratories run the procedure. Two consequences follow and both belong with any result. A falsification carries the full weight of its confidence statement for the first reagent tested against this cut and less for each one after it, so how many reagents have been tested against it, and in what order, is part of the result rather than context for it. And nothing corrects in the other direction either: a reagent that clears the cut among many has not thereby been shown to discriminate, only not to have falsified.

The reagents of §4.1 all hold gap-level margin 3, so as a set they measure the level of selectivity at the top margin rather than whether selectivity orders by margin. One optional arm supplies that contrast, at *EWSR1* exon 12: 5'-GCATATCATCAAACCA-3' is the margin-1 register of the same junction at the same geometry. Margin is not the only variable that moves with the register, and the others are named here rather than left for a reader to find: GC falls from 43.8% to 37.5%, outside the 40–60% window §2.10 audits, so this arm would fail a conventional rule the lead reagent passes; gap-paired near-matches fall from 123 to 34 while single-mismatch off-targets rise from 1 to 22; and its longest mature-parent duplex is eight base pairs against wild-type *FUS*, which is the same length the lead reagent carries against wild-type *TFG*, so the two do not separate on that screen at all. A difference between the arms is therefore not attributable to margin alone.

One asymmetry decides what the arm can show. Its five parent-paired gap nucleotides are *EWSR1*, the donor, not *NR4A3* — so the liability its low margin creates is invisible to a readout defined on wild-type *NR4A3*, and interpreting this arm requires the wild-type *EWSR1* measurement alongside it. The mirror register at the same seam, 5'-CAGGGCATATCATCAA-3', is the one whose five fall on *NR4A3*, and it carries an eleven-base-pair duplex against wild-type *NR4A3*. That is not a reason to prefer one mirror over the other: it puts that register in the do-not-order class of Box 1. No table here prints that 16-mer — the tables select per junction and per screen — so the canonical file carries its verdict and is where it must be read. This arm is available in one direction only.

*TAF15* exon 6 cannot supply the same arm: all four of its lower-margin registers pair a parent through the whole gap at eleven or twelve base pairs, two of them against wild-type *NR4A3*. Lower-margin registers do clear the parent screen at *EWSR1* exon 13 and *TCF12* exon 5 (Table 2), so the arm is placed at *EWSR1* exon 12 because that is where the lead reagent sits, not because it is the only junction that could carry one. A contrast at one junction tests the ordering at that seam and not the ranking across the panel, and the comparison it supports is between two arms rather than of one ratio against the cut, for which this section states no rule.

#### 4.5 · A design procedure for a breakpoint outside this panel

The reagents named above do not reach every patient: the two leads address the published breakpoints of roughly two thirds of molecularly confirmed cases, and the panel is bounded by what has

been sequenced rather than by what can be designed. The deliverable is therefore the procedure as well as the reagents, and it is the procedure that produced this paper's 190 designs, released unchanged with the artefacts.

Its input is the breakpoint at nucleotide resolution, by RNA sequencing of the tumour, which §4.4 requires in any case; an exon pair inferred from a break-apart assay is not sufficient. The builder itself consumes the exon pair, and the nucleotide resolution is what establishes that the pair is the right one and that the seam falls where the model puts it. Three things about running it are not guessable from the module names given below, so the invocation is given rather than described. The breakpoint is declared in the environment and the audit gate is a flag:

```
python3 junction_aso.py --audit
FUSION_JUNCTION_MODE=real DONOR_GENE=TCF12
DONOR_EXON_END=5 NR4A3_EXON_START=3 \
python3 junction_aso.py
```

The audit run grades every declared breakpoint and designs nothing; a panel may only be emitted for a pair it grades emittable. All three breakpoint variables are required and the builder refuses to run without them, because a default would emit one junction's panel under the name of whichever junction the caller believed they were designing for. And a non-coding acceptor is refused by default: the *NR4A3* exon-2 and cryptic-exon seams of §2.6 are reachable only with `PUBLISHED_BREAKPOINT_JUNCTION=1` set and the seam present in the builder's whitelist of already-published non-coding acceptors, so a new patient's exon-2 acceptor needs an entry added before the builder will emit for it — which is the one acceptor class §2.6 calls most consequential, and the one place the released procedure does not simply run. Given a declared exon pair, `junction_aso.py` retrieves the parent transcripts, builds the modelled fusion, grades the pair for frame and tiles the junction-spanning gapmers, emitting each with its GC, its gap-level margin and a check that it complements neither parent perfectly. The five screens then run on that panel: `junction_aso_offtarget.py` the alignment screen against human RefSeq RNA, classifying each near-match by whether the catalytic gap is paired; `aso_insilico.py` the exhaustive transcript scan, the target-site accessibility fold and the sequence-liability filters; `aso_premrna_offtarget.py` the parents' unspliced sequence; `aso_parent_gap_pairing.py` the mature-parent screen; and `aso_genome_offtarget.py` every position of GRCh38. A design is *certifiable* here where all five could be run on it and it cleared all five; a design one screen cannot address is *uncertifiable* whatever the other four return. The parent screens matter most: pairing a parent through the whole catalytic gap is this paper's central negative and surrenders the only advantage the modality has. Where the acceptor half is not exonic in the mature transcript, the un-rearranged-allele scan of §2.6 applies as well, and it lives in two further modules rather than in a step within the five screens: `aso_taf15_intron2_designs.py` holds the single implementation of that scan, which grades a design gap-paired on the un-rearranged allele's unspliced transcript — a hybridisation test, not an establishment of cleavage — when it pairs the unspliced *NR4A3* sequence with no mismatch inside the catalytic gap, and `aso_noncoding_acceptor_designs.py` calls that same implementation at the exon-2 acceptor, where the register runs across the intron-1/exon-2 boundary instead of within intron 2.

What that yields is a candidate, not a validated reagent. Nothing designed this way has been synthesised or tested, the procedure has been run at the junctions reported here and at no others, and a design it returns is ordered by the ranking §4.4 states a falsification threshold for.

## 5 · Bounds on every claim, and the conditions for falsification

The conditions for falsification this heading names are stated with the experiment that would test the ranking (§4.4); what this section adds are the bounds under which any such test would be read. Seven of them sit under everything above, ordered by how much a decision would change if one of them bit. None alters the ranking by gap-level margin; each alters what a count taken from it means. The block that closes the section is not one of them: it records a released artefact that had no reading to give.

**Counts are lower bounds, and a zero is not exempt.** The alignment screen is heuristic and stores at most 50 hits per query, so 136 of the 183 filtered designs carry right-censored counts: 35 at the cap, 101 more past the 15 hits retained. Re-screening at ten times the ceiling raised the count for 164 of the 180 designs screened at both depths, 129 of which had never approached the cap, so reaching the cap is not what censors a count. Truncation is not the mechanism: of the 47 designs whose default hit list the pipeline records as complete, 37 returned more at the deeper ceiling and none returned fewer. One reporting 9 near-matches returned 34, one reporting 10 returned 110, and one reporting exactly the 15 hits retained — a list the pipeline marks uncensored — returned 204, with a second at that same boundary returning 374. Depth then moves the headline result: six of the nine clean designs lose the property at the deeper ceiling, three having returned no near-match at all at the default one (§2.4). Only those 47 of the 183 hit lists are short enough to assess for cleanliness, so nine is a floor over that subset and an over-count of it at once, and their own raw counts, zero to eight hits each, are not a measurement either. Retention alone withheld a verdict on seven further records — a different seven from the seven whose default-depth query failed at the remote service (§2.4) — and the deeper pass decided six of the seven; none of the six is clean, and the seventh re-screen did not return, so that record remains undecided. The nine are untouched by that test. BLAST's sensitivity at  $\geq 14/16$  is bounded rather than unquantified, and the bound is not one: at a word size of seven, two mismatches split a 16-mer into at most three segments whose longest is at least five, and 15 of the 120 two-mismatch configurations leave no exact seven-mer and cannot be seeded at all — about an eighth of the class, unreachable at any depth. So "no sense-strand near-match" is a property of this search and not of the transcriptome; the exhaustive transcript scan is complete for substitutions only within its one-mismatch budget, so it corroborates the exact and single-mismatch part of the claim, and no screen here establishes the absence of two-mismatch sense-strand near-matches.

**The parent threshold is stated, not measured.** The mature-parent counts — 87 of 190, 61 of them against wild-type *NR4A3* — are taken at a contiguous duplex of ten base pairs, a criterion adopted here rather than measured for this architecture, so each is a floor at that choice: at the seven-base-pair end of the same cited range the screen returns 175 of 190. Nor does the criterion transfer cleanly, ten being a whole-duplex length where the source counts RNA:DNA

nucleotides, which this architecture holds at six (§6). Nor does the rate transfer to a wider search space. Every count at this criterion, and every null rate beside it, is a rate over the 19,921 sixteen-nucleotide windows of six parent transcripts (20,011 nucleotides, 0.0028% of the windows of the transcript corpus screen 2 measured), and the criterion is saturating over that corpus: independent uniform bases predict about  $2.7 \times 10^3$  qualifying sites per 16-mer there, so an arbitrary 16-mer would meet it transcriptome-wide with probability one, as would a scramble and as would a chimera. What the screen measures is therefore self-liability rather than an off-target rate. Of the 87 liable designs, 85 are paired by one of their own two parent genes and the other two by a FET paralogue of their donor, and this work claims no transcriptome-wide rate, nor could one be taken from it. The threshold also sets how much of the liability the nulls account for, and ten is not where they account for least: the exon-terminus chimera reaches 40.6% against 45.8% observed, which is 88.6% of the observed rate and the third-lowest such share on the eight-cut ladder of §2.5. At four of those eight cuts, six, eight, nine and thirteen, the strongest null stands above the observed rate rather than below it, so the share the nulls account for is not monotone in the criterion and changes sign with it. One cut does leave a share to attribute to the reported breakpoints themselves, and it is not the cut reported here: at eleven the strongest null falls below the observed rate's interval rather than inside it, the only cut on the ladder at which the observed rate nominally stands clear of every null (§2.5).

**The patient may not carry these junctions.** Which exon pair a patient carries is not decidable from exon structure. The coverage figure of §4.1 is arithmetic over two published cohorts and not a measurement of anything — no patient was screened with any sequence named here, and §4.1 sets out the four bounds on it. It prices *TAF15* at 3 of 3 on a three-tumour series, while a functional study reports a second isoform at a cryptic exon within *NR4A3* intron 2 and calls the two "the two major *TAF15-NR4A3* isoforms detected in human tumors" without counting either,<sup>27</sup> so that arm's coverage is an upper bound and 68.4% is optimistic on it by an unmeasured amount; no design in the 38-junction panel crosses that seam (0 of 190; the designs §2.6 reports at it sit outside the panel), and the two isoforms share no sequence 3' of the breakpoint, so it is a different target and not a near-miss. The multi-partner result is conditional on *TAF15* and *FUS* breaking at the homologous exons, which is not established here, and the five partners are not the catalogue: *ACTB*,<sup>3</sup> and others are reported, and 2% of one cohort carried no identified partner.<sup>9</sup> Three of the 15 *EWSR1*-rearranged tumours of the primary breakpoint series carry transcript types the retrieved record does not name, so retrieval is an upper bound on what would open that block. At least one named variant is reported to arise from a genomic breakpoint interior to *EWSR1* exon 12 rather than between two exons, in a source that carries a citation marker on that sentence and is therefore restating an earlier report.<sup>36</sup> That marker, and the 1995 report of this fusion it resolves to — reference 1 here — were read from the paper's scanned page images through an optical-character layer, the only form of its full text obtained here, and are recorded with the retrieval in `lit-targets-aso-type3-designability.json`; two later attempts to re-fetch the full text returned no open-access copy, so the reading stands on that layer and is marked unverified beyond it. Such a breakpoint is not undesignable: handing the same builder a donor model cut inside its exon returns five builder outputs, all fusion-specific against both parents, three gap-centred and a best gap-level margin of 3. What

such a junction lacks is an exon index, which is how every design here is specified, and a published nucleotide position, which no retrieved source states, so it is out of reach for a stocked panel while remaining designable for a named patient whose breakpoint has been sequenced. How many tumours this accounts for is not established by any source.

**One geometry.** Every screened count here is for one architecture, a 16-mer at 5-6-5, except those §2.9 reports and the ones §4.2 and Table 5 carry over from it for the 5-8-5 control. Across those three panels the ten-base-pair criterion is not one criterion. It is an absolute hybrid length that does not scale with the gap, so a run counted at it is six RNA:DNA pairs plus four locked-wing pairs at 5-6-5, eight plus two at 5-8-5, and ten RNA:DNA pairs at 5-10-5, where every window pairing the whole gap clears it by construction. §2.9's liable count is flat at 87, 88 and 87 across a criterion whose RNA:DNA content rises from six to eight to ten, so that flatness is not a reading at a constant substrate. The genome scan is unavailable at 18 and 20 nucleotides by construction rather than merely unrun, so the nesting bound on a longer design's genome liability is a next step and not a result, and no RNase-H1 assay distinguishes these geometries here.

**Hybridisation, not cleavage, and not exposure.** All five screens address hybridisation-dependent liability only, and the free-energy calculation speaks to duplex formation rather than to cleavage. Within that they grade cleavage-competent hybridisation, so a liability that needs binding alone falls outside every one of them: an oligonucleotide occupying the wild-type *NR4A3* 3' splice acceptor — the intron-2/exon-3 boundary §2.5's pre-mRNA class concentrates at — can alter that transcript's splicing with no catalytic gap paired, and no screen here reads occupancy. It is unmeasured rather than absent. The headline parent counts require the catalytic gap paired in full, an inclusion criterion adopted because no retrieved measurement grades a partly-paired parent duplex; the class it excludes is counted separately, as the 21 designs of §2.5. Nothing here establishes that a matched gene is transcribed in the

organs a systemic dose reaches: 13 of the 46 loci returned no reading and carry 52 of those 649 hits, so there the exposure question is unanswered rather than answered negatively, and reference bulk medians describe a population's normal tissue rather than a dosed patient's organ (Table 6). Two liabilities of this chemistry are separated here because they are usually conflated. The phosphorothioate-class effects — protein binding, and toxicity that tracks backbone content rather than sequence — are not a function of any feature graded here. Hepatotoxicity is the second, and this paper grades neither. Nothing in this work establishes which mechanism drives it, and no screen here is a readout on it: the only pre-mRNA arm, screen 3, is bounded to six parent genes rather than to the transcriptome, while the genome-wide arm, screen 5, grades hybridisation and not cleavage. The affinity-linked component that the architecture (§6) would raise is not graded either.

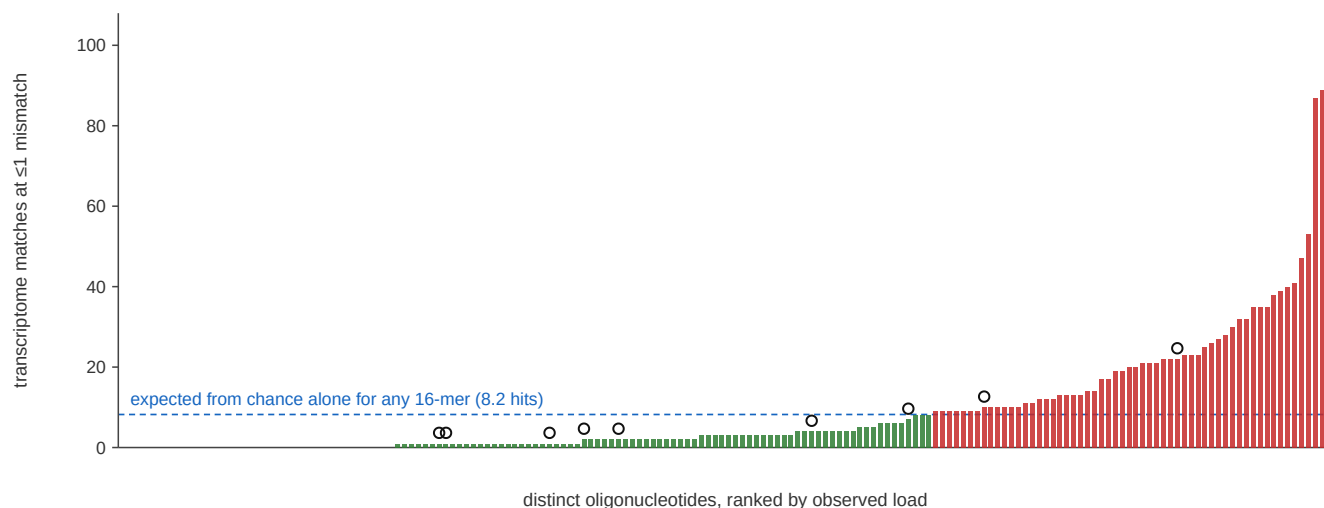
**The chance null is crude.** It assumes independent uniform bases, where real transcript sequence is composition-skewed and repetitive. An arbitrary position matches a given 16-mer at  $\geq 14/16$  with probability  $2.6 \times 10^{-7}$ , or 189 near-matches for any 16-mer whatever over the exhaustive transcript scan's measured span of 718,571,139 nucleotides across 186,185 transcripts — a figure the alignment screen's 50-hit cap cannot test at all. That span is screen 2's transcript corpus and is under a quarter of the  $3.10 \times 10^9$  nucleotides of assembly the genome scan covers, so the two chance figures this section quotes stand over different denominators and are not comparable to one another. The scan itself can, and on the mean it comes in at chance: at  $\leq 1$  mismatch the same span predicts 8.2 per design against an observed 9.2 over the 176 distinct oligonucleotides, a ratio of 1.12. The shape disagrees in both tails, 40 of the 176 returning no match where a Poisson null of that mean predicts fewer than one, and a right tail reaching 100 matches on one design (Supplementary Figure S1), so the null separates "more than chance" from "at chance" and nothing finer and does not license reading the zeros as cleanliness.



### Transcriptome load per distinct oligonucleotide against chance expectation

176 distinct oligonucleotides at real exon junctions, ranked; exact plus  $\leq 1$ -mismatch matches over 186,185 transcripts.

○ the 9 designs spanning two or three partners' seams, each one molecule plotted once.



One bar is one molecule and not one design record: the 190 design records at real exon junctions are 176 distinct oligonucleotides, because the 9 marked designs each span more than one partner's seam and enter the series once rather than once per seam. The chance expectation assumes independent uniform bases; the transcriptome span it is computed over is the exhaustive scan's measured one. It separates "more than chance" from "at chance" and is not a significance test. 118 of 176 bars fall at or below that line. Counts are predictions from sequence search, not measured off-target activity. Bars are green at or below the line and red above it, which repeats each bar's own height and carries no information of its own. 58 of the 176 plotted designs sit above it; on the mean the set comes in at chance — 9.2 observed against 8.2 expected, a ratio of 1.12, while the median is 3, so the excess is a long right tail rather than a shift of the whole set. 40 of the 176 bars are zero and therefore draw nothing at all, so 22.7% of the abscissa is blank. The series begins at the left edge, not where the first visible bar is. Not plotted: 10 designs at 2 modelled breakpoints rather than real exon junctions; 4 of those sit above it, at 16–95 matches and 75.0–81.2% GC.

**Supplementary Figure S1. Transcriptome load per molecule against chance expectation.** Per molecule, not per design record: the 190 records are 176 distinct oligonucleotides and each is plotted once, so this series is 176 bars and not 190. Bars are green at or below the chance line and red above it, which is redundant with each bar's height against that line and carries no information of its own. Each bar is one distinct oligonucleotide's count of exact plus  $\leq 1$ -mismatch matches over 186,185 transcripts, ranked. The 190 design records at real exon junctions collapse to 176 molecules, because nine of the 16-mers are junction-spanning at more than one partner's junction at once — five at three junctions and four at two — and each of those is one physical oligonucleotide, plotted once rather than repeatedly (marked). The line is the number of such matches expected for an arbitrary 16-mer under an independent-uniform-base null, 8.2, computed against the scan's measured 718,571,139-nucleotide span; 118 of the 176 fall at or below it and 58 exceed it. Ten further designs from modelled breakpoints not built from a spliced transcript model are excluded, and are released with the artefacts; four of those ten carry loads ABOVE the chance line, at 16 to 95 matches, so the excluded set is not a set of quiet designs and the panel understates the corpus rather than flattering it. Forty of the 176 return no match at all; their bars have zero height and so draw nothing, which is why the left of the panel reads as empty rather than as the start of the series. It is an expected count: the observed mean is 9.2, a ratio of 1.12, while the median is 3, so real transcript sequence produces a long right tail the null cannot rather than a uniform shift away from it. The line separates "more than chance" from "at chance" and is not a significance test; the counts are predictions from sequence search, not measured off-target activity.

**Records are not genes, and one assembly is not a genome.** RefSeq carries one accession per annotated variant, so a match to a constitutive exon is counted once per variant: over the 44 designs of the 38 junction screens whose hit lists permit a locus recount the median inflation is 2.25 records per locus and the maximum 11.0 at the default search depth, more than doubling the apparent number of distinct genes. Tables 2 to 4 are built at the deeper ceiling, where the same recount gives a median of 5.52 and a maximum of 28.0, so the tables' locus counts are inflated by more than this paragraph's figures and not less. Where a load sits matters as much as its size: NM\_/NR\_ records are curated, XM\_/XR\_ computationally predicted. Both compound on 5'-GGGCATATCATCAAAC-3', whose 123 gap-paired near-matches at the deeper ceiling recount to six loci: 82 of the 123 are predicted models and the other 41 are curated records, and those 41 are themselves inflated, 32 of them *ANKS1B* accessions and three *ZNF667*. The curated share moves with depth as well. At the default ceiling that design carried a single curated sense-strand hit, *H2AP* (NM\_012274), mismatched inside the catalytic gap and so counted in full under the pessimistic bound; at the deeper ceiling it carries 43 curated sense-strand hits, 41 of them gap-paired. The genome scan decides sense-strand membership against an annotation, so unannotated transcription is uncounted; it is one

assembly, silent on a patient's private variation; and every exhaustive screen here is complete for substitutions by construction and blind to insertions and deletions.

**An earlier genome-wide attempt is released, is uninterpretable, and is not screen 5.** Before the exhaustive scan was built, one genome-wide query per design was run against a public mixed corpus of assemblies, genomic clones, patent sequence and transcripts rather than against a genome reference. Its output is released with the artefacts, as `aso-premRNA-offtarget-genomic.json`, so that nobody repeats it. It could not have yielded an interpretable number under any outcome, for two structural reasons. That corpus has no defined nucleotide span, so no chance expectation can be formed against it, and at this threshold an off-target count that cannot be referenced to chance is not a reading at all. And its retrieval ceiling sat far below what chance alone predicts, so every query that returned came back at or above the ceiling: of nine designs queried, one failed at the remote service, seven returned exactly 50 records and the eighth 52.

The per-design rows in that file therefore look like off-target findings and are none. Of its 402 retained rows, six are exact 16-of-16 matches, four of them to genomic clone records and two to annotation records on chromosome 6, and 55 are immunoglobulin



variable-region records, every one of them at 14 or fewer of 16. A corpus of this kind carries one stretch of sequence once per record that contains it, so those rows are largely redundant copies of one another, and there is no denominator against which to say whether any of them is more than arithmetic. No count, ratio, load or cleanliness statement in this paper is taken from that file. Nor does it bear on screen 5, which is a different object that shares the word genome-wide: screen 5 is the exhaustive GRCh38 scan of §2.7 and §6, whose expectations are computed from the 2,948,609,696 windows it actually searched over a measured  $3.10 \times 10^9$  nucleotides of assembly, and which caps nothing at scan time. The earlier attempt is retained as the record of an instrument that had no reading to give, and it should be read as nothing else.

## 6 • Methods

**Transcript models.** Canonical transcripts for the five partner genes and for *NR4A3* were obtained from Ensembl.<sup>40</sup> Each model was self-checked before use: exon lengths must sum to the spliced cDNA, the coding sequence (CDS) must occur exactly once within it, and translation of the CDS must reproduce the annotated protein. Per-exon coding content was additionally cross-checked against an independent exon audit for *EWSR1* and *NR4A3*; for the other four partners that audit does not exist, and the weaker check is recorded per gene in the released artefacts. Every exon number, coordinate and length in this paper is relative to one specific model per gene, and the canonical transcript of a gene can change between Ensembl releases, so the accessions are given here rather than left to the artefacts: ENST00000397938 (*EWSR1*), ENST00000605844 (*TAF15*), ENST00000333725 (*TCF12*), ENST00000254108 (*FUS*), ENST00000240851 (*TFG*), ENST00000395097 (*NR4A3*) and ENST00000325455 (*PGR*), the last for the single seam of §2.6 that lies outside the five modelled partners. Each is the Ensembl canonical transcript for its gene at the release read, which is what the pipeline selects; no MANE status was checked for any of them here. Two things this argument needs were not captured and cannot be recovered from the artefacts: the Ensembl release, and the version suffix of each accession. The transcript-model record carries the instant of its fetch, 2026-08-12T13:41:57Z, and the REST endpoint it used; the unspliced-sequence cache carries neither a release nor a date. An unversioned accession under an unstated release does not by itself pin sequence, so what pins it here is the committed sequence caches, which travel with the archive and are what a reproducer should compare a refetch against rather than assume agreement with. The release stated for the genome scan below governs that scan's annotation and is not evidence about these models.

Exon numbers are transcript exon indices, counted from the transcript 5' end of the model named and including non-coding exons; they are not coding-exon indices. The distinction is not cosmetic, and it is the axis on which an earlier version of this work was withdrawn: *TCF12* carries 21 transcript exons and 19 coding, *TFG* eight and seven, *NR4A3* eight and six. The count difference is not the index shift, and for *TCF12* the two differ: its two non-coding exons are the first and the last, so only one of them precedes the coding sequence and the index shifts by one rather than by two. *TCF12* transcript exon 5 is coding exon 4 under the other convention, with a different 3' terminus; *TFG* shifts by one and *NR4A3* by two, their non-coding exons all preceding the coding

sequence. *EWSR1*, *TAF15*, *FUS* and *PGR* have a coding first exon and the two conventions coincide for them.

**Chimera construction.** Chimeras were built from transcript sequence rather than by joining coding sequences. A fusion keeps the whole *NR4A3* acceptor exon, so any bases of that exon lying ahead of the *NR4A3* start codon are still present in the fusion transcript, and they are the first bases an oligonucleotide meets on the *NR4A3* side of the junction. At the exon-3 acceptor, the only one the 38-junction panel yields designs at — §2.6 designs and screens at *NR4A3* exon 2 and at the intron-2 cryptic exon besides — there are two. Joining coding sequences alone would omit them, shifting every design by two positions. A pair of exons is *in frame* when the partner's coding bases, plus those retained bases, sum to a multiple of three. Every declared exon pair was graded by that rule before any design was emitted, and only the in-frame pairs were carried forward, since only those describe a fusion that could exist.

**Design.** Junction-spanning 16-mer gapmers were tiled in a 5-6-5 β-D-oxy-LNA/DNA/β-D-oxy-LNA architecture, which is the chemistry the design rules below assume. Every one of the 15 internucleoside linkages is a phosphorothioate — wings and gap alike, not the wing-only variant — and that backbone is specified as stereorandom, the ordinary case: a 16-mer carries 15 chiral phosphorus centres and is made as a mixture. The bicycle is named because α-L-LNA, 2'-amino-LNA, ENA and cEt are also sold as locked analogues and behave differently in a gapmer wing. Nucleobase modification is specified here rather than left to a vendor default, because two suppliers filling the same base string with different conventions would ship two molecules: locked cytosines are 5-methylcytosine, and the cytosines of the DNA gap are unmethylated 2'-deoxycytidine. The second half of that is what makes §2.10's CpG audit read as written — 5-methylation of a gap cytosine removes the motif while leaving the base pair, the target window and the gap-level margin untouched, so a paper that left the state open would be auditing a liability the reader could already have removed at no cost. Neither choice is modelled by the free energies below, which are computed for an unmodified DNA:RNA hybrid. Termini are free 5'-hydroxyl and 3'-hydroxyl, with no cap and no 5' phosphate, and the sodium salt is intended. Backbone stereochemistry is not modelled anywhere in this work, and nothing here distinguishes one diastereomer from another — which bears on §4.2's gap-length arm, since a six-nucleotide gap offers few cleavage positions to redistribute among. Each way of sliding that 16-mer along the transcript is a *register*, and only registers placing the junction inside the six-nucleotide DNA gap were retained, since RNase-H1 cleaves within the DNA:RNA duplex of the gap and needs a minimum run of contiguous DNA to do so.

The gap length is a compromise and is treated as one. Reported minima for that run are five to six nucleotides — at least five for cleavage to occur,<sup>41</sup> or a DNA segment of "six or more bases" to activate the enzyme<sup>42</sup> — and for LNA/DNA/LNA gapmers specifically a six-nucleotide gap gives noteworthy but incomplete activity, with seven to ten reported as optimal.<sup>39</sup> None of the three figures is a titration in this architecture, and SI §S3 gives the provenance of each. Six therefore sits at the short end of the usable range and below the reported optimum. Each wing is five contiguous locked residues. That is a high locked-residue content for a 16-mer, against the two to four locked residues per wing this work takes to be usual; and the 5-6-5, 5-8-5 and 5-10-5 ladder tiled here is adopted

from the geometries commonly reported for 2'-O-methoxyethyl gapmers rather than for locked ones. No source was retrieved into this work for either statement. Both are premises adopted to justify a specification, neither is evidence for any result here, and the three citations in this paragraph are attached to the gap-length figures above and not to either of them. The geometries were held fixed so that gap length is the only variable across the three panels, which is what §2.9 measures; the consequence is that these are not conventional locked-nucleic-acid reagents, that their matched-duplex melting temperatures are correspondingly high, and that this work takes high affinity to carry a risk of sequence-dependent hepatotoxicity. That last is a premise adopted here and not a retrieved finding: no source for it was retrieved into this work. It is stated because it bears on any decision to synthesise, and none of it is graded here. It was retained as the panel's fixed baseline rather than as an optimum, and the two longer geometries were tiled and screened beside it precisely so the trade it makes is measured rather than assumed (§2.9, Table 7). The five junction-spanning registers per junction it admits are a consequence of that gap length, not a reason for it: 5-8-5 and 5-10-5 admit seven and nine. No claim is made that a short gap improves fusion-versus-parent discrimination: one series that shortened a 5-10-5 gapmer to 5-6-5 reported lower off-target knockdown but also lower on-target activity and lower allele selectivity.<sup>41</sup> Within that same series 5-8-5 was the one shortened design reported to give a small increase in activity or allele selectivity in some cases, and it also increased off-target knockdown relative to 5-10-5 for several of the genes tested, so the exception is not a free one. Those gapmers carry thiomorpholino rather than LNA wings and are directed at a single-nucleotide polymorphism distinguishing two alleles rather than at a fusion junction, so neither the rule nor its exception is evidence about gap length in this architecture. Because that trade is the modality's central one, 5-8-5 and 5-10-5 were tiled over the same junctions by the same rule and carried through four of the five screens below — the exhaustive genome scan is unavailable at 18 and 20 nucleotides by construction, as §5 and Table 7 both state — wings held at five nucleotides so that only the gap changed and LNA affinity enters every parent duplex identically (§2.9).

**Ranking.** What separates the fusion from a parent is the junction-unique bases inside the gap, not identity across the whole oligonucleotide, because the gap is where the enzyme cuts. Designs were therefore ranked by their *gap-level margin*: the junction-unique bases inside the gap on the shorter side of the junction. That is the panel-level statistic, which compares designs across junctions. Selecting within one junction is a different question, and Table 2 — from which the reagents of §4.1 are taken — orders designs by parent liability first, then pre-mRNA sites, then distinct gene loci, with the margin breaking ties. Each candidate was screened against all six parent transcripts rather than the two of its own fusion, because a design's liability need not fall on its own two parents: §5 records the designs a FET paralogue of their donor pairs instead.

**Specificity screening.** Five screens were applied. Each is named below and referred to by that name throughout, because each reaches a compartment the others cannot and each is blind to something another catches. No single screen supports any claim here on its own.

**1. The alignment screen.** Each target window was queried against human RefSeq RNA<sup>43</sup> through the NCBI BLAST URL

interface<sup>44</sup> rather than a local installation, so the database is a live service rather than a pinned snapshot. When it was queried is not recorded either: no screen artefact released here carries a run date, a date range or a database build identifier, so the divergence between the counts reported here and what the same query returns today cannot be bounded from the record. Each screen stores the request identifier the service returned, which is not a date. The parameters decide what a ≥14/16 screen can return and are therefore given in full: `PROGRAM=blastn, DATABASE=refseq_rna, WORD_SIZE=7, EXPECT=1000, MEGABLAST=off, FILTER=F, ENTREZ_QUERY=txid9606[ORGN]`, and a hitlist of 50 per query raised to 500 on the deeper pass, with ≥14/16 identity applied on the returned alignments. That 500 is not on the same footing as the rest of the list: it is stored in a parameters block for the longer-geometry screens, while for the 38 panel screens it is inferred from retention exceeding what a default run can produce rather than read back from the request. What does not depend on the inference is that no deep hit list was truncated — the largest holds 374 records against a ceiling of 500, and no design's list is recorded as incomplete. A transcript window matching a design at 14 or more of its 16 positions is a *near-match*, classified by whether the six-nucleotide gap is itself base-paired: one that pairs the gap is *gap-paired* — the paper uses that one term throughout, and a sense-strand gap-paired near-match is what it calls a gap-paired sense-strand match — and RNase-H1 could cleave there; one pairing only the wings could not. That definition is written in substitutions, but `blastn` returns gapped alignments and an identity filter does not reject them, so this screen alone — unlike screens 2 to 5, which are substitution-only by construction — admits a near-match carrying an insertion or a deletion. Across every 5-6-5 alignment screen released here 110 retained alignments carry an indel and 28 of those are counted as gap-paired sense-strand matches. Admitting them can only raise a design's count, so no cleanliness statement rests on their exclusion; the point is that this screen's counts are not substitution-only and should not be read as though they were. This is a heuristic search retaining only a limited number of hits per query, so every count it yields is a lower bound — an effect §5 measures. Records of a design's own two parent genes, its donor and NR4A3, matched by the symbol and the aliases each screen records in its parameters, are counted separately and excluded from this screen's near-match counts, since each of those two pairs one wing by construction and would otherwise dominate the list; the parents are assessed instead by the gap-level margin, by screen 4 and — for their unspliced sequence — by screen 3, whose own near-match counts are counts of parent sites and are reported as such (§2.5). The exclusion is not taken over the six parent genes of the panel: a record of a partner gene that is not the design's own donor is retained, because engagement of a gene the fusion does not involve is a specificity finding at that junction rather than a parent pairing. The distinction is measured and not hypothetical. 5'-GCATATCTGAATACTG-3' at *TFG* exon 7 retains nine near-matches to wild-type *FUS* in its deeper re-screen, every one of them gap-paired on the sense strand. A substitution-only gap-paired near-match at 14 of 16 carries both its mismatches in the locked wings, since the six gap positions are paired by definition; the gapped alignments named above are the exception, their two non-identical positions not being substitutions in a wing at all. A

mismatch opposite a locked residue is more destabilising than one opposite DNA, and no locked-nucleotide mismatch penalty is applied anywhere in this work. These counts are therefore an upper bound on hybridisation as well as a lower bound on search depth, and the architecture is invoked for the free-energy margin (§6, duplex thermodynamics) while being left out of the off-target counts. Both directions are stated rather than composed.

2. **The exhaustive transcript scan.** A seed-and-extend scan searched 186,185 transcripts (GRCh38.p14) for exact and  $\leq 1$ -mismatch matches. It is complete for substitutions by construction, reads the sense orientation only, and does not detect insertions or deletions.
3. **The pre-mRNA screen.** Screens 1 and 2 search mature transcript, so a third covers the nuclear compartment they cannot reach. Unspliced sequence and exon coordinates for all six parents were retrieved from Ensembl, and every target window was scanned against them in both orientations at  $\leq 2$  mismatches. That is the threshold the alignment screen admits, which keeps the two comparable: a stricter one here would return a cleaner pre-mRNA result for that reason alone. This arm is exhaustive for substitutions, seeded on three blocks of the 16-mer so that a hit within the threshold must match one block exactly. Each hit is classified as wholly intronic, wholly exonic, or spanning an intron–exon boundary, since only the exonic class could have been visible to a transcript screen.
4. **The mature-parent screen.** This addresses the parent transcripts screen 1 excludes, in the compartment screen 3 cannot reach. Mature parent transcripts were spliced from the same Ensembl records, and every target window was compared to every 16-nucleotide window of all six — *EWSR1*, *FUS*, *NR4A3*, *TAF15*, *TCF12* and *TFG*; *PGR* lies outside this set — forward orientation only. The query is the target window, which is the reverse complement of the antisense oligonucleotide the tables print. A window counts only if all six gap positions are paired. Its size is the longest contiguous run of perfect pairing containing the whole gap. It is not the duplex the enzyme acts on: exactly six of its ten to thirteen base pairs are the RNA:DNA gap, and the rest are LNA:RNA wing pairs. That RNase-H1 does not cleave those pairs is why a gapmer has a gap at all: no RNase-H-mediated cleavage was observed for a fully modified 11-mer locked oligonucleotide or an 11-mer locked/DNA mixmer, where an LNA/DNA/LNA gapmer with a six-nucleotide DNA gap elicited substantial activity.<sup>39</sup> That the catalytic footprint reaches past the DNA stretch is retrieved too: a footprint model places sugar-modified, non-DNA residues at footprint positions flanking a five-base DNA gap and reports cleavage retained there.<sup>45</sup> Whether the hybrid-binding domain reads a locked pair as hybrid is a third statement and is not one of these: no measurement retrieved into this work bounds that domain's affinity for a locked duplex, so that it does not is a premise adopted here, and the ten-base-pair threshold below is stated on it. The caveat below states the consequence. Runs shorter than ten base pairs are not treated as plausible substrates. That is a stated threshold, not a measured one, so every design's longest run is released. Ten is the strict end of a figure its source gives as a possible explanation of its own observation — "this could be because RNase H1 requires a minimum length of 7 to 10 RNA:DNA hybridized nucleotides to bind with its hybrid binding domain" — rather than as a

measured minimum.<sup>46</sup> The qualifier does not transfer cleanly: the run counted here is the whole contiguous duplex, of which exactly six nucleotides are the RNA:DNA pairs of the gap and the rest are LNA:RNA wing pairs the source's wording would not count. Ten here is therefore a total-duplex length, while the source's 7 to 10 is a count of RNA:DNA nucleotides that this architecture holds at six in every design, below the range the source states; what relation the one criterion bears to the other is not established here. The count it produces is a floor at that choice, and what the choice costs is stated beside the count in §2.5.

**The null for this screen.** Bounded by six transcripts, this screen has no chance expectation of its own, and a count against six transcripts is not interpretable without one. Six ensembles of arbitrary 16-mers were therefore pushed through the identical screen, 200 draws per design in each: each design's target window shuffled to preserve base composition; shuffled to preserve dinucleotide composition by an Eulerian-path shuffle; drawn from uniform bases; drawn from the panel's pooled base composition; and two arms holding either the catalytic gap or the wings fixed while shuffling the other. A seventh is structural rather than a chance null: a random window of a real donor parent joined to a random window of real *NR4A3* at one of the junction offsets the panel's own registers use, which keeps the whole design rule and randomises only where in each transcript the two pieces come from. Three further structural arms, on the same 200 draws per design, hold the exon boundaries fixed as well, because a real design's halves do not sit at arbitrary interior positions: the donor half ends at an exon 3' terminus and the acceptor half begins at an exon 5' terminus. Exon boundaries were taken in mature coordinates by cumulative exon length from the same committed record the transcripts are spliced from, and the first and last boundaries of each transcript were excluded, being its ends rather than splice sites. The arms are: the donor half drawn to end at a real donor exon terminus with the acceptor half left at a uniform interior window; both halves drawn at real exon termini; and the same with the *NR4A3* exon-3 acceptor that every junction tiled here uses excluded from the draw altogether, so that arm cannot sample the disease's own acceptor even by chance. §2.5 reports all three. Proportions carry Wilson 95% intervals. The pseudo-random stream is a splitmix64 written out in the released module rather than taken from the interpreter's own, and it is seeded at 20260815, each ensemble drawing its own stream from that seed and its own name so that adding or removing an ensemble cannot shift the draws of the ones either side of it. Those two facts are what make the artefact reproducible bit for bit, and the artefact records both. The module is `aso_parent_null.py` and its output `aso-parent-null.json`; the screen it re-runs is `aso_parent_gap_pairing.py` unchanged, so between the observed arm and every null only the query differs. §2.5 reports the result.

5. **The genome scan.** Screens 1 to 4 are bounded either by an annotation or by six transcripts. The fifth removes that SEARCH-SPACE bound, and only for the pre-mRNA class: it runs at two mismatches, and a contiguous run of eleven or twelve base pairs inside a 16-mer leaves five or four positions unpaired respectively, so the mature-parent duplex class of §2.5 stays bounded by the same six transcripts it was bounded by before. A reader who takes this bullet alone gets a wider clearance than the screen supplies (§2.7). Each distinct target window and its reverse complement were placed in one membership set and



every plus-strand position probed once, which covers both orientations without a second pass, at  $\leq 2$  mismatches, exhaustively: windows containing an N were excluded, leaving 2,948,609,696 of 3,099,747,808 over a measured  $3.10 \times 10^9$  nucleotides, with no seed, no word size and therefore no search sensitivity to quantify. The assembly is `Homo_sapiens.GRCh38.dna_sm.primary_assembly.fa.gz` and the annotation that decides every strand-agreement and gene call is Ensembl release 116; the primary assembly matters, because the full GRCh38 with alternate contigs would not reproduce this denominator. §2.7 reports it.

**The un-rearranged-allele scan.** A sixth instrument sits outside those five and is scoped to the seams where it can bite. Screens 1 to 5 ask what else a design can pair; this one asks whether a design pairs the patient's own un-rearranged *NR4A3* allele, which can only arise where a design's acceptor half is *NR4A3* sequence the wild-type allele presents in the same register: the exon-2 and intron-2 cryptic-exon seams of §2.6, not the exon-3 acceptor the 38-junction panel uses. Each target window was compared to every window of the committed *NR4A3* unspliced sequence, the same cache screen 3 reads, in the forward orientation only, which is screen 4's convention rather than screen 3's. A site is retained at two or fewer mismatches over the whole 16-mer and is called cleavage-competent only where none of those mismatches falls within the six gap positions, which are positions 6 to 11 of the target window. Where the cache is absent, or where the cryptic exon is not a substring of it, the scan refuses and records an absent reading rather than a clean one. This is one locus and not a genome-wide result; §2.6 reports it, and three designs are condemned by it and by nothing else, which is why they carry a flag no parent-screen column accounts for.

**Strand orientation.** A match matters only if an antisense oligonucleotide could base-pair with it, which means the sense strand; a window carrying the reverse complement is not a liability at all. `blastn` searches both strands, so such a hit passes an identity filter unless orientation is parsed, and screens produced before that parsing was added recorded them as gap-paired sense-strand matches. Orientation is now parsed and filtered in all 38 junction screens and the 183 designs they hold, and therefore in every cleanliness statement made here. Only two released screens are unfiltered, and neither carries a junction from the 38-junction panel or supports a claim here (SI §S5). The same rule governs pre-mRNA, which is transcribed in transcript orientation: a forward match can be base-paired and a reverse-complement match cannot.

A design is called clean where it carries no sense-strand near-match, and that is always a statement about a complete hit list at a stated search depth. Both qualifications matter. A hit list the cap truncated is not complete, so no verdict is available for that design, and a design clean at one depth need not be clean at another. §2.4 and §5 report both effects.

**Target-site accessibility.** Accessibility was estimated as the mean unpaired probability over a local fold of up to 180 nucleotides, computed with the ViennaRNA partition function,<sup>47</sup> and it spans 0.160 to 0.707 across all 190 designs at real exon junctions, with a median of 0.477. The window is not a fixed 180: it spans the union of a junction's candidate target windows plus 80 nucleotides either side, on the modelled fusion transcript, clipped at the transcript ends. The partition function was called at the package's own defaults, and no version, temperature or energy-parameter set was captured for it,

so the released values are reproducible only against the same installation. It is released with the artefacts and ranks nothing here. That omission is deliberate — accessibility bears on potency, which is not claimed for any sequence, rather than on the discrimination this work is about — and SI §S1 gives the three reasons in full.

**Expression of the off-target loci.** No screen above says whether a matched gene is transcribed in the organs a systemic dose reaches. For four of the five junctions of the 38-junction panel with a published exon-resolved breakpoint — those Table 6 covers — the gene loci their deeper screens return in the gap-paired class were read against GTEx v8 median TPM.<sup>48</sup> No such reading was taken at *TFG* exon 7, so that junction carries no expression reading rather than a negative one. The readings are in two blocks, reported separately and never combined, against two cuts used for legibility rather than as thresholds of concern: a lower cut of 1 TPM, below which a reading is taken as below detection, and an upper cut of 10 TPM. The first block is liver and both kidney compartments, the organs a systemically dosed phosphorothioate gapmer distributes to — a premise taken from the chemistry, for which no measurement or citation was retrieved here; the second is six soft-tissue types: skeletal muscle, subcutaneous adipose, tibial nerve, cultured fibroblasts, tibial artery and sun-exposed skin of the lower leg, standing in for the compartment EMC arises in, since no atlas contains the tumour itself. Both blocks are read from the same v8 gene-level release, `GTEx_Analysis_2017-06-05_v8_RNASeQCv1.1.9_gene_median_tpm`, whose cells are medians across a tissue's donors and are gene-level rather than transcript-level. NCBI Gene supplied locus identity, so a locus with no reading is attributed rather than left blank, and the Human Protein Atlas<sup>49</sup> was read as a transport check only, its consensus incorporating GTEx rather than confirming it independently.

**Discrimination model.** The binary assumption that any mismatch inside the gap abolishes cleavage is not supported by the primary literature and is not used for any claim of cleanliness. The field's general figure for single-nucleotide discrimination by a gapmer carrying no positional modification in its gap is approximately five-fold, given in its source as prior work restated rather than measured there,<sup>33</sup> and at 16-mer length one study reports no efficient discrimination at all.<sup>50</sup> Both underlying measurements are against a single-nucleotide substitution rather than a fusion junction, and the pessimistic one used unmodified antisense DNA, so they are used here as bounds for unmodified chemistry rather than as a property of this architecture. Neither is resolved at a gap of six. The five-fold figure carries no geometry at all, being prior work restated rather than measured where it is stated, and the pessimistic one has no gap, an unmodified antisense 16-mer presenting a sixteen-nucleotide DNA:RNA hybrid. Both are applied as a scalar per gap mismatch, so the re-score treats a mismatch at the gap's edge and one at its centre alike, where the surviving contiguous run does not — five nucleotides for a mismatch at the first of six gap positions against three for one at the third (§2.9). The classification records how many gap positions mismatch and not which, so no count reported here is positional; the alignment strings are released, so such a re-score is possible and has not been done. Gapmer-specific work points the same way, which is why the bounds are not narrowed: across more than 120 gapmers spanning five single-nucleotide changes, three achieved preferential cleavage of the mutant allele both in vitro and in cells, two at one substitution and one at another, while several more did so in vitro only,<sup>51</sup> and in the one retrieved campaign that

achieves allele selectivity on purpose it is engineered by a chemical modification at a gap position that restricts cleavage of the near-match, rather than obtained from the mismatch itself: a single 2'-O-methyl at gap position 2, in locked and 2'-O-methoxyethyl gapmers directed at one substitution in one gene, and reported by its own authors as preliminary proof of concept.<sup>52</sup> That is one gene, one substitution and one modified position, and it is not evidence about a fusion junction.

Every screen that resolves the gap was therefore re-scored under both bounds as a graded residual cleavage load, holding the hit set fixed so that only the scoring changed: all 38 junction screens, and 39 of the 93 screens released in total (SI §S4). Two distinct bounds follow, and they run in opposite directions. Where a hit list is truncated, the strand of the remainder is unrecoverable, so some designs keep a strand-blind count; that over-counts liability, because it includes matches no antisense oligonucleotide can hybridise, and is an upper bound. The same truncation also means fewer hits are recorded than the search returned, so the count of hits is itself right-censored: a lower bound on how many exist. Each design records which bounds apply to it.

**Duplex thermodynamics.** A base count is a proxy for discrimination, and a free energy is the field's standard instrument for it, so each design was also scored thermodynamically. A junction gapmer is a perfect complement of the fusion across all 16 positions, while a parent transcript can pair only the half of the oligonucleotide it contributes. The comparison is therefore the full 16-mer duplex against the donor-side and acceptor-side runs alone. Nearest-neighbour enthalpies and entropies for a DNA:RNA hybrid were taken from Sugimoto and colleagues,<sup>53</sup> as packaged in `Bio.SeqUtils.MeltingTemp.R_DNA_NN1` (Biopython 1.88) and read from the installed package rather than entered by hand, and  $\Delta G^\circ_{37}$  computed as  $\Delta H^\circ - T\Delta S^\circ$  at  $T = 310.15$  K, with that table's own initiation term included in the sum and  $\Delta S^\circ$  converted from cal to kcal. Those four specifications decide the absolute free energies below; the parameter values alone do not. The arithmetic was checked against an independent implementation, which agreed exactly; that check verifies the summation and not the choice of strand, and the strand concentration it uses enters no reported free energy (SI §S2). No salt correction was applied. A nearest-neighbour set is determined at the ionic strength of the measurements behind it rather than at a physiological one, and the correction is length-dependent, so it does not cancel in a  $\Delta\Delta G^\circ_{37}$  taken between a 16-base-pair duplex and the seven- to ten-base-pair parent run it is measured against. That term is unstated in the artefact and uncorrected here, and it is one of at least three omissions running against the floor direction stated below rather than with it. The other two are chemistry this work specifies and does not model. The backbone is a phosphorothioate throughout, wings and gap alike, and no phosphorothioate parameters were applied, so a substitution carried by every linkage of both duplexes is absent from a comparison that sets 16 base pairs against seven to ten and therefore does not cancel between them. And a parent seam is not the isolated short duplex the calculation treats it as: the rest of the same molecule, five locked residues among them, stays covalently attached and stacks on that seam, where the fusion duplex has no unpaired remainder to stack. Both are taken here to narrow  $\Delta\Delta G^\circ_{37}$  rather than widen it, and both directions are premises adopted for this work rather than measurements: no source retrieved into it bounds either term.

These designs carry LNA wings and the table is for an unmodified hybrid, so what is computed is the duplex the DNA backbone would form. Because the junction lies inside the gap, each parent pairs one of the two five-nucleotide LNA wings while the fusion pairs both, so LNA should widen the free-energy difference rather than narrow it, and every reported  $\Delta\Delta G^\circ_{37}$  is a floor in that sense. That direction follows from the architecture and was not computed: no LNA parameters were applied.

That floor is a statement about the difference and not about selectivity, and it must not be read as one. What decides whether a parent duplex forms in a cell is its absolute stability at the exposure concentration, not its distance from the fusion duplex. Five contiguous locked residues per wing raise duplex melting temperature substantially, and the same modification that widens the difference therefore also stabilises the parent seam it is measured against. Every one of the 38 top-margin designs divides its 16-mer eight and eight, and every one of them carries the same acceptor-side seam — the shared *NR4A3* exon-3 octamer — at  $-7.77$  kcal/mol as an unmodified hybrid. That figure is therefore a structural floor of the class rather than one weak outlier in it, and it is the more stable of the design's two seams, the one the free-energy margin is measured against, for 25 of the 38. Within the top-margin class the best parent seam runs from  $-7.77$  to  $-12.60$ ; the 190-design panel across all three margins runs to  $-17.51$  with a median of  $-8.66$ , which is a wider distribution but a different denominator. That median coincides numerically with a different quantity §2.9 and §4.2 print, the lead reagent's own best parent seam at 5-8-5; the two are unrelated and the repetition of  $-8.66$  is arithmetic coincidence rather than one cell copied twice. A seam that weak melts far below  $37^\circ\text{C}$  at the strand concentrations these parameters are defined at, which is not the same as saying it is unoccupied at the concentration a cell sees, and it is exactly this class that LNA wings would push toward occupancy. No modified-duplex melting temperature or dissociation constant was computed for any parent seam, so discrimination at the binding step is not resolved here — it is uncomputed, and an occupied but catalytically incompetent parent duplex is the steric-block liability §3 names as unevaluated.

**The four design rules.** Every design was separately audited against four rules commonly applied in antisense design and adopted here without a retrieved source: GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, and no CpG dinucleotide. The set is this work's own, declared in the released module that computes it, and no design-guidance source was retrieved for any of the four; §2.10's disagreement is therefore between this rule set and the gap-level margin, and not between the field's convention and either. The quadruplex rule tests for four separate runs of two or more guanines, which is what makes it silent on this panel (§2.10). The audit is not there to grade the designs, but to ask whether conventional triage and the gap-level margin would select the same molecules.

**Sequences. Do not order an oligonucleotide by copying it out of this PDF.** Every sequence named here travels with the archive as `fusion-junction-aso-sequences.csv` and `fusion-junction-aso-sequences.fasta`, both under `research/manuscripts/asos/` in the repository named above. They are the canonical record: generated from the same artefacts as the tables, carrying each design's geometry, junction and gap-level margin, and flagging every design this paper names as one not to be



carried forward. A design's `junction` is one of its junctions and not all of them — nine of the 16-mers span two or three, which is why 190 design records are 176 distinct molecules — so `also_tiled_at_junctions` carries the rest and both columns must be searched before concluding that a junction has no reagent.

**Which column to select on, because the obvious one is wrong.** The record's `role` column carries the paper's own answer: `best available at this junction` marks the design §4 would name there. Ranking by gap-level margin instead — the only ranking this paper states, and the one a reader naturally reaches for — is not the selection rule and does not reproduce it. The file keys a row to 43 distinct junctions, 42 of which carry a 5-6-5 register; over those 42 the top-margin register is a design the file condemns at eight. At three of those every register is condemned and no rule could do better, so what the margin rule actually costs a reader is a clean design at the other five, where one was available and it picked a design pairing a wild-type parent through the whole catalytic gap — four of those against wild-type NR4A3. Those rows carry `do_not_order`, as does every one of the 249 records whose `mature_parent_duplex_through_gap_bp` reaches the criterion applied throughout; three further records carry the flag for the separate un-rearranged-allele reason of §2.6, so 252 of the 780 carry it in all. An empty `do_not_order` is not a clearance: the flag is set at ten base pairs, while 175 of the 190 panel designs pair a parent through the whole gap at seven and 181 do so at any length (§2.9). A typeset table cell is not a machine-readable record — whether a sequence and the column beside it stay separate on extraction is a property of the reader's software — and the sequences here are 16 to 20 bases in which a single substitution changes what the molecule does. The bases alone are also not the reagent: the geometry column denotes locked-nucleic-acid wings around a DNA gap on a phosphorothioate backbone (§6), and unmodified DNA of the same sequence is a different molecule about which nothing reported here holds.

**Availability.** All code, graded artefacts and per-design tables are public at the repository [github.com/trimcrae/Rare-cancers](https://github.com/trimcrae/Rare-cancers), and are to be released as a single archived version deposited from it under [doi:10.5281/zenodo.22028916](https://doi.org/10.5281/zenodo.22028916). Every result reported here is re-derived, without network access or credentials, from the committed artefacts in that repository, which is what a reader can check today. That claim is meant to be checked rather than accepted: `./scripts/regenerate_aso_chain.sh` re-derives, in dependency order, the artefacts its own step list names: the duplex thermodynamics and the chance baseline, the locus collapse, the per-junction and non-canonical-acceptor tables, the four figures and their provenance record, the submission tables, references and metrics, the canonical sequence file, the prior-art evidence and the archive manifest. It re-runs the consistency, citation and style gates in about two and a half minutes on four cores with no network. Four producers are not in that list and have to be run separately to re-derive the central negative and its null: the design panel (`junction_aso.py`), the mature-parent screen (`aso_parent_gap_pairing.py`), the pre-mRNA screen (`aso_premrna_offtarget.py`) and the null ensembles (`aso_parent_null.py`). Two of the four read committed caches with no switch; the design panel takes its transcript models from the committed cache under `TRANSCRIPT_SOURCE=cache`, and the pre-mRNA screen scans the committed fetch under `--offline`. `ASO CHAIN OK` is a statement about the artefacts the script regenerates and not about those four. The artefacts

are current where it reports `ASO CHAIN OK` and rewrites no artefact but one: the archive manifest records the commit it is committed in, which a committed file cannot hold, so that single row moves on every run and is not evidence of staleness. The chain additionally needs the repository's `literature-cache` branch, which carries the prior-art retrieval indices and which a clone fetches but a file archive does not; `git fetch origin literature-cache` supplies it. A fuller guard suite in the same repository re-derives each reported number from its artefact and fails if the two diverge; its invocation is documented there rather than here, since it is repository tooling and not a step in reproducing a result. Which file holds which number is given here rather than left to a search. Under `research/modalities/`: the design panel is `nr4a3-fusion-junction-atlas.json`; the transcript models `emc-construct-inputs.json` and the unspliced sequence and exon coordinates `aso-premrna-sequences.json`; screen 3 `aso-premrna-offtarget.json`, screen 4 `aso-parent-gap-pairing.json` and its nulls `aso-parent-null.json`, screen 5 `aso-genome-offtarget.json`; the alignment screens one file per junction under `junction-aso-offtarget-*.json`; the free energies and the four design rules `junction-aso-thermo.json`; the chance baseline `offtarget-chance-baseline.json`; the expression readings `aso-offtarget-tissue-expression.json`; and the gap-length comparison `aso-gap-length-tradeoff.json`. The *TCF12* base-level breakpoint assignment of §2.3 is `research/manuscripts/aso/tcf12-breakpoint-assignment.json`. Regenerating the specificity screens from scratch is not offline, because the alignment screen queries NCBI BLAST and the exhaustive transcript scan downloads the GRCh38.p14 RefSeq RNA set, but no reported number requires it: each screen's hit set is archived and the re-scores hold it fixed. The pre-mRNA and mature-parent screens are fully offline against the repository, since the retrieved unspliced sequence and exon coordinates are committed to it and will travel into the archive unchanged.

## Declarations

### Research use only, and not for administration to any person or animal.

Every oligonucleotide sequence in this manuscript, its Supplementary Information and its tables — the two lead reagents and the two named for coverage, the three named as not to be used, the second-geometry control and any scrambled control drawn by the procedure of §4.4, every design in the released panel, and the non-canonical-acceptor designs reported beside that panel rather than pooled into it — is a research reagent intended solely for laboratory investigation. None is a medicine, an investigational medicinal product or a candidate drug. None has been synthesised, formulated, tested in any cell, tissue or animal, or evaluated for potency, purity, stability, immunogenicity or toxicity by anyone. None may be administered to any human being or animal, compounded or formulated for such use, or supplied to any person for such use, and nothing in this manuscript should be read as recommending, authorising or enabling that. Custom oligonucleotide synthesis is commercially available, so the sequences here are orderable by anyone; that is a fact about the supply chain and not a statement about what they are fit for. A person with extraskeletal myxoid chondrosarcoma seeking treatment should be directed to a clinical trial or a sarcoma centre.

**Supplementary Information.** The sections cited above as SI §S1 to §S6 are in `fusion-junction-aso-research-article-supplementary-information.pdf`, rendered to PDF from the same builder as this manuscript and deposited as a separate file beside it, and included in the archive below.

**Data and code availability.** The archive is that single deposited version, under [doi:10.5281/zenodo.22028916](https://doi.org/10.5281/zenodo.22028916), from [github.com/trimcrae/Rare-cancers](https://github.com/trimcrae/Rare-cancers). `fusion-junction-aso-archive-manifest.json` lists every archived file with its SHA-256; it is generated with the archive and travels with the deposit. Two renderings of this manuscript travel with it and their text is the same: `fusion-junction-aso-research-article-manuscript.pdf` in submission format is the version of record, the one to cite and to deposit onward, and the typeset preview beside it is the same text set as a printed article. `fusion-junction-aso-submission-tables.md` is the machine-readable copy of Tables 1 to 7, whose content is printed in both renderings. Artefacts include the graded junction atlas, per-junction design panels, all five screens, the per-junction reagent table behind Table 2, the graded re-scores under both discrimination bounds, and the retrieval records for every literature claim.

**Provenance and corrections.** An earlier version of these analyses placed the acceptor junction incorrectly through a coding-versus-transcript exon indexing error and was withdrawn in full; all panels were rebuilt and verified against two independent transcript acquisitions. The complete correction record, including every superseded value, is released with the archive.

Because that is the failure a reader will reasonably assume could recur, the two instruments the paper's conclusions rest on were reimplemented a second time and the two implementations compared. The reimplementation shares no code with the original and differs from it on four axes: it splices each mature transcript out of the genomic record rather than reading the cDNA record; it locates each gene's coding start by open-reading-frame search rather than by reading the annotated 5' untranslated length, which is the class of value the retracted error turned on; it grades the reading frame by arithmetic on exon coding-length vectors rather than by translating the chimera; and it computes the mature-parent screen by substring search over the design's gap-containing substrings rather than by scanning every parent window and extending outward from the gap. The two agree on all 231 graded exon pairs, field by field and not only on the grade, and on the longest parent duplex of all 190 designs, giving the same 87 and the same 61 against NR4A3. The two transcript acquisitions agree base for base for all six genes, and the annotation-free coding start reproduces the annotated one for all six. Both implementations, the comparison and its deliberate-corruption tests are in the archive.

That check should not be read as either of two things. It is not external review: the same author prepared both implementations. And it bounds implementation error only — two implementations of a specification that is itself wrong will agree with each other and both be wrong, so agreement here is not evidence that the longest contiguous run containing the catalytic gap is the right quantity to compute. Independent review of the code by another group remains wanted and is not claimed.

**Competing interests.** The author declares no financial competing interests: he holds no position, equity, consultancy or patent relating to any gene, sequence or agent named here, and no oligonucleotide described in this manuscript has been synthesised, licensed or offered for sale. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

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**Ethics.** No human subjects, human material or animals were involved. All clinical figures are taken from published reports and are cited.

**Use of AI tools.** A large language model (Claude, Anthropic) was used throughout this work: to write the analysis code, to run the graded design and screening pipelines, to draft and revise this manuscript, and to conduct internal critical review of earlier drafts. The work ran from 21 June 2026, the first commit of the design code, to the commit this version was built from, whose date the repository's commit record carries; an end date typed here would exclude the deposited version from its own declared span, since preparing that version is itself part of the work. That record names two model versions over the span and no other: Claude Opus 4.8 for the design and screening code committed between 21 June and 3 July 2026, and Claude Opus 5 for everything committed from 6 August 2026 onward, which includes every version of this manuscript. Every quantitative statement derived from sequence or from a screen is produced by code in the released archive and is reproducible from it, while the clinical figures are transcribed from the publications cited for them; no numerical result was generated by a language model directly. Every literature identifier was checked against a retrieved bibliographic record, and identifiers that could not be so anchored were removed rather than retained. The three references dated 2026 were additionally checked for live resolution against two independent registries, Europe PMC and Crossref, which return the same title and the same digital object identifier for each; those records travel with the archive. The frame grading and the mature-parent screen were additionally reimplemented and cross-checked as described under Provenance, which bounds implementation error but is not independent review. The author takes full responsibility for the content, including for the correctness of the code and for the interpretation of the results.

**Author contributions.** T.D.M. is the sole author and directed all work reported here: conception, design, analysis, code and manuscript. The use of a language model in producing the code and the manuscript text is declared above and is not restated here as authorship.

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53 entries, numbered by first citation in this manuscript. Metadata is read from retrieved bibliographic records; an unretrieved field is left absent rather than completed. 9 data sources follow, unnumbered.

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